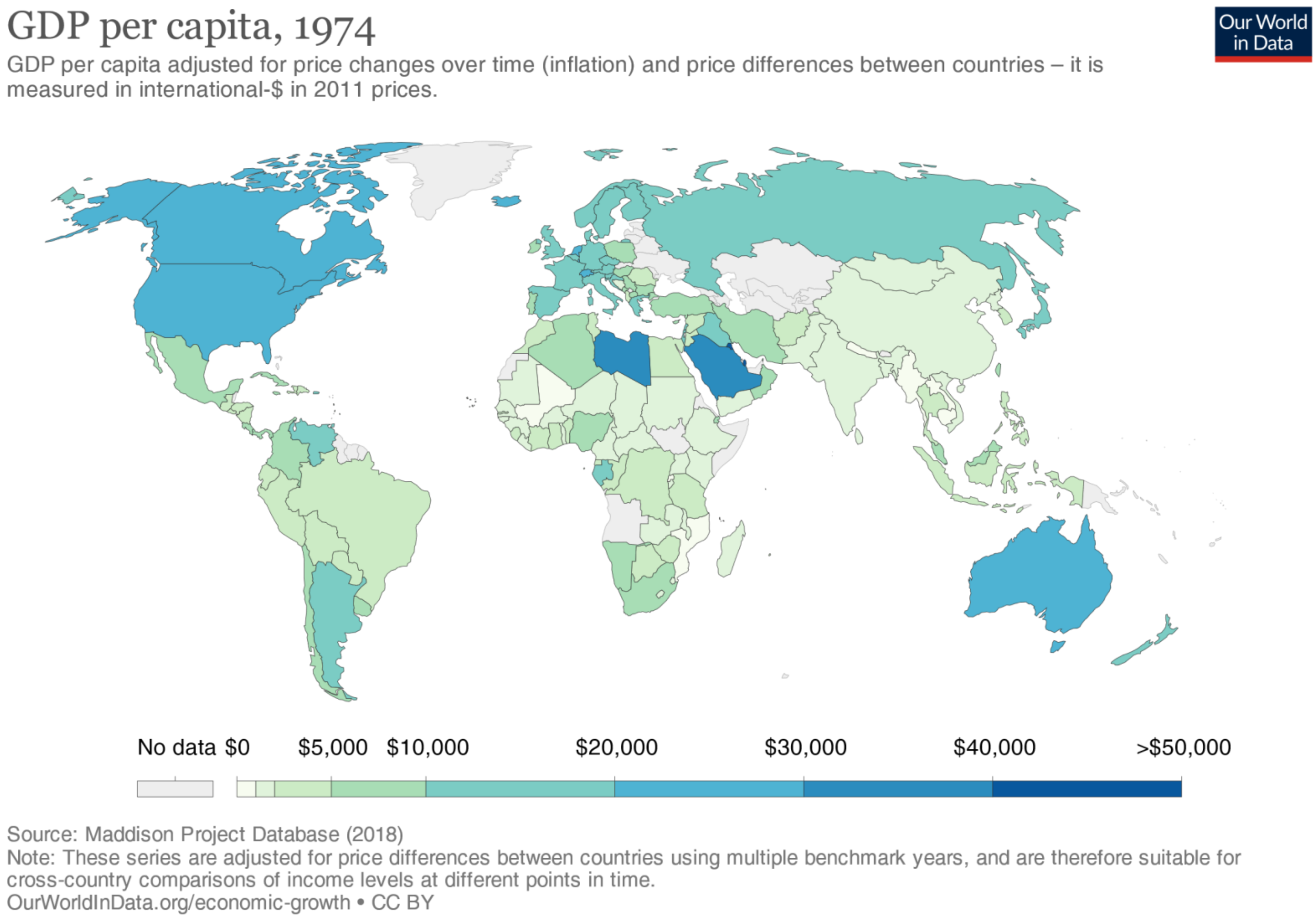
Environmental Management (HS200)

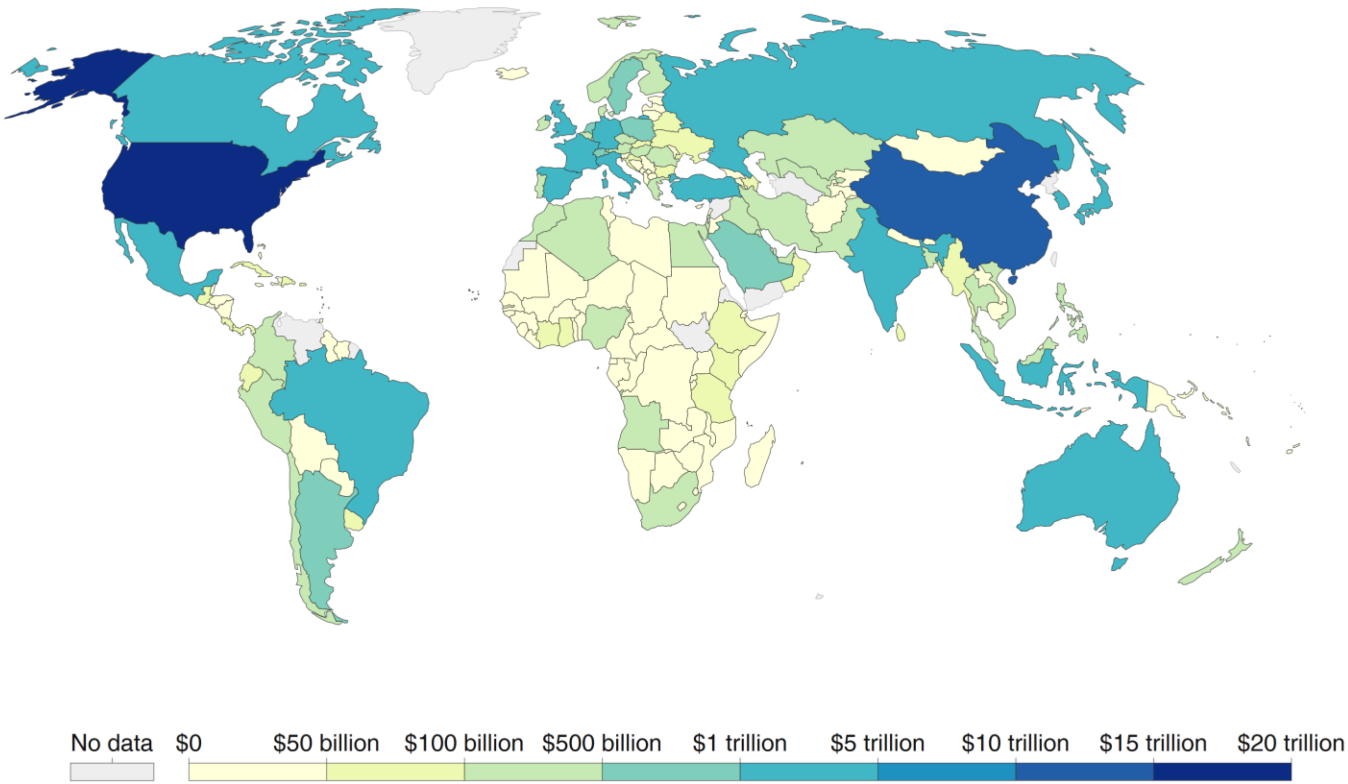
Prof. Haripriya Gundimeda

World then

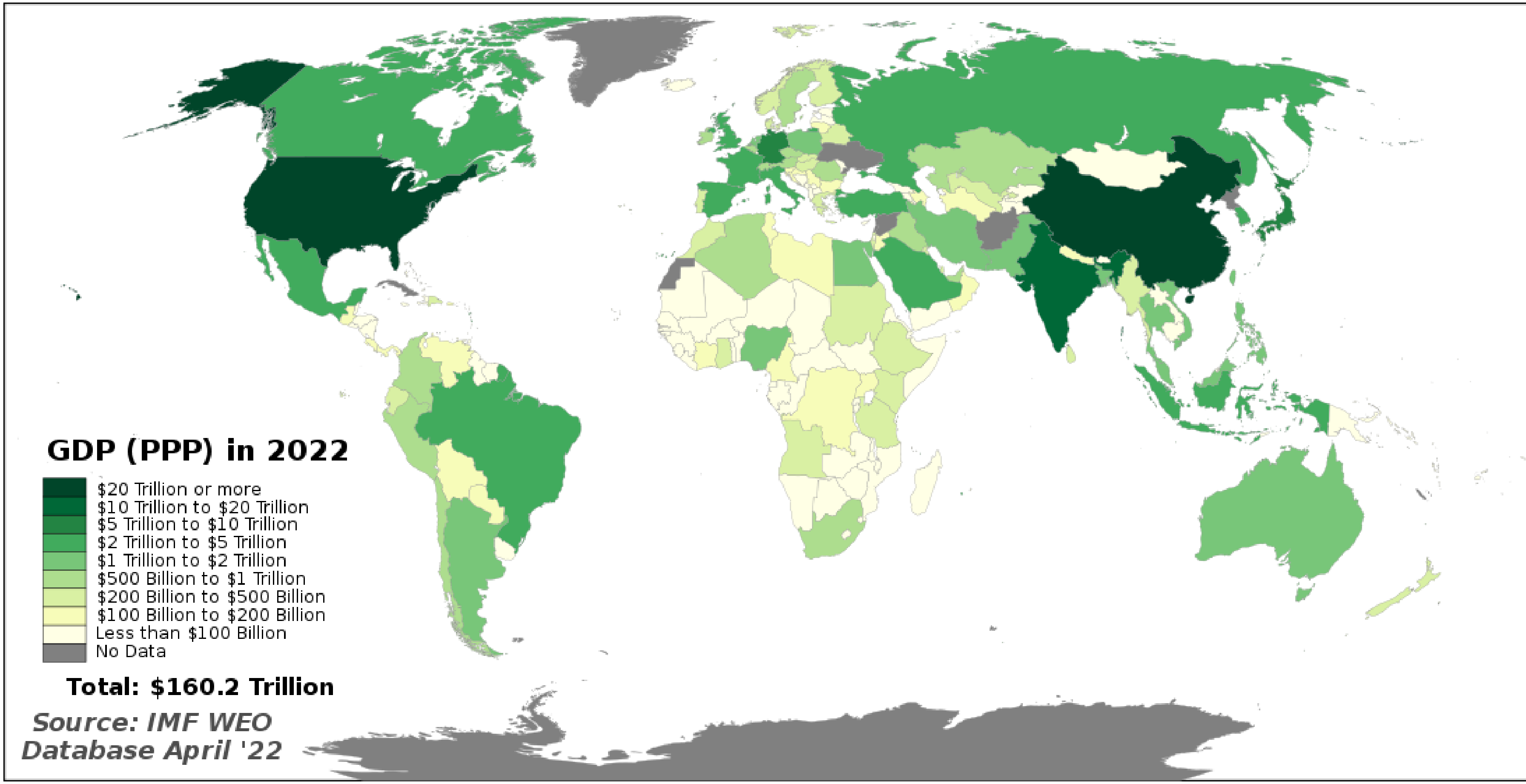

GDP per capita, 1974
GDP per capita adjusted for price changes over time (inflation) and price differences between countries – it is measured in international-$ in 2011 prices.
Our World in Data
No data
$0
$5,000
$10,000
$20,000
$30,000
$40,000
>$50,000
Source: Maddison Project Database (2018)
Note: These series are adjusted for price differences between countries using multiple benchmark years, and are therefore suitable for cross-country comparisons of income levels at different points in time.
OurWorldInData.org/economic-growth • CC BY

Gross domestic product (GDP), 2020

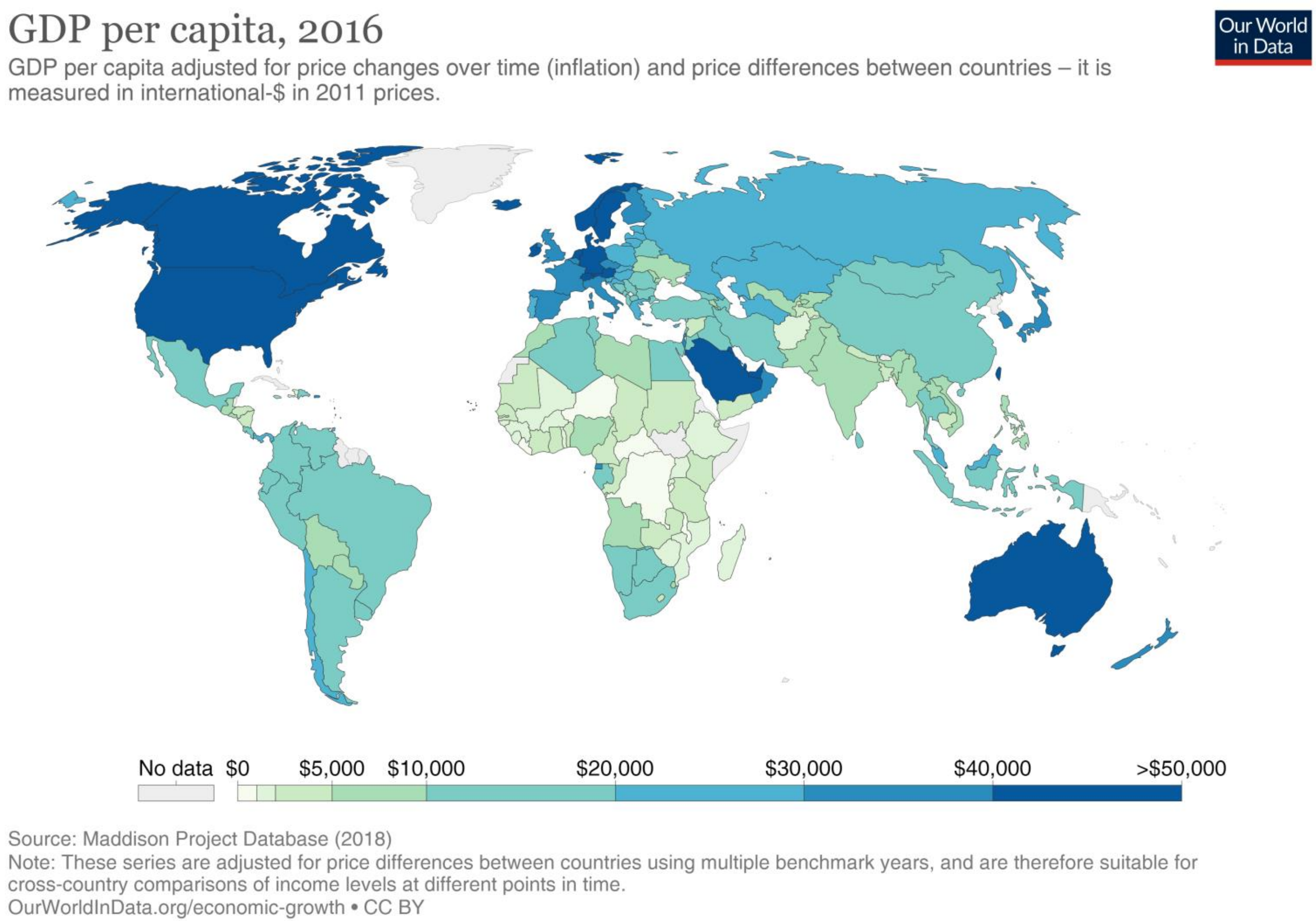
Gross domestic product adjusted for price changes over time (inflation) and expressed in US-Dollars.



Source: World Bank and OECD

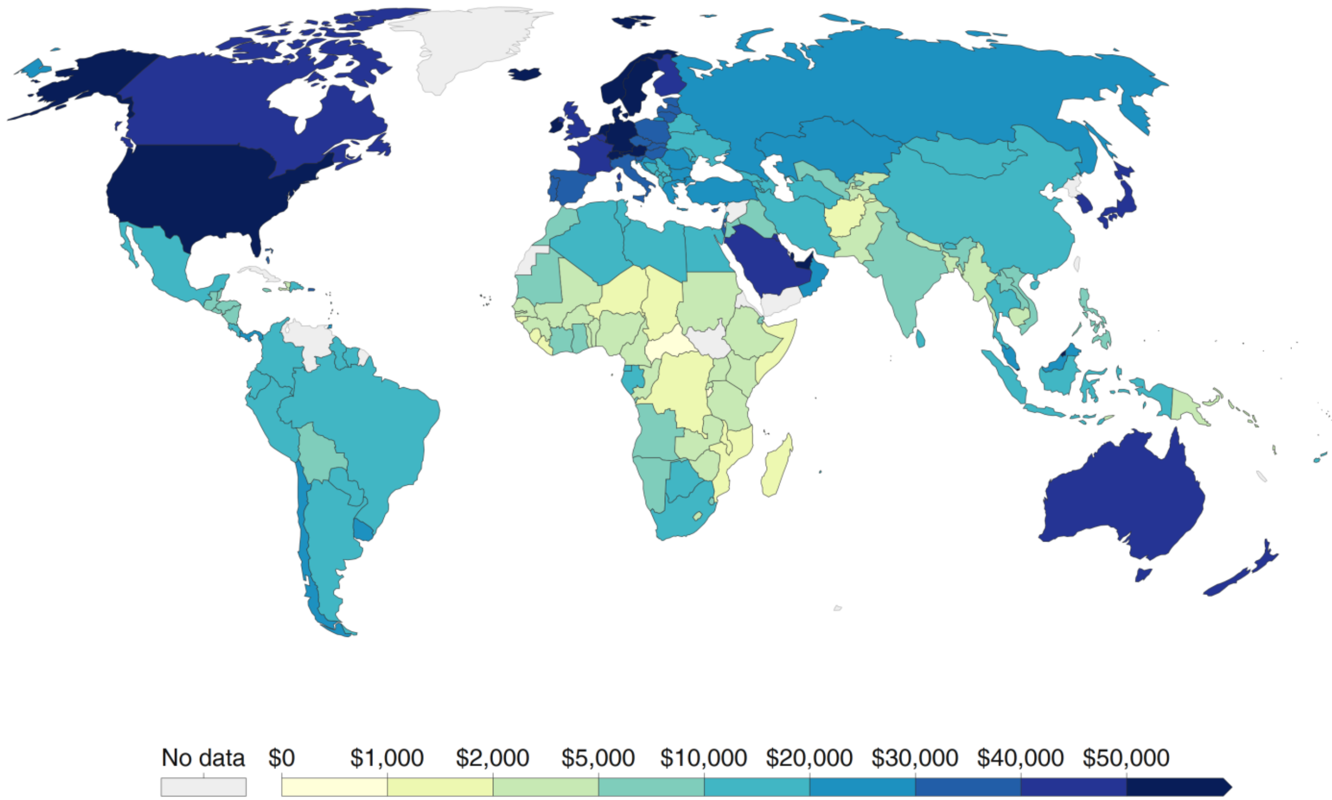

GDP (PPP) in 2022
$20 Trillion or more
$10 Trillion to $20 Trillion
$5 Trillion to $10 Trillion
$2 Trillion to $5 Trillion
$1 Trillion to $2 Trillion
$500 Billion to $1 Trillion
$200 Billion to $500 Billion
$100 Billion to $200 Billion
Less than $100 Billion
No Data
Total: $160.2 Trillion
Source: IMF WEO Database April '22

Economic Development – improvement in living standards



GDP per capita, 2020

Measured in constant international-$.



Source: Data compiled from multiple sources by World Bank

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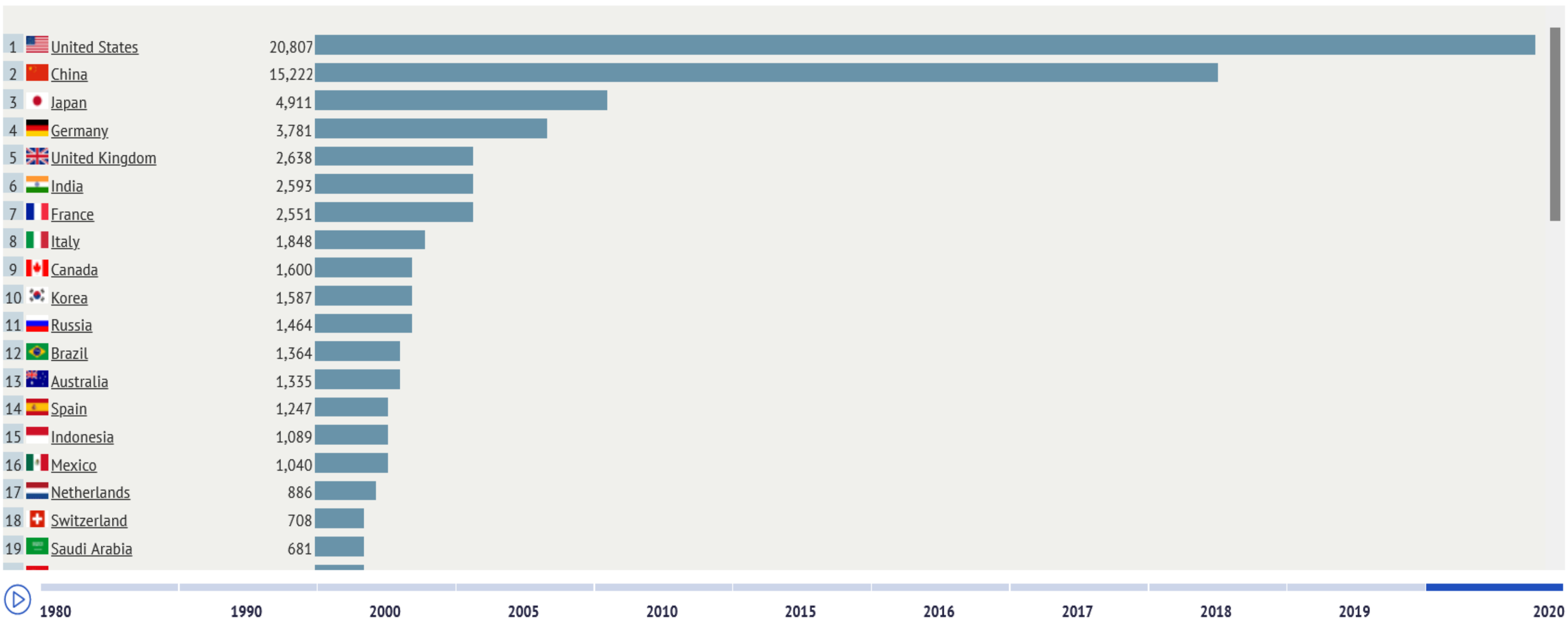
GDP in Current Prices

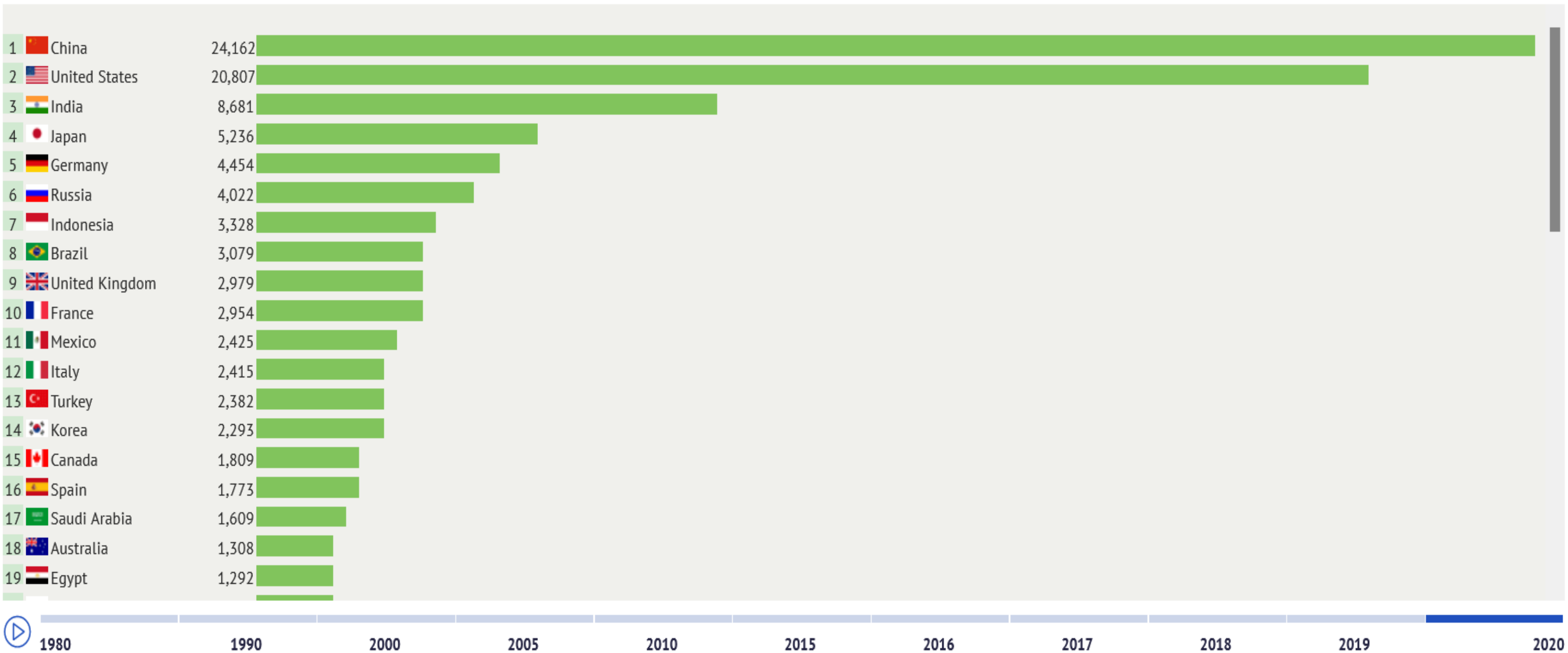
Billion USD



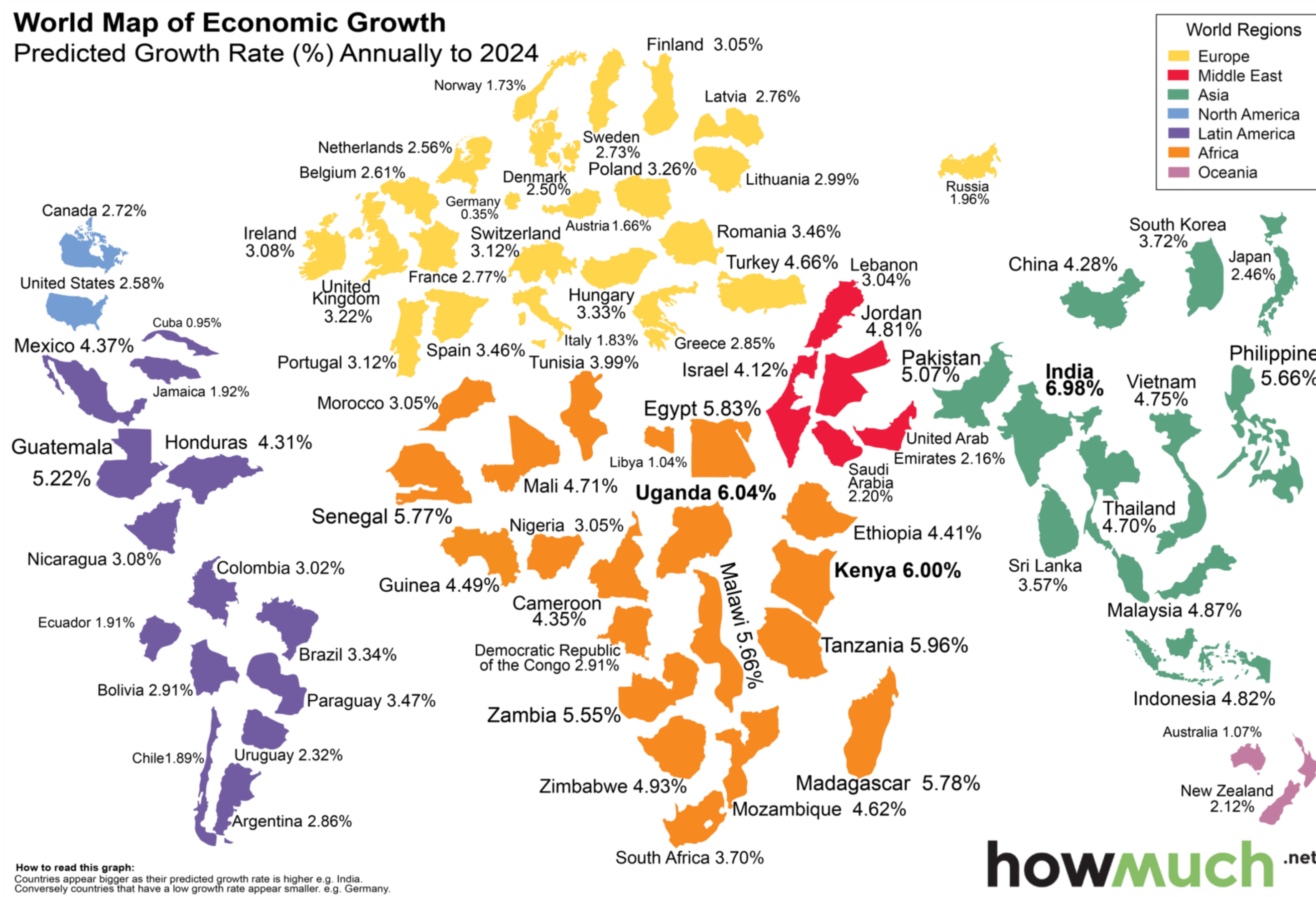
OK

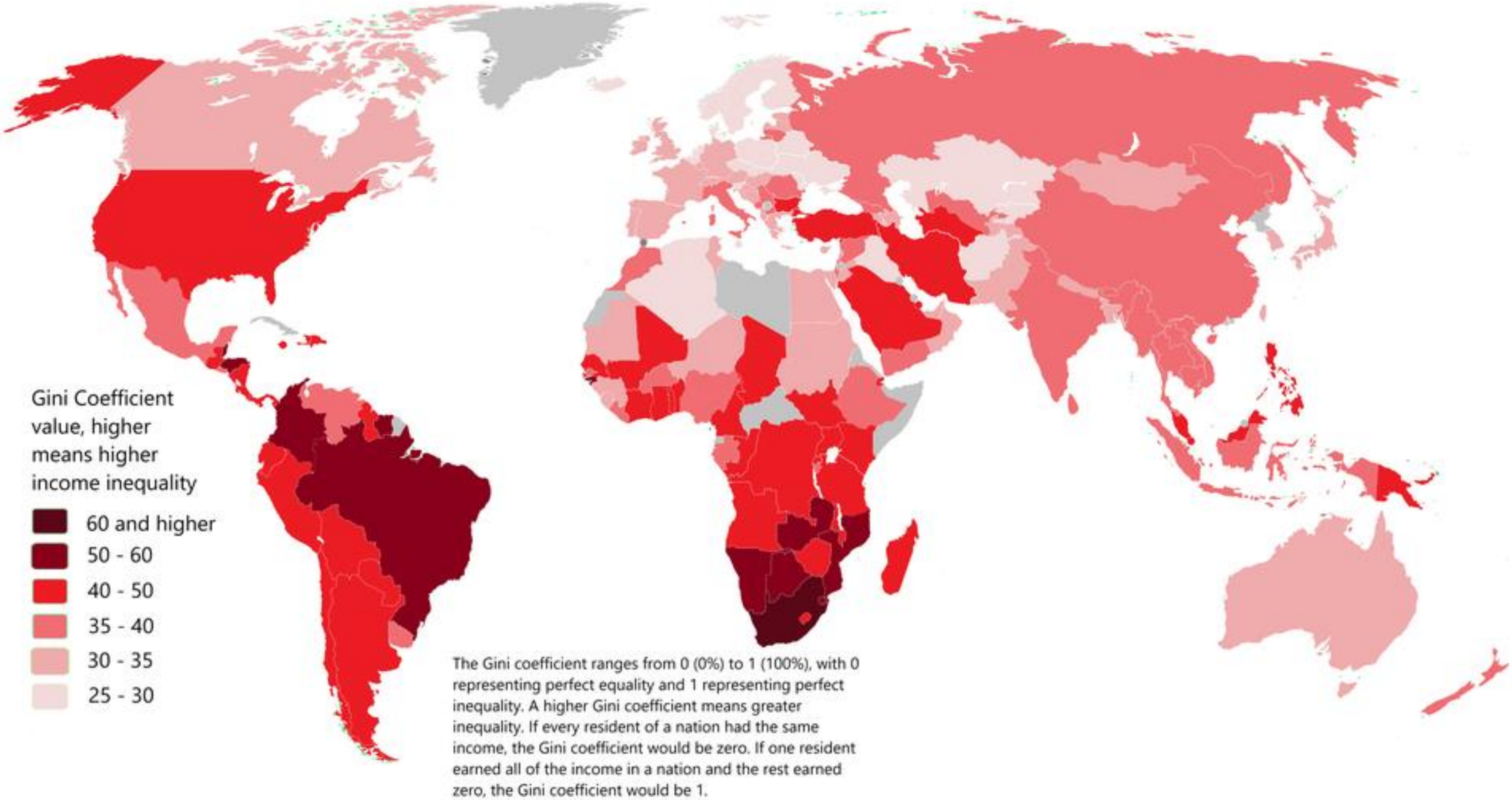
GDP Based on PPP Valuation

Billion current international dollars



OK

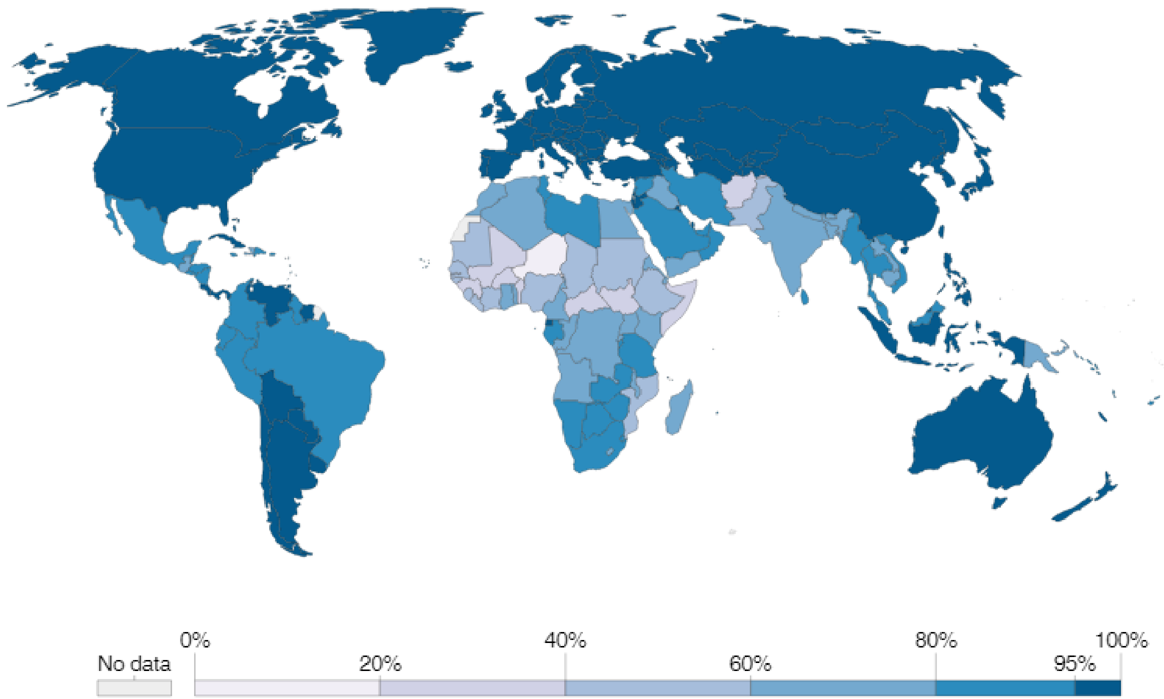

World Map of Economic Growth
Predicted Growth Rate (%) Annually to 2024
World Regions
Europe
Middle East
Asia
North America
Latin America
Africa
Oceania
Finland 3.05%
Norway 1.73%
Latvia 2.76%
Sweden 2.73%
Netherlands 2.56%
Belgium 2.61%
Denmark 2.50%
Poland 3.26%
Lithuania 2.99%
Russia 1.96%
Germany 0.35%
Austria 1.66%
Canada 2.72%
Ireland 3.08%
Switzerland 3.12%
Romania 3.46%
South Korea 3.72%
Japan 2.46%
United States 2.58%
France 2.77%
Turkey 4.66%
Lebanon 3.04%
China 4.28%
United Kingdom 3.22%
Hungary 3.33%
Jordan 4.81%
Cuba 0.95%
Mexico 4.37%
Portugal 3.12%
Spain 3.46%
Italy 1.83%
Greece 2.85%
Pakistan 5.07%
India 6.98%
Philippines 5.66%
Tunisia 3.99%
Israel 4.12%
Vietnam 4.75%
Jamaica 1.92%
Morocco 3.05%
Egypt 5.83%
United Arab Emirates 2.16%
Guatemala 5.22%
Honduras 4.31%
Libya 1.04%
Saudi Arabia 2.20%
Mali 4.71%
Uganda 6.04%
Senegal 5.77%
Thailand 4.70%
Nigeria 3.05%
Ethiopia 4.41%
Nicaragua 3.08%
Kenya 6.00%
Sri Lanka 3.57%
Colombia 3.02%
Guinea 4.49%
Malawi 5.66%
Cameroon 4.35%
Malaysia 4.87%
Ecuador 1.91%
Brazil 3.34%
Tanzania 5.96%
Democratic Republic of the Congo 2.91%
Bolivia 2.91%
Paraguay 3.47%
Indonesia 4.82%
Zambia 5.55%
Australia 1.07%
Chile1.89%
Uruguay 2.32%
Madagascar 5.78%
New Zealand 2.12%
Zimbabwe 4.93%
Mozambique 4.62%
Argentina 2.86%
South Africa 3.70%
howmuch.net
How to read this graph:
Countries appear bigger as their predicted growth rate is higher e.g. India.
Conversely countries that have a low growth rate appear smaller. e.g. Germany.


Gini Coefficient value, higher means higher income inequality
60 and higher
50 - 60
40 - 50
35 - 40
30 - 35
25 - 30
The Gini coefficient ranges from 0 (0%) to 1 (100%), with 0 representing perfect equality and 1 representing perfect inequality. A higher Gini coefficient means greater inequality. If every resident of a nation had the same income, the Gini coefficient would be zero. If one resident earned all of the income in a nation and the rest earned zero, the Gini coefficient would be 1.

Literacy rate, 2015

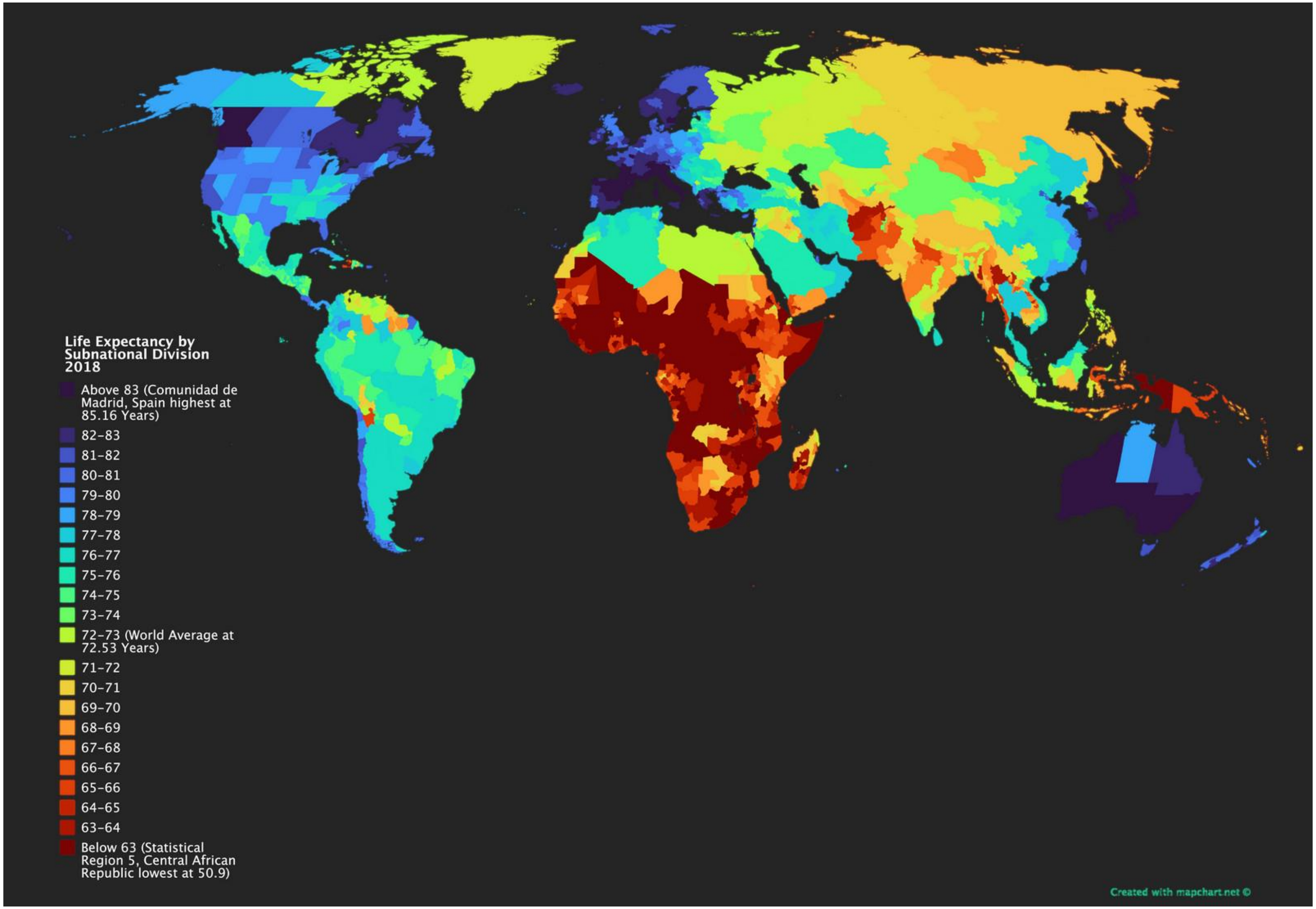
Our World in Data

Estimates correspond to the share of the population older than 14 years that is able to read and write.



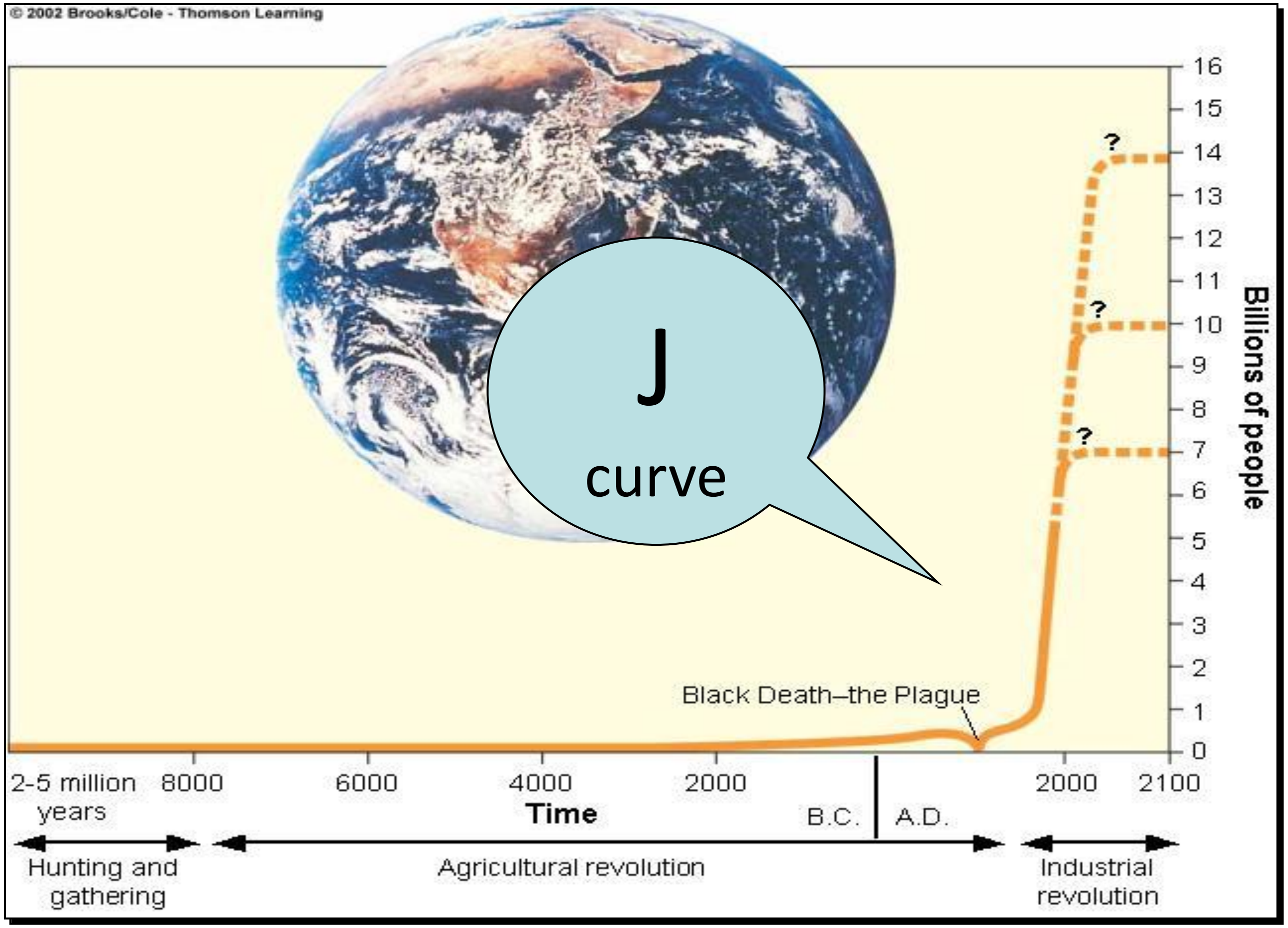
Source: WDI, CIA World Factbook, & other sources

Note: Specific definitions and measurement methodologies vary across countries and time. See the 'Sources'-tab for more details.

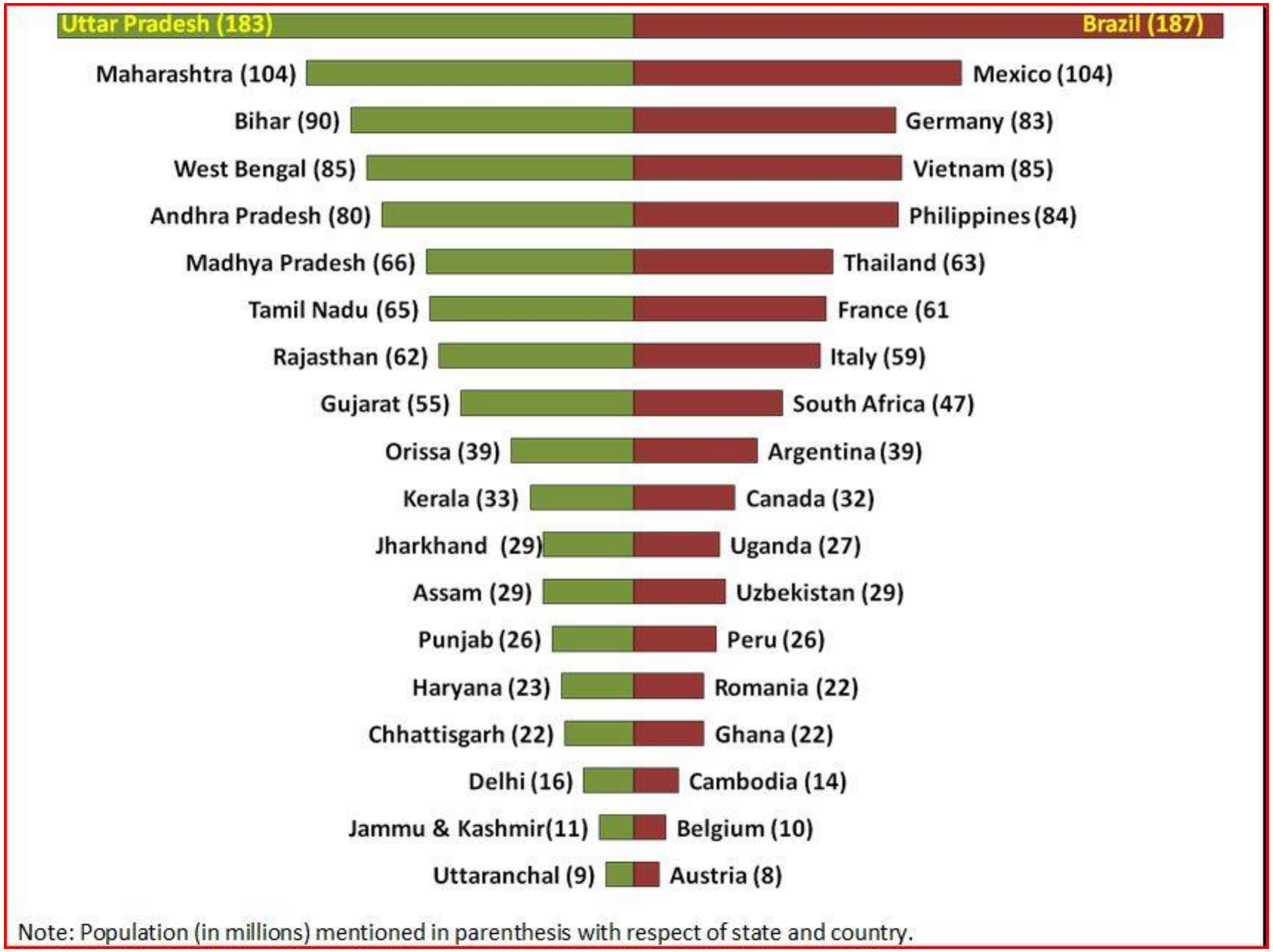


Life Expectancy by Subnational Division 2018
Above 83 (Comunidad de Madrid, Spain highest at 85.16 Years)
82-83
81-82
80-81
79-80
78-79
77-78
76-77
75-76
74-75
73-74
72-73 (World Average at 72.53 Years)
71-72
70-71
69-70
68-69
67-68
66-67
65-66
64-65
63-64
Below 63 (Statistical Region 5, Central African Republic lowest at 50.9)
Created with mapchart.net ©

World Population



Comparison of population in Indian states with that of different countries



World’s most polluted cities (air pollution)

Rank	City	2021
1	Bhiwadi, India	106.2
2	Ghaziabad, India	102
3	Hotan, China	101.5
4	Delhi, India	96.4
5	Jaunpur, India	95.3
6	Faisalabad, Pakistan	94.2
7	Noida, India	91.4
8	Bahawalpur, Pakistan	91
9	Peshawar, Pakistan	89.6
10	Bagpat, India	89.1

Rank	City	2021
11	Hisar, India	89
12	Faridabad, India	88.9
13	Greater Noida, India	87.5
14	Rohtak, India	86.9
15	Lahore, Pakistan	86.5
16	Lucknow, India	86
17	Jind, India	84.1
18	Gurugram, India	83.4
19	Kashgar, China	83.2
20	Kanpur, India	83.2

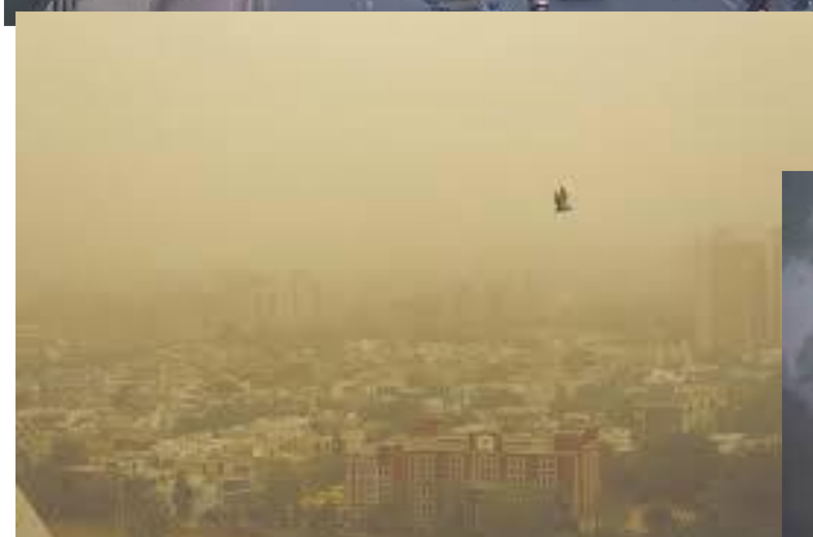
Source: Hindustan times, March 22, 2022

Bellandur Lake, Bangalore

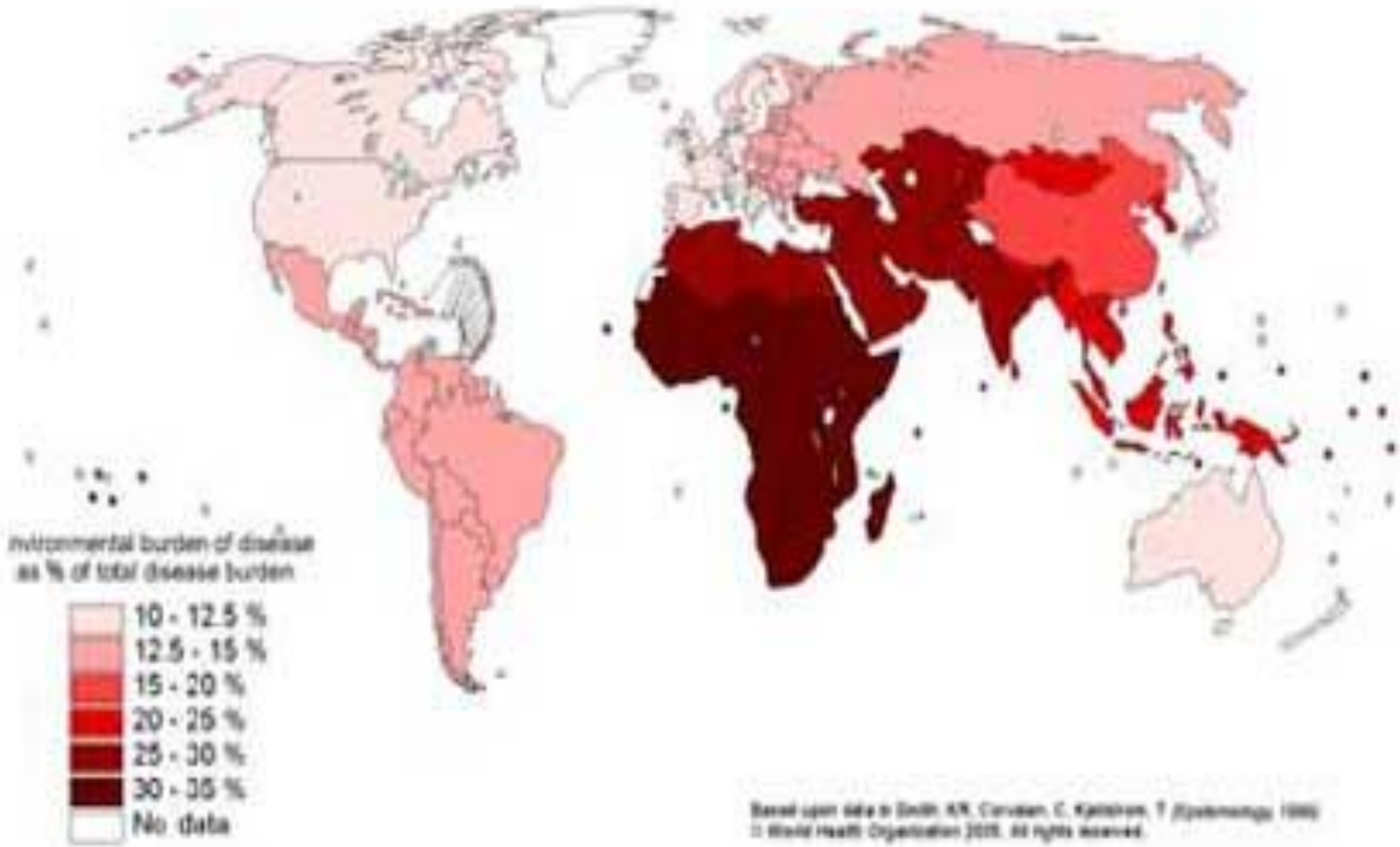
City of burning lakes: experts fear Bangalore will be uninhabitable by 2025

River pollution

E-waste

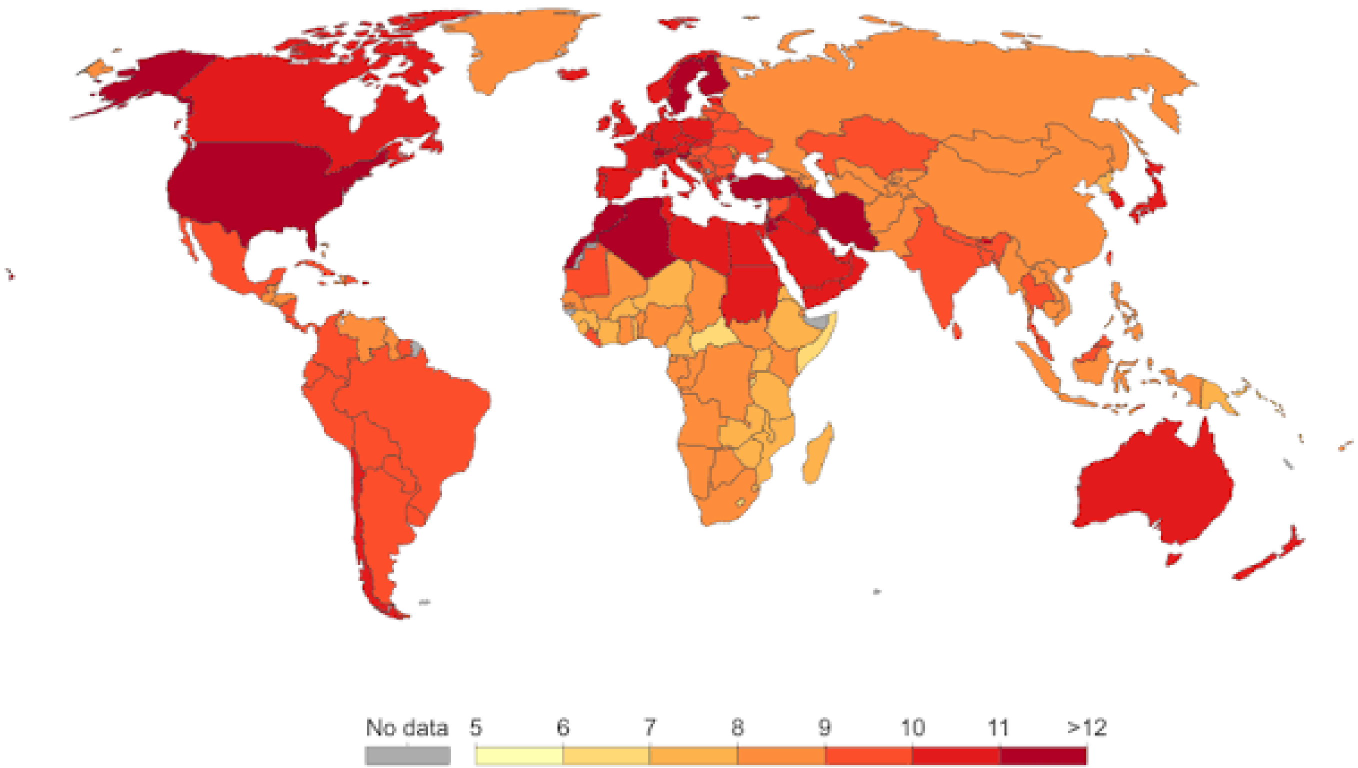


Environmental burden of disease globally



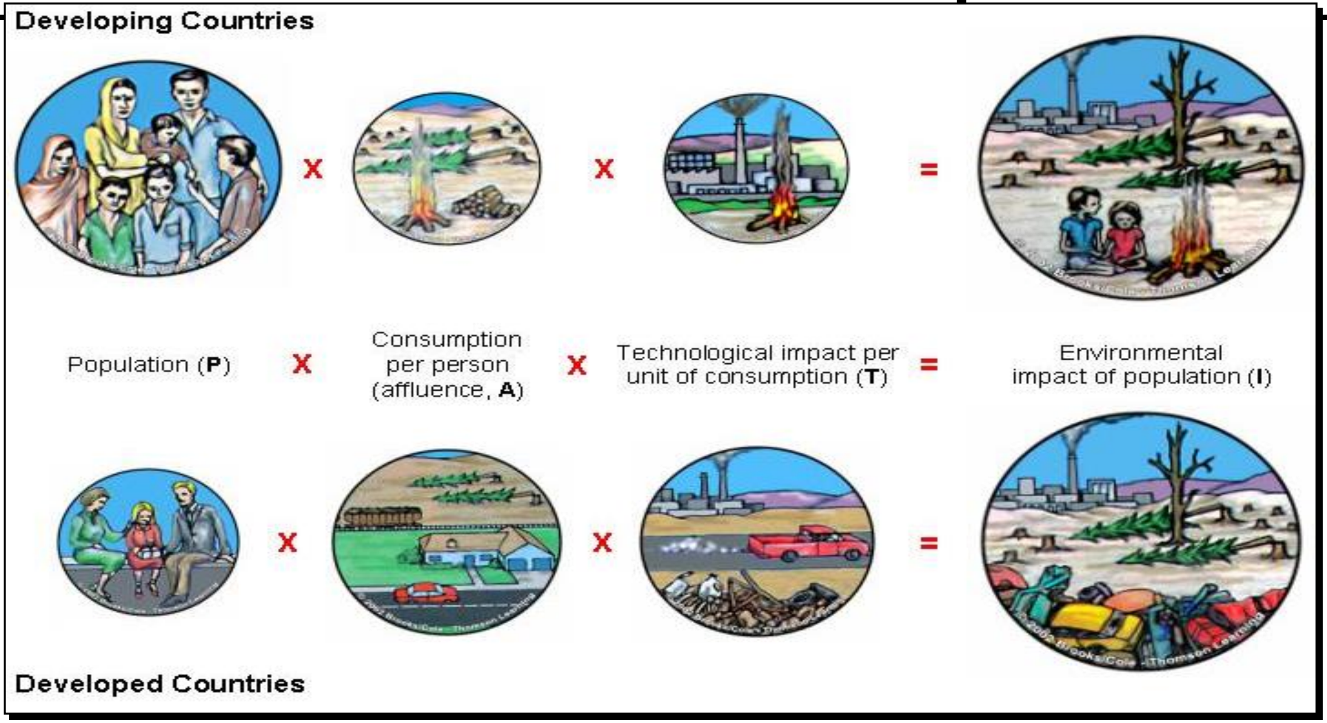
Expected years of living with disability or disease burden, 2016

Average number of years with disability an individual born in the respective year can expect to experience. This is calculated as the difference between total and healthy life expectancy.



Source: IHME, Global Burden of Disease

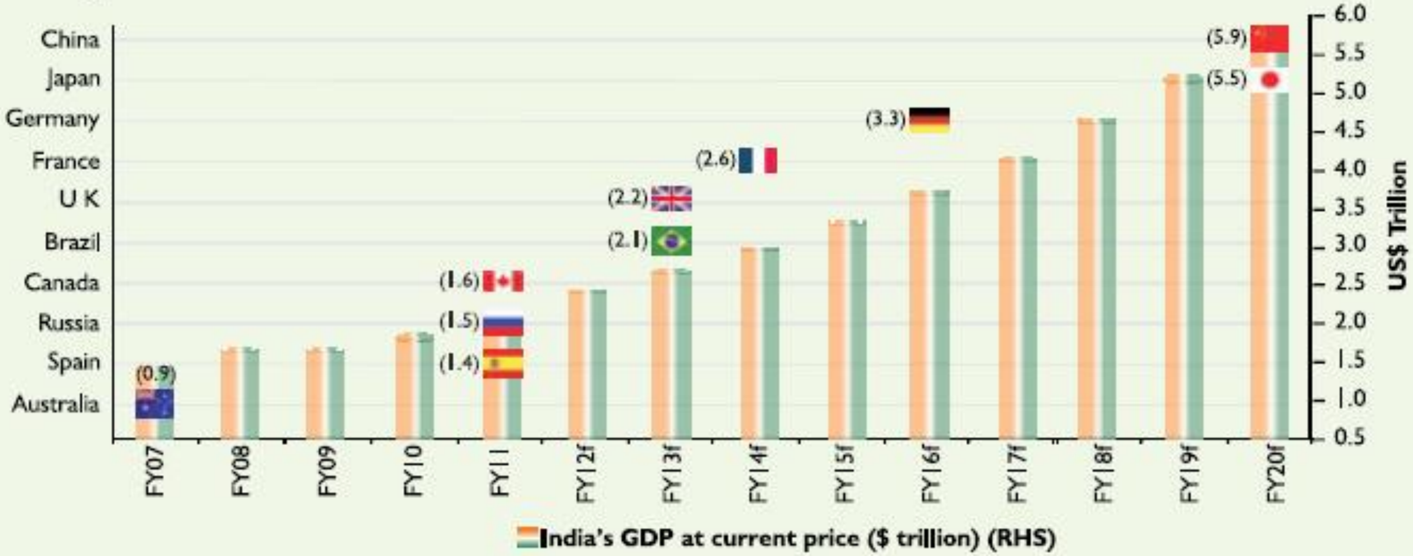
Environmental Impact



Fi. 1-11 p. 13, Tylor

India shining

Chart 1.3: India to cross the GDP (as of 2010 in current US$) of major developed countries during the current decade



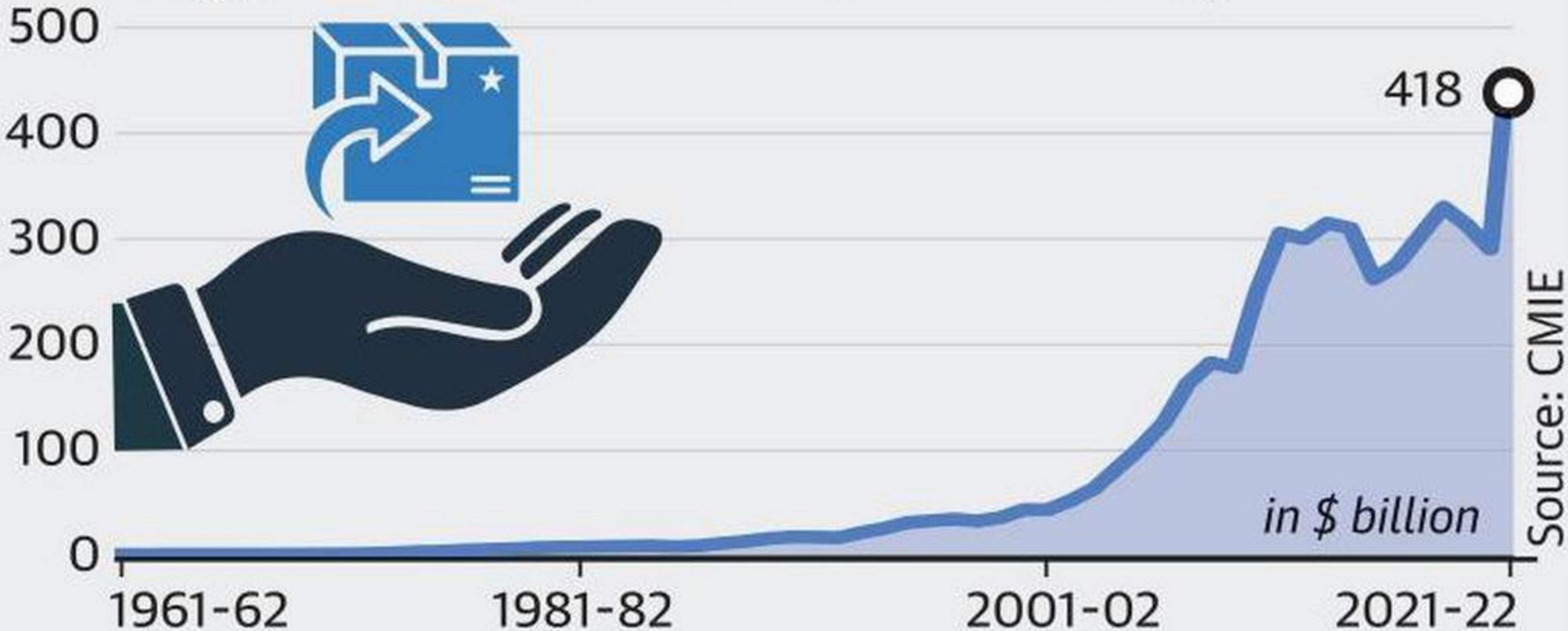
According to D&B's estimates, in the journey during the current decade as India traverses its high growth path it would eventually achieve and surpass the GDP levels (in 2010 at current international $) of some of the major developed countries in the world.

- India has already crossed Australia's GDP (in 2010 at current US$) in FY07 and Spain, Russia and Canada in FY11.
- It is likely to cross Brazil and UK in FY13 and Germany in FY16.
- It is expected to cross Japan by FY20. India will be close to China in FY20.

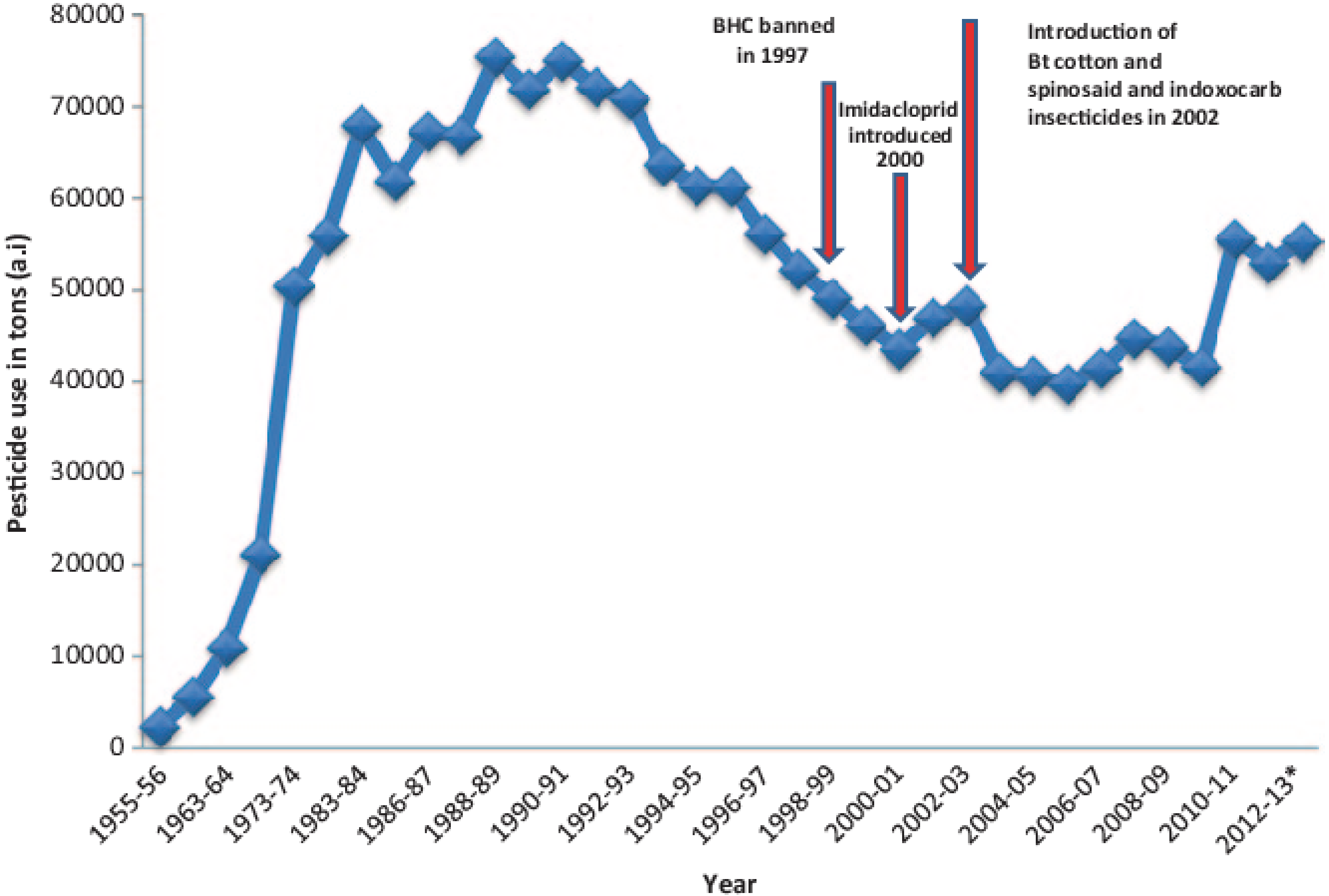
Note: Figures in the parentheses are GDP of respective countries in current US$ trillion
Value of GDP of all the countries, except India, taken as of 2010
For Australia, GDP is for 2009 (at current US$)
Source: World Bank, CSO, D&B India

Record-high surge in exports

India's exports spurted to $418 billion in the 2021-22 fiscal, crossing $400 billion for the first time in history

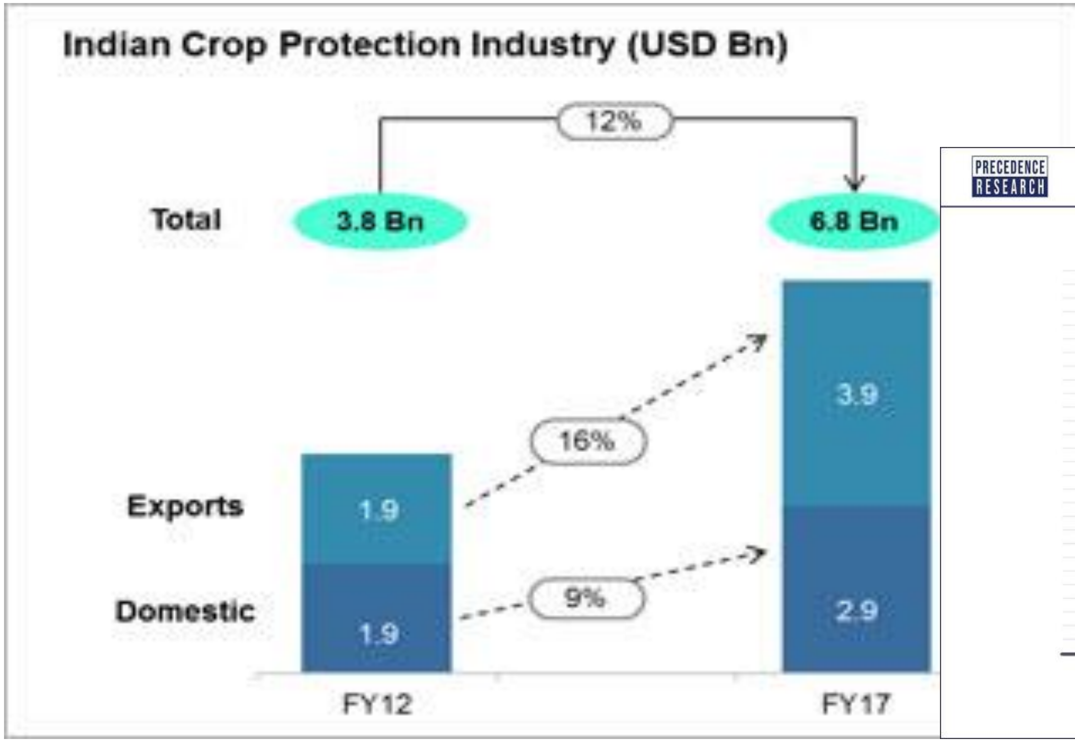


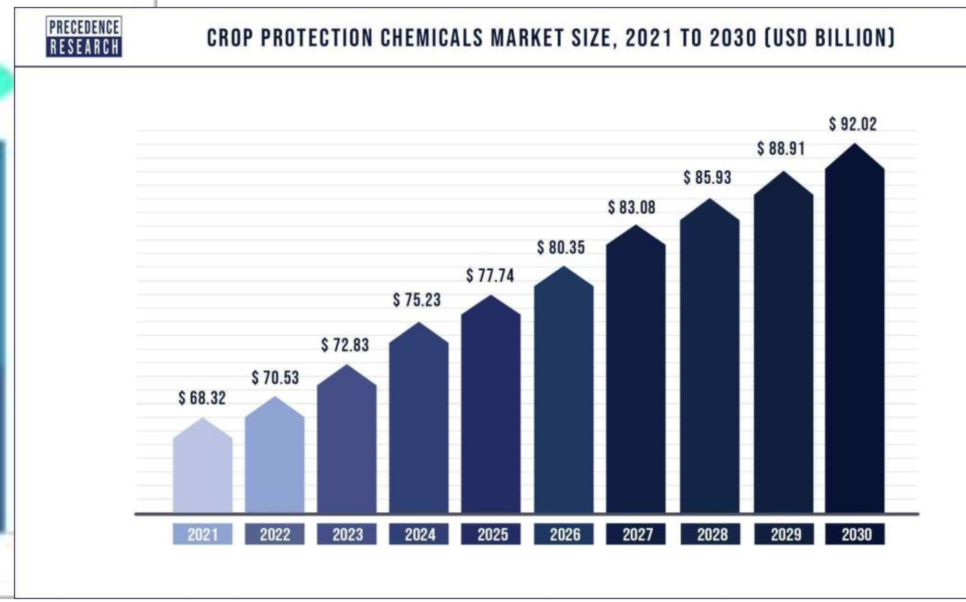
Pesticide Use in India



Source: Granthalayam

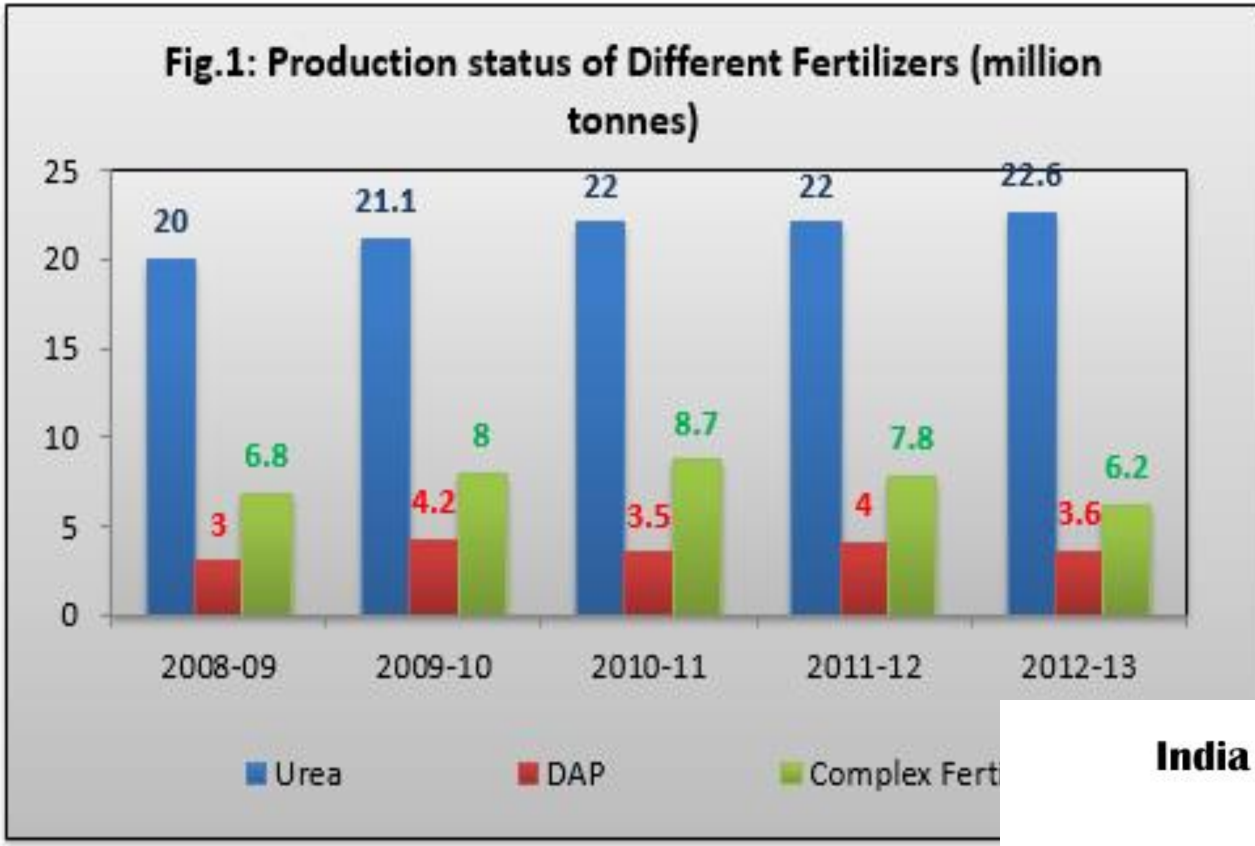
India Development Inc…

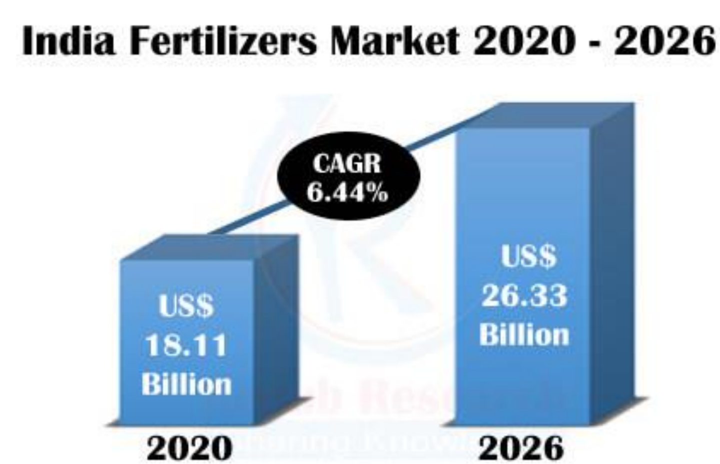




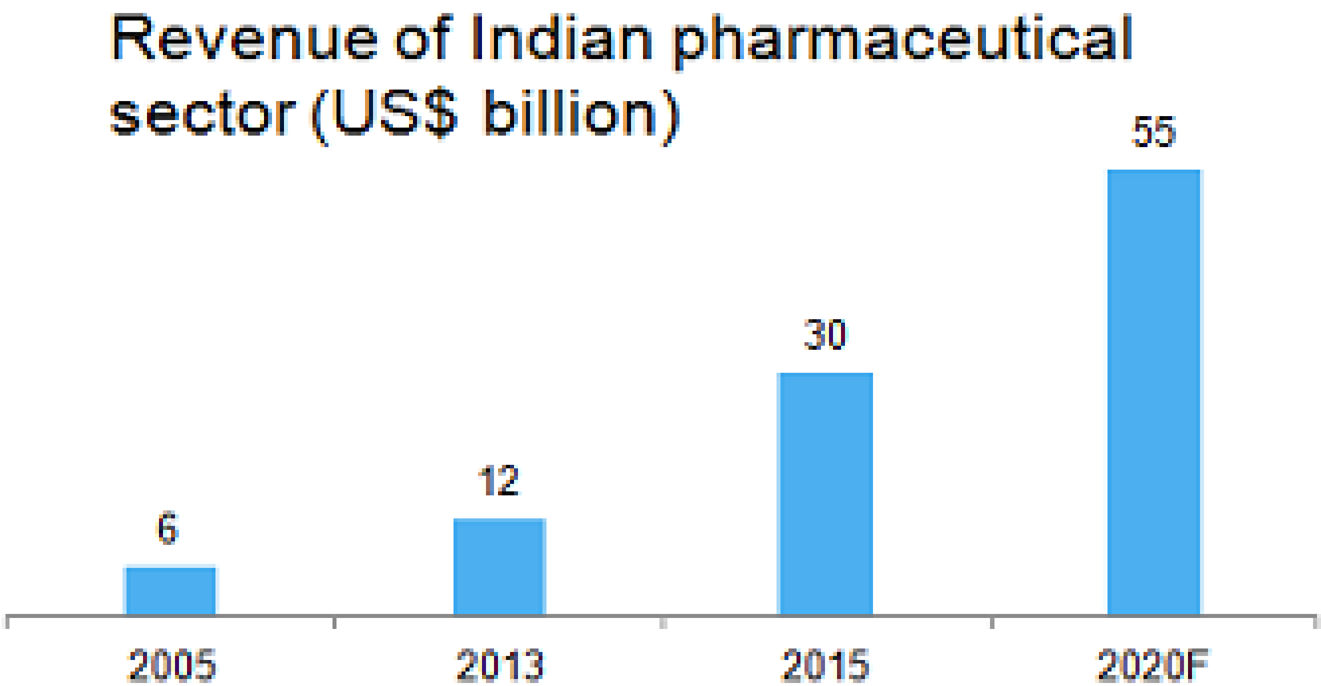
Source: Business standard

Good times for Indian fertiliser makers



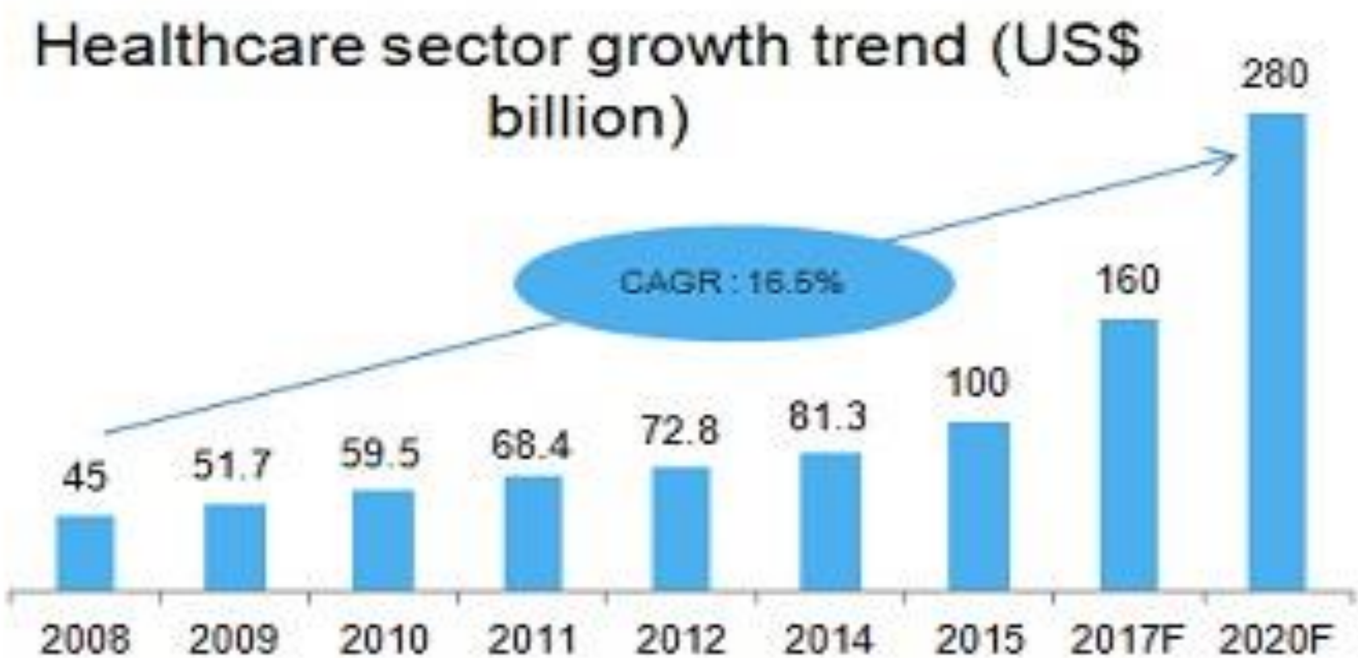


India’s booming health story



Source: Department of Pharmaceuticals, PwC, McKinsey, TechSci Research
Notes: F - Forecast, CAGR - Compound Annual Growth Rate

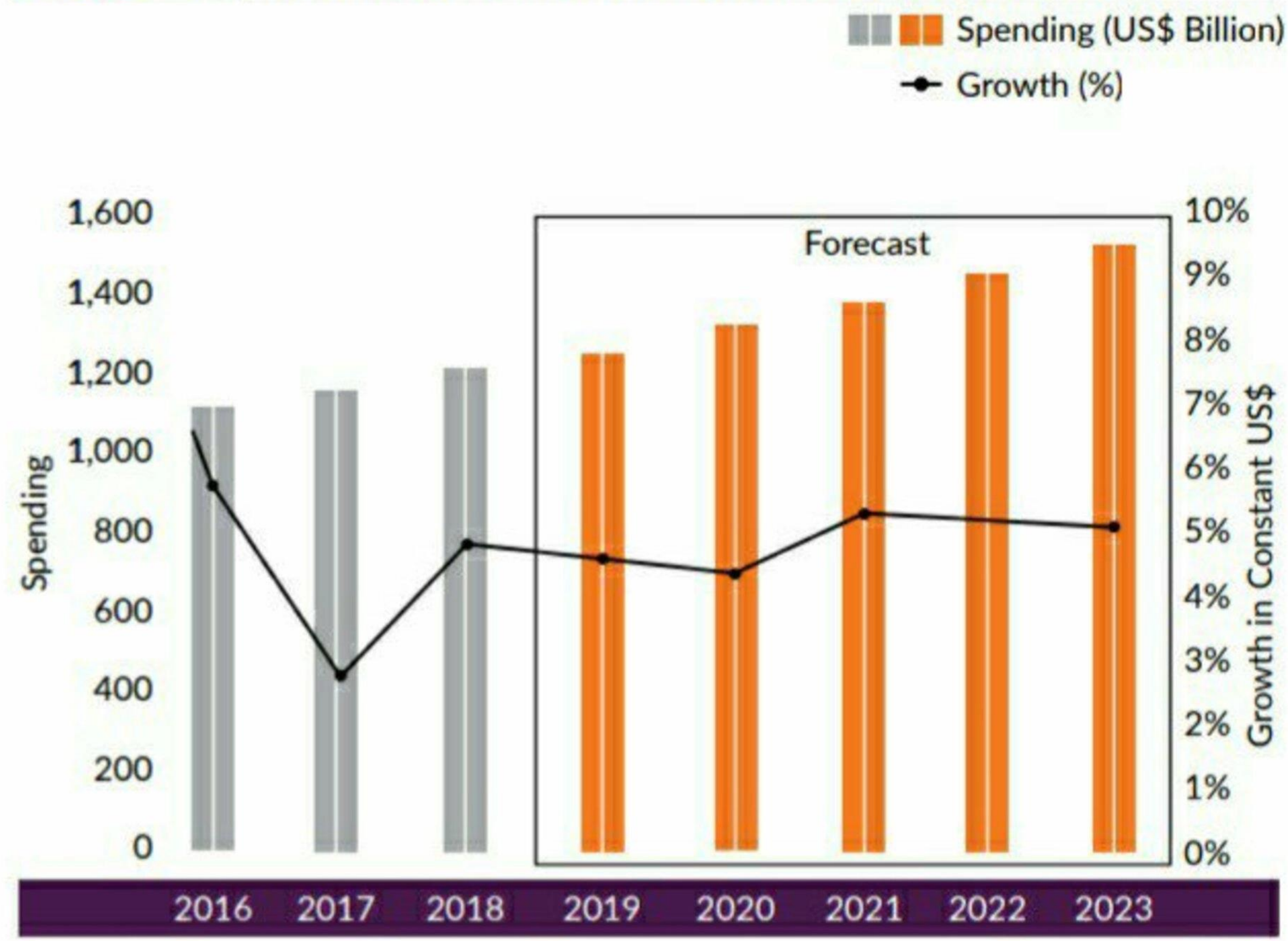
Increasing health conscious citizens



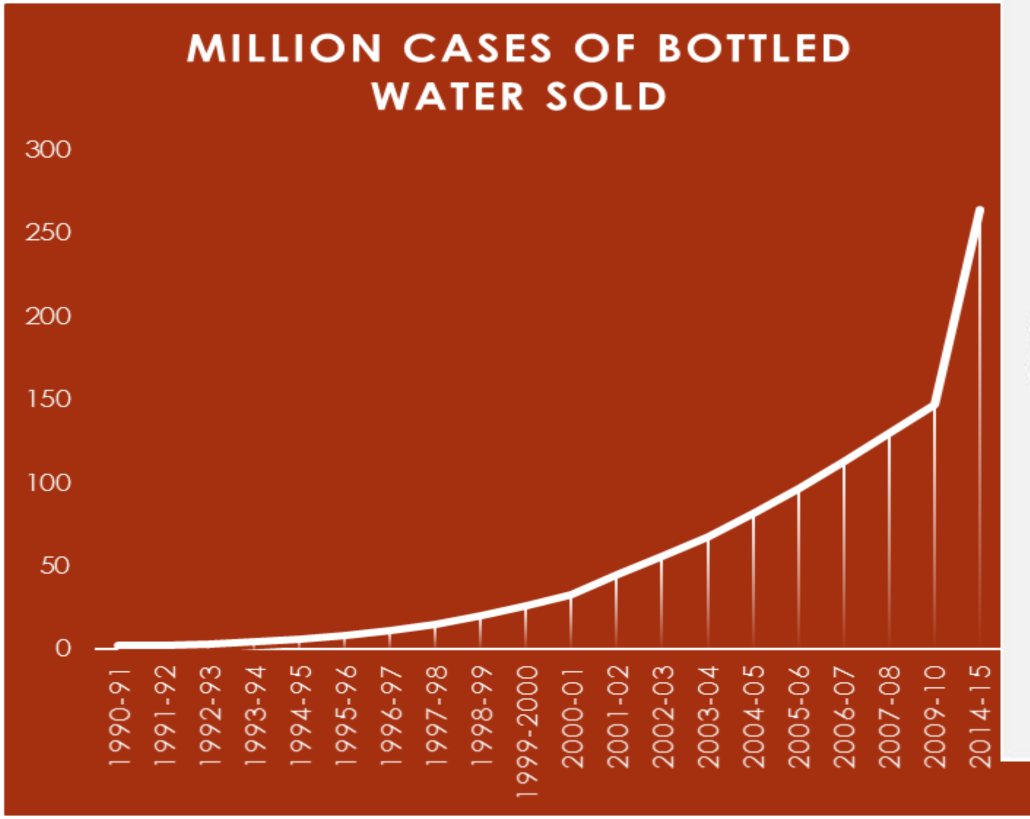
Source: Frost & Sullivan, LSI Financial Services, Deloitte, TechSci Research

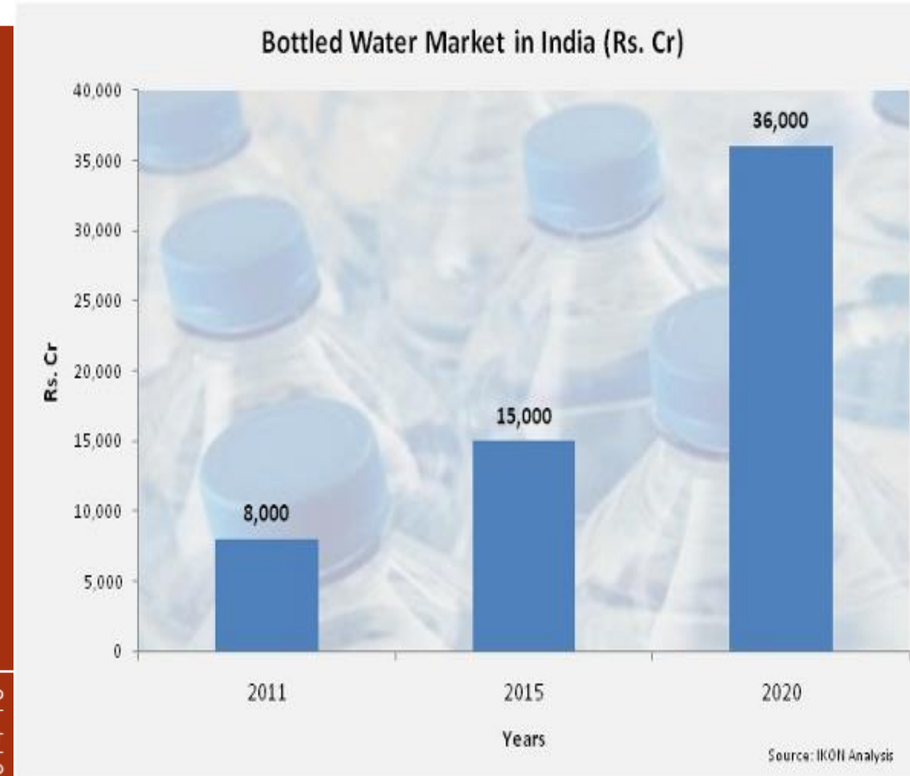
Notes: E - Estimate, F - Forecast, CAGR - Compound Annual Growth Rate

Chart 1 Global medicine spending and growth, 2016-23[1]



Water – driving growth





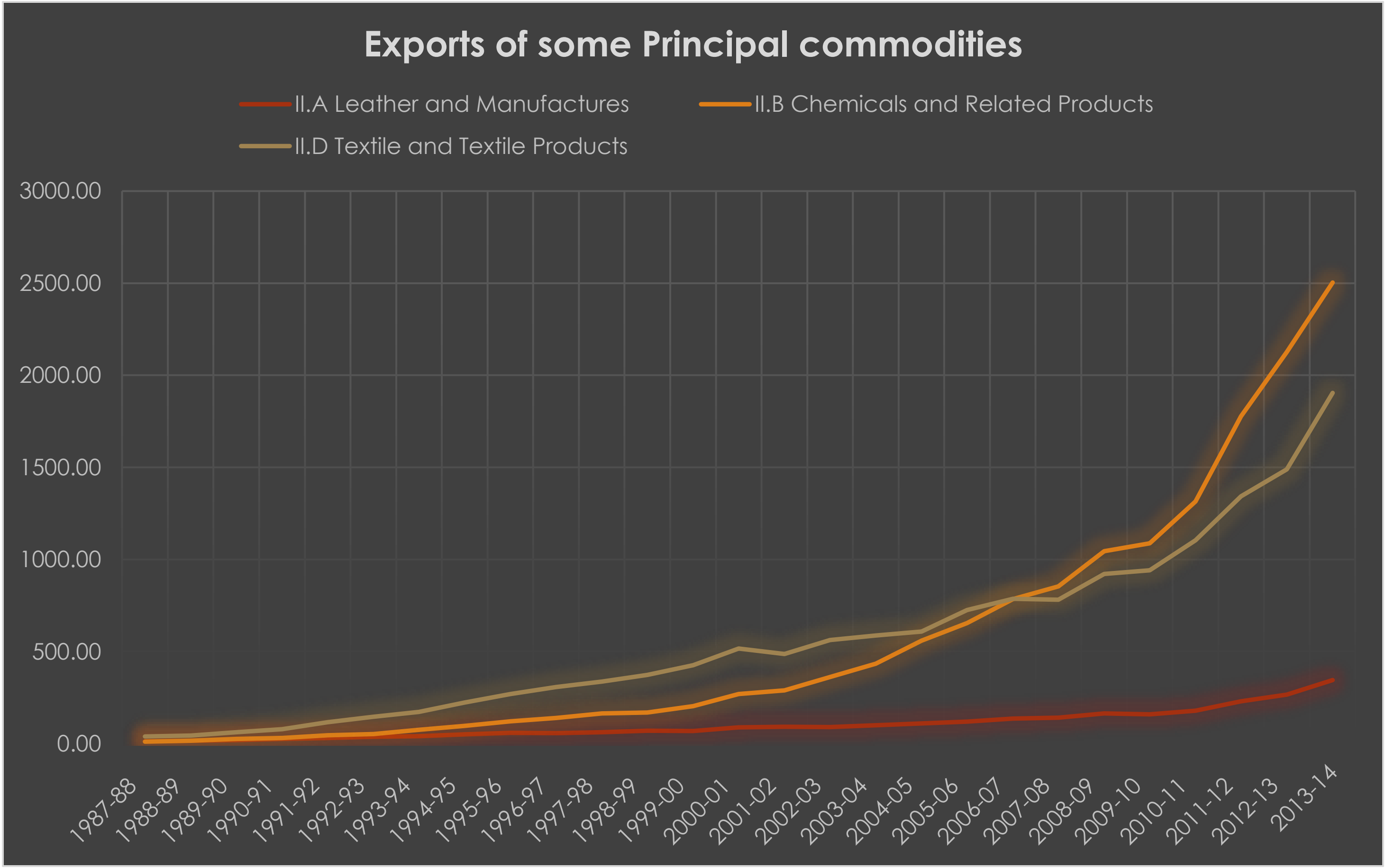
- Growing at 40 percent, 5000 crores business
- More than 200 brands, 80 percent local brands
- Market shares: Bisleri – 40%
- Kinley – 25% Aquafina – 10%

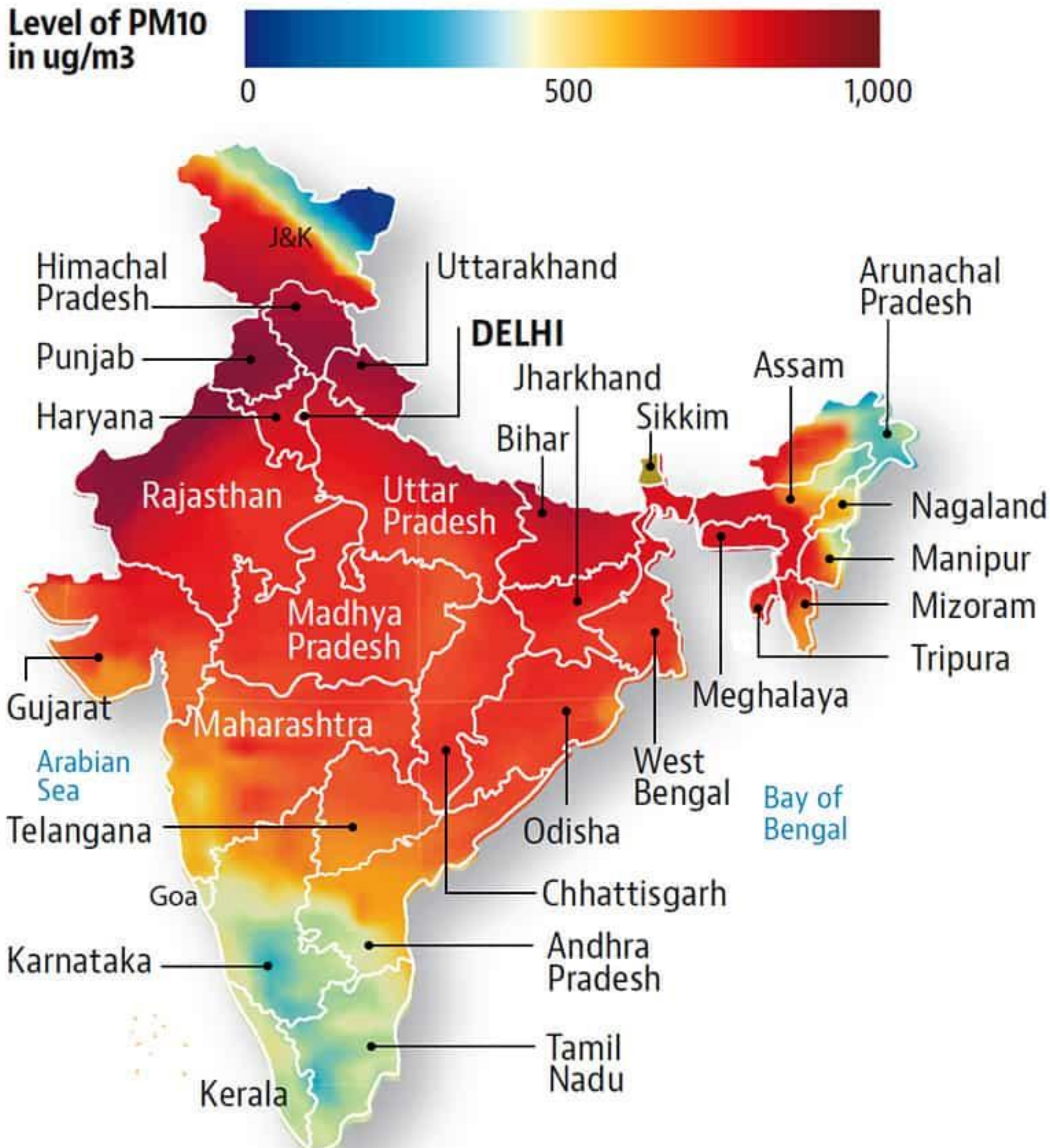
Chart 1.5: Rising share of services sector in India's GDP

All figures are at factor cost constant prices

Source: CSO, D&B India

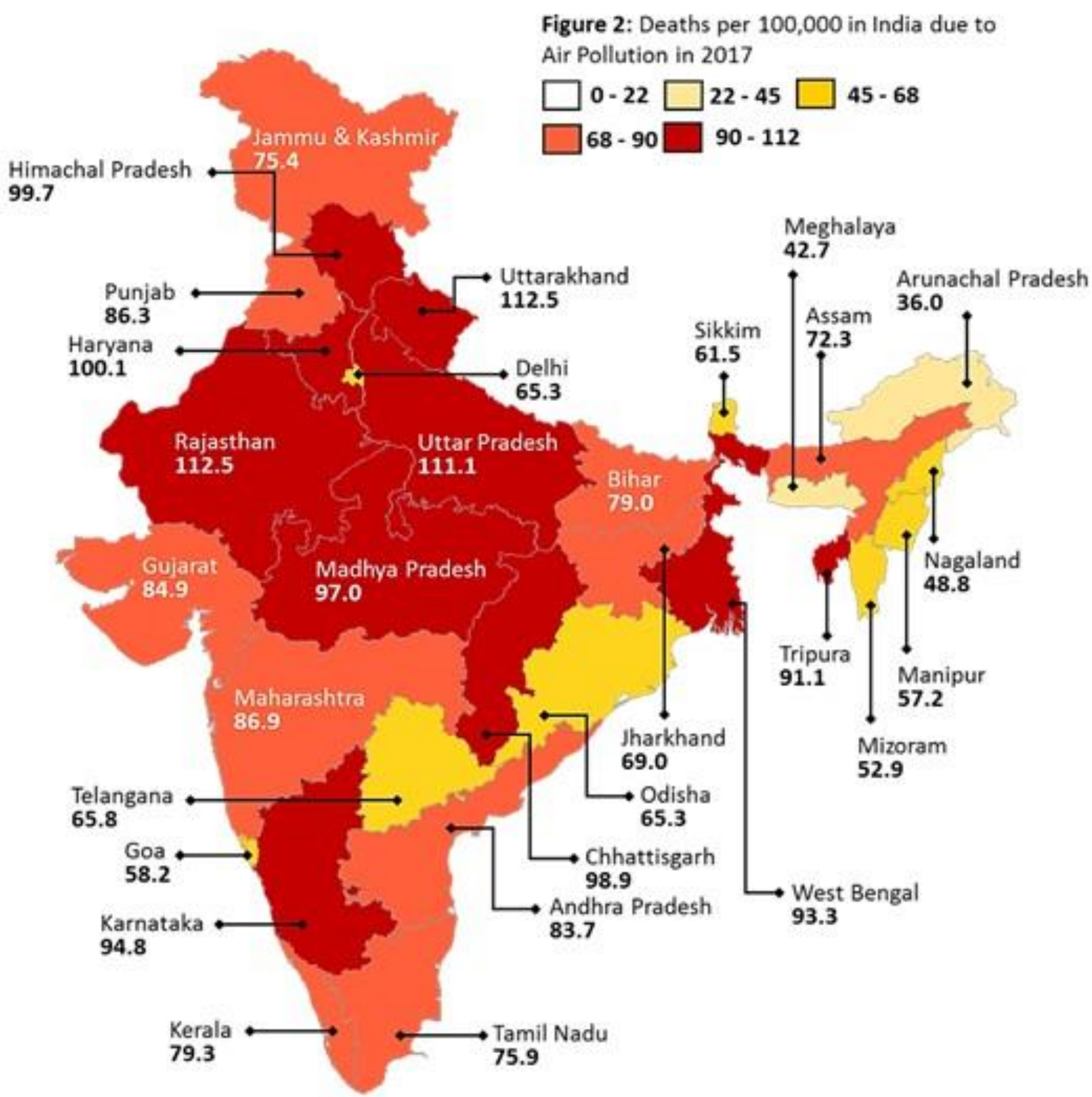
India's booming exports





Source: earth.nullschool.net

Deaths due to air pollution

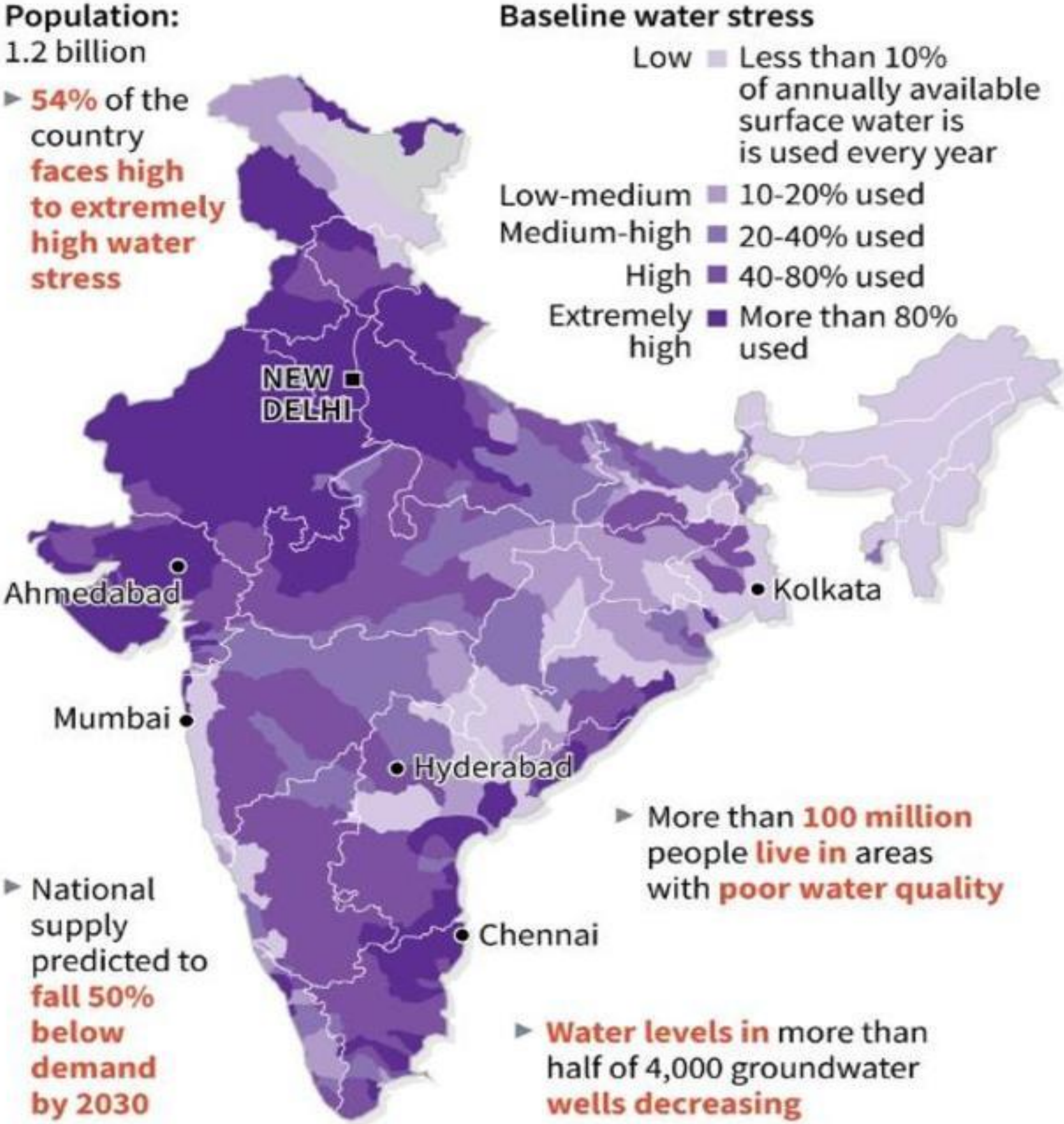


Water woes in India

Decades of population growth and uncontrolled urbanisation have created a water crisis in the country

Population:
1.2 billion

- **54%** of the country **faces high to extremely high water stress**

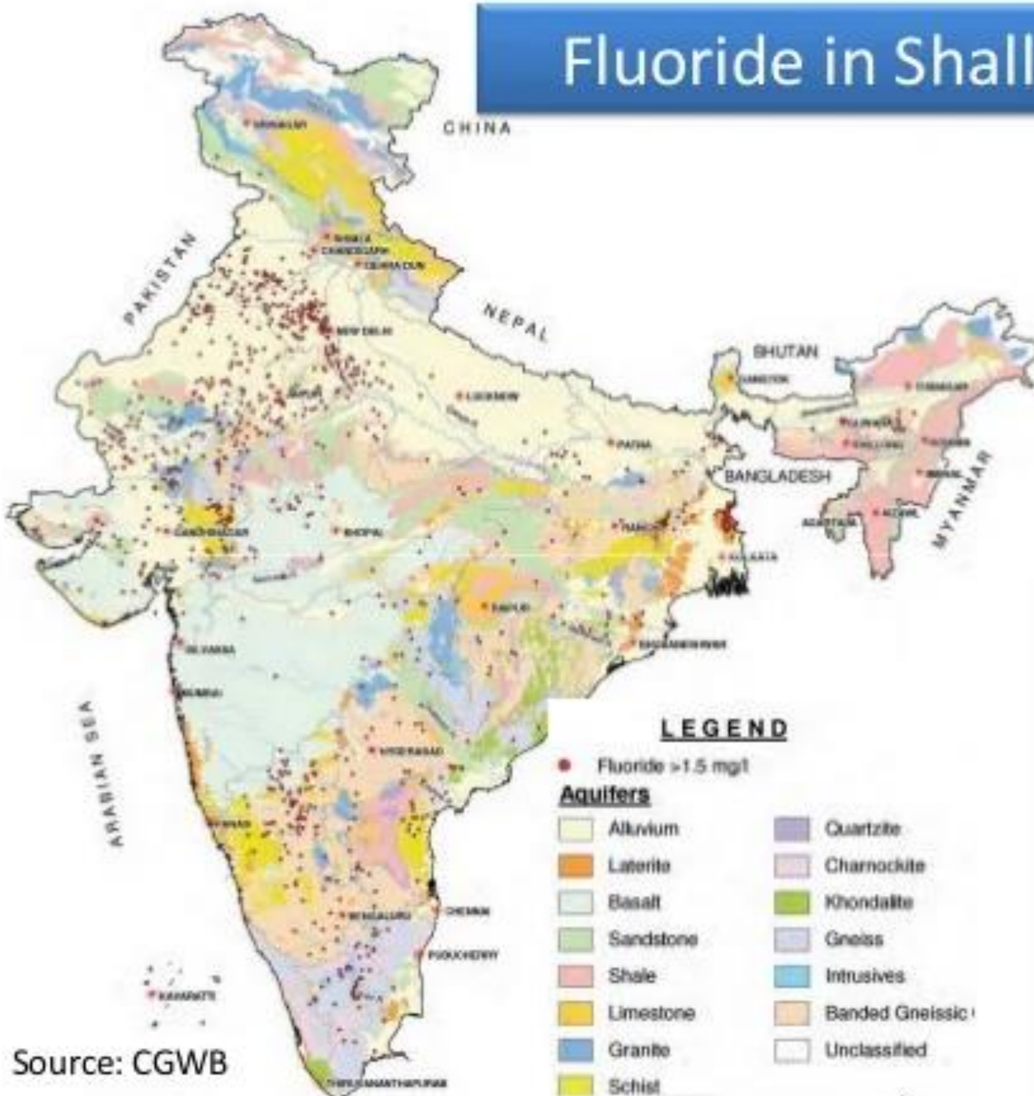


- More than **100 million** people **live in** areas with **poor water quality**
- National supply predicted to **fall 50% below demand by 2030**
- **Water levels in** more than half of 4,000 groundwater **wells decreasing**

Source : Water Resources Institute/UN

AFP

Fluoride in Shallow Aquifer



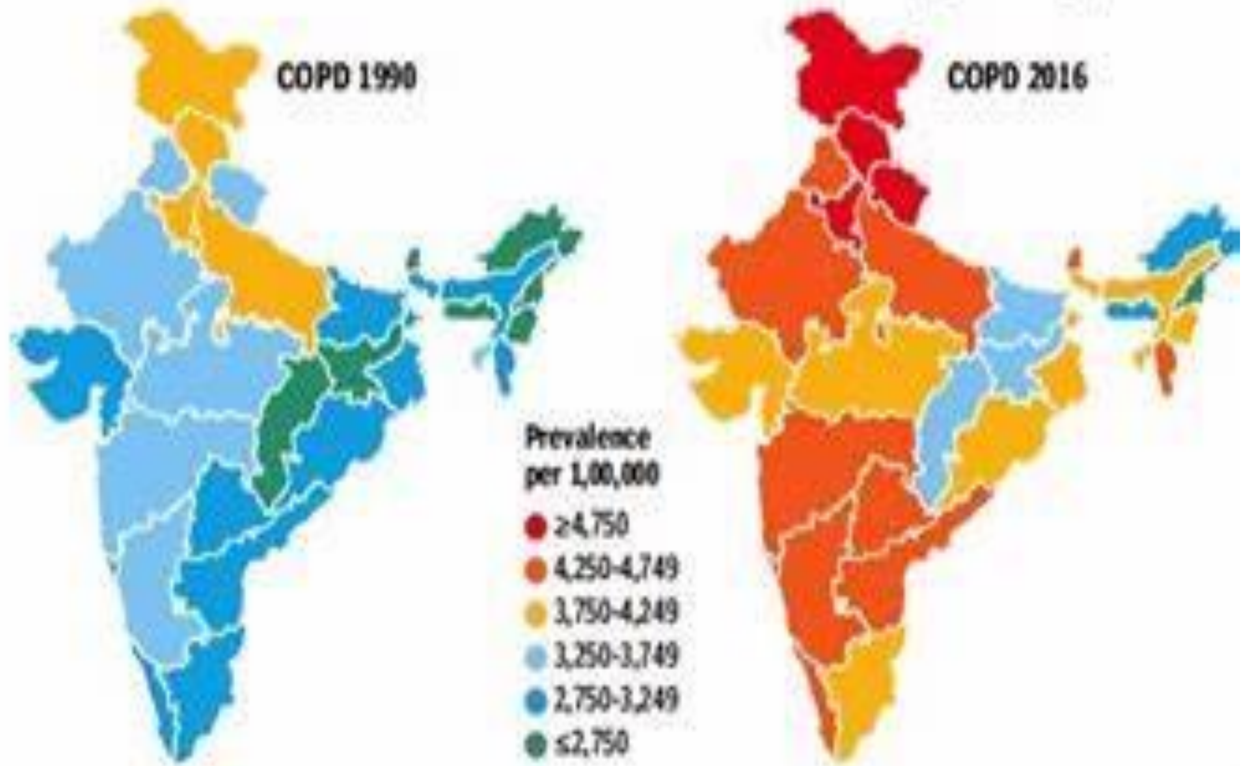
Source: CGWB

Water - the real story

COPD SECOND IN DISEASE BURDEN IN INDIA

Chronic respiratory diseases, including chronic obstructive pulmonary disease (COPD) and asthma, were responsible for 10.9% of the total deaths in India in 2016 compared with 9.6% in 1990

- COPD is one of the leading non-communicable causes of death globally, as well as in India
- After ischaemic heart disease, COPD was the second leading cause of disease burden in India, contributing 8.7% of the total deaths
- The number of COPD cases in India increased from 28.1 million in 1990 to 55.3 million in 2016 while the number of COPD deaths went from 6,24,000 in 1990 to 8,48,000 in 2016



So what does this tell us?

- So what does this tell us?
- Unsustainable
- Indicators of economic growth are incorrect
- Increasing populations contributes to accelerated degradation of resources
- Limited population size can be sustained with the earth's finite resources
- Potential for new technologies may alleviate the strain on the resources
- Growing populations in the LDC have led to land, water, and wood shortages in rural areas, and sanitation and water in urban areas

Growth versus the Environment

- Question of whether or not it is possible to achieve growth without environmental damage
- The worst environmental damage by the richest billion and poorest billion of the world
- Therefore idea that increasing incomes of the poor would decrease environmental damage
- Increasing consumption while keeping environmental degradation low is difficult

Why pollution occurs?

- Environment a public good
- Lack of well defined property rights for environment
- Market failure
- Externalities

What is Externality?

Externalities are unintended (and uncompensated) side effects of one person’s or firm’s activities on another.

Examples:

1) Health effect of smoke emissions from vehicles
2) Health effect of smoke emissions from Thermal power plant

3) Health effect of discharging untreated effluent from.

Why these side effects?

Because of interdependence in production or consumption.

The utility of individual i depends not only on his consumption but also on the consumption of another individual:

$$Ui = Ui(Xi, Xj) \quad \& \; Pi = Pi(Xi, Xj)$$

NOTES: A) These damages are unintentional *per se* as they are typically difficult to avoid.

B) This interdependence must also be a non-market dependence to qualify as an externality.

Examples

1) Congestion caused by a vehicle on other drivers

- unintentional and difficult to avoid.

2) Air Pollution caused by power plants

- unintentional and difficult to avoid

3) If many people are in queue to buy water or medicine
$\rightarrow$ this may lead to $\uparrow$ in price.

Can this be Called as EXTERNALITY?

No

$\because$ it is perpetuated through market mechanism.

Hence not an external effect.

Another Approach to define externality

This approach supposes that ownership rights or markets are missing for the particular resource.

Example: If there were private ownership of air, then people would have to buy the right to pollute it with smoke.

If ownership rights of a city's air (e.g., Delhi or Chennai or Kanpur etc.) is given to (or owned by) few individuals and then vehicle owners will have to pay to these individuals to pollute.

Types of externality

Two Types:

1) Depletable

2) Non-depletable

• Manure from horse is depletable. Because if one person takes, other cannot.

• Odour of Manure is non-depletable. Because exposure of one does not reduce exposure of other individuals.

(Vehicular air pollution or Odour from Solid waste are non-depletable)

When Externality arises?

Two Types of COSTS:

1) Private Costs

2) Social Costs

Ex. - w.r.t. Vehicular Air pollution in Delhi/Chennai

Private Costs - Cost of Owning a vehicle, Fuel (petrol/diesel/LPG), maintenance cost etc.

Social Costs - Effect of pollutant on other people, my vehicle causing congestion for others + Private costs.

When SOCIAL COSTS > Private Costs

$\Rightarrow$ **Negative Externality**

Other Examples of Externality?

Examples of Negative Externality

1) Pollution of Ganges or Yamuna by upstream activities

2) Transboundary Pollution

3) Soil Erosion caused by excessive and type of agriculture

4) Floods caused by Deforestation

5) Garbage (NIMBY)

6) Use of pesticides in agriculture - affecting downstream water bodies

Positive Externality

Positive Externalities:

When Social Benefit > Private Benefit

<u>Examples of Positive Externality</u>

1) R&D by a firm for Cancer / SARS drug

2) New filter for cars to reduce emissions

3) Forestation in hilly regions affecting water supply in the plains.

Where is the Problem?

Since Social Costs are more than Private Costs - if the persons causing social costs are asked to bear part of the social costs (i.e., their contribution of social costs), this externality problem will be solved.

i.e., Internalising the Externality

HOW DO WE Internalise the externality?

$\Rightarrow$ Using Policy INSTRUMENTS (CAC, EI/MBI or Suasive)

In case of Vehicular Pollution - By asking them to switch to cleaner fuel (Diesel run - buses in Delhi to CNG or – diesel to CNG run auto rickshaws in Delhi)

How policy instruments work?

Standard externality argument provides the answer.

If polluters are forced to pay this environmental cost also, this would internalise the externality.

• For example: if users of polluting transport are forced to pay an amount equivalent to the environmental cost, it will result in cleaner vehicles and smaller traffic volume and hence reduced air pollution.

How to limit negative externalities?

- Traditionally, environmental regulation instruments are classified according to
  - How much pollution to abate?
  - Need for regulator to monitor emissions

- Former category is concerned with CAC and MBI. The latter involves Direct and Indirect instruments.

Direct instruments are emission standards, emission fees and market permits; and

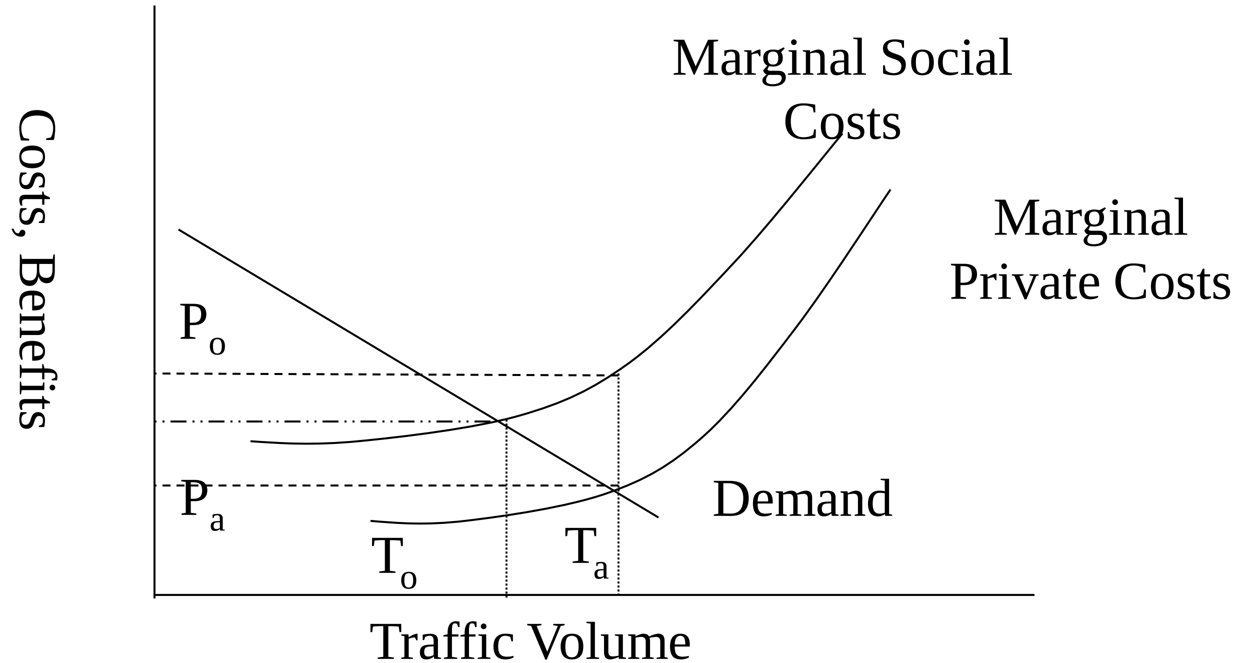
Indirect instruments are environmental taxes, technology standards etc.

Classification of Instruments

3 Types -

- Command and Control (CAC) or Environmental Regulations
- Market based instruments (MBI / EI)
  - creating markets
  - using existing markets
- Engaging the Public / Involving Participation

Economics of Vehicular Pollution Control – Externality Diagram



Instruments for correcting negative externalities

CAC or Regulations	Economic or Market Based Instruments		Involving Participation
	Using Markets	**Creating Markets**	
1) Standards	1) *Charge Systems* – effluent charges; user charges; product charges; administrative charges; impact fees	1) Property rights / decentralisation	1) Voluntary agreements
2) Bans	2) *Fiscal Instruments* or *Environmental Taxes* – pollution taxes; input taxes; aid in installing new technology; subsidies for environmental R&D	2) Tradeable permits / rights	2) Public participation
3) Permits/ quotas	3) *Financial Instruments* – financial subsidies; soft loans and grants	3) International offset systems	3) Information disclosure
	4) *Deposit-refund systems* and *environmental performance bonds*	4) Liability insurance legislation	4) Two-tier monitoring

Pollution control - Standards

- MINAS standards (Minimum national standards) – water (ten industries) and air (twelve industries)
- Product Standards (e.g. lead content of petrol, quality of drinking water)
- Process standards (e.g. chrome tanning process)
- Emission standards (e.g CFC standards, standards on total emissions, national emission standards)
- Environmental Quality goals and standards (quality of air, water etc)
- Residue standards
- Critical Loads (amount of substance deposited that may be deposited in a given area in a given time without adverse effects)
- Exposure standards (primary protection standard, maximum concentration permitted)
- Biological standards (e.g. lead in human blood - to control the combined effect from all levels)
- E.g. Lead can reach humans through multiple sources (from soil, water, plumbing, paints, petrol etc)

an standards

Human tolerance for different noise levels

Noise level	Maximum duration of exposure
75dB	Comfortable for human hearing
90dB	2 and half hours
100dB	15 minutes
110dB	30 seconds
120dB	9 seconds
140dB	1 second

Source: WHO

ode	Category of area/zone	Limits in dB(A) leq*	
		Day time	Night time
	Industrial area	75	70
	Commercial area	65	55
	Residential area	55	45
	Silence zones	50	40

q denotes the time weighted average of the level of sound in decibels on scale A elatable to human hearing. Source: Central Pollution Control Board, India

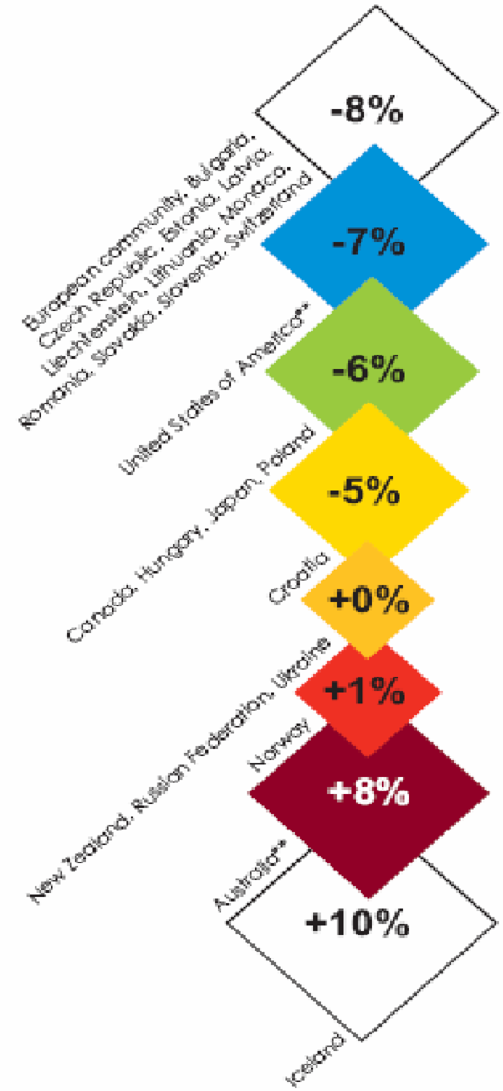
	USEPA	WHO	ISI	ICMR	CPCB
	6.5-8.5	6.5-8.5	6.5-8.5	6.5-9.2	6.5-8.5
	-	-	10	25	10
g/l)	-	-	-	-	2000
	-	-	-	-	600

es	ISI limit (μg/l)	WHO limits (mg/l)
	42	
	17	0.0003
	17	0.0003
	1	-
	3	-
	56	-
	18	-
	35	0.020
	18	-
	100	-
	5	-
	100	-

	USEPA	WHO	ISI	ICMR	CPCB
		0.01	-	-	No relaxation
		0.05	0.05	0.05	No relaxation
Lead (mg/l)	-	0.05	0.10	0.05	No relaxation
Zinc (mg/l)	-	5.0	5.0	0.10	15.0
Chromium (mg/l)	0.1	-	0.05	-	No relaxation
E. coli (MPN/100 ml)	-	-	-	-	No relaxation

Permits/quotas (non-transferable)

- Fishing permits
- Individual fishing quotas
- emission quotas
- Input controls
- Technology controls (best available technologies)
- Output quotas or prohibitions
- Emission licence
- Location controls (zoning, planning controls, relocation)



* The base year is flexible in the case of EIT countries (see page 17)
** Countries which have declared their intention not to ratify the Protocol

Selection of Policy Instrument

Number of criteria -

- Static Cost Efficiency – Is the PI least costly from firm's point of view?
- Dynamic Cost Efficiency ($\delta AC/\delta t$) – Does it give adequate incentive to a firm to continuously innovate so as to $\downarrow$ pollution?
- Goal Fulfillment – Will the PI fulfill the goal for which it is intended? – e.g., mercury reduction
- Administrative Costs – What will be the administrative costs of implementing?

Selection of Policy Instrument

Criteria contd. -

- Barrier to Entry – Will the implementation of PI create barriers for entry of new firms
  - e.g., standards are usually for newer firms/units, thereby protecting older plants.
  - Older vehicles not covered under EURO II norms.
- Polluter Pays Principle – Is the polluter paying for the externality?
- Politics of Implementation (Acceptability) –Is it acceptable to all the parties concerned?

Policy Instrument Selection Matrix

Criteria	CAC Regulation	MBI	
		Tax or charge	**Tradable Emission Permits**
Static cost efficiency	No – neither static nor dynamic	Yes (specially with many agents with distinct technology)	Yes – if market not too thin
Dynamic cost efficiency	No – Perverse incentives	Yes	Yes – Depends
Goal fulfillment	(No) may be at firm level but not aggregate level	No – optimal tax level unknown. Inflation may be	Yes – The regulator can be fairly certain of reaching target
Administrative costs	Best – control needed but CAC often	Depends	Hard to administer with many / few agents
Barrier to entry	Standards for new plants; protect old	Neutral	Can be used by established firms
Polluter Pays principal	Yes	Yes	Yes (if auctioned)
Politics of implementation	Risk for rent-seeking behaviour	Risk of opposition if not refunded	Neutral (at least if grand-fathered)

Instruments and sample applications

	Policy Instrument	Natural Resource Management (Water, fisheries, agriculture, forestry, minerals & biodiversity)	Pollution Control (Air, water, solid waste & hazardous waste)
1	Direct Provision	Provision of parks	Waste Management, Timer at intersection
2	Detailed Regulation	Zoning, Regulation of fishing (e.g., dates and equipment), Ban on ivory trade to protect biodiversity	Catalytic Converters, traffic regulations etc., Ban on chemicals (Azo dyes)
3	Flexible Regulation	Water quality standards	Fuel quality, CAFÉ
4	Tradable quotas or rights	Individually tradable fishing quotas, Transferrable rights for land development, forestry or agriculture	Emission permits
5	Taxes, fees or charges	Water tariffs, park fees, fishing license	Waste fees, congestion pricing,, gas taxes, industrial pollution fee
6	Subsidies and subsidy reduction	Water, fisheries, reduced agricultural subsidy	Energy taxes, reduced energy subsidies

	Policy Instrument	Natural Resource Management (Water, fisheries, agriculture, forestry, minerals & biodiversity)	Pollution Control (Air, water, solid waste & hazardous waste)
7	Deposit-refund schemes	Reforestation deposits or performance bonds in forestry	Waste management, used vehicles, vehicle inspection
8	Refunded emission payments		NOx abatement in Sweden
9	Creation of property rights	Private national parks, property rights and deforestation	
10	Common property resources	CPR management	
11	Legal mechanisms, liability	Liability bonds for mining or hazardous waste	
12	Voluntary agreements	Forest Products	Toxic Chemicals
13	Information	Labeling of food,	PROPER and other

Environmental Disclosure China

- Green Securities Policy
- 14 highly polluting industries to report required environmental information

U.S. Market for SO2 Allowances

- *Sulfur dioxide (SO2) is a primary product of coal-burning power plants.*
- *SO2 pollution lowers the pH scale of rainfall, leading to the problem commonly known as acid rain*.
  - *Increased acidity of lakes and rivers.*
  - *Slower growth, injury or death of forests.*
  - *Visibility impairment*.
- *SO2 can impair respiratory function and is linked to a variety of respiratory ailments.*

U.S. Market for SO2 Allowances

What Are Allowances?

An allowance authorizes a unit within a utility or industrial source to emit one ton of SO2 during a given year or any year thereafter. At the end of each year, the unit must hold an amount of allowances at least equal to its annual emissions, i.e., a unit that emits 5,000 tons of S02 must hold at least 5,000 allowances that are usable in that year.

Allowances are fully marketable commodities. Once allocated, allowances may be bought, sold, traded, or banked for use in future years. Allowances may not be used for compliance prior to the calendar year for which they are allocated.

U.S. Market for SO2 Allowances

- *The process of setting up a market for pollution emissions involves three steps:*
  - *Establish an overall cap on pollution (i.e. Q')*
    - *For the U.S. SO2 market the cap was set in the 1990 Clean Air Act amendment.*
    - *Reduce SO2 emissions from fixed sources by 50% from 1980 levels (17.3 million tons).*
  - *Allocate permits (allowances) to polluters*
    - *In the U.S. SO2 market, a 'grandfathering' scheme was used.*
      - *Allocated to polluters based on their 1985-87 pollution.*
  - *Facilitate trading between polluters*
    - *In the U.S. SO2 market the trading was generally unrestricted.*
    - *Stiff penalties to those firms who exceeded their allowances.*

U.S. Market for SO2 Allowances

How Are Allowances Allocated?

Allowances are allocated for each year beginning in 1995. In Phase I, EPA allocates allowances to each unit at an

emission rate of 2.5 pounds of SO_2/mmBtu (million British thermal units) of heat input, multiplied by the unit's baseline mmBtu (the average fossil fuel consumed from 1985 through 1987)

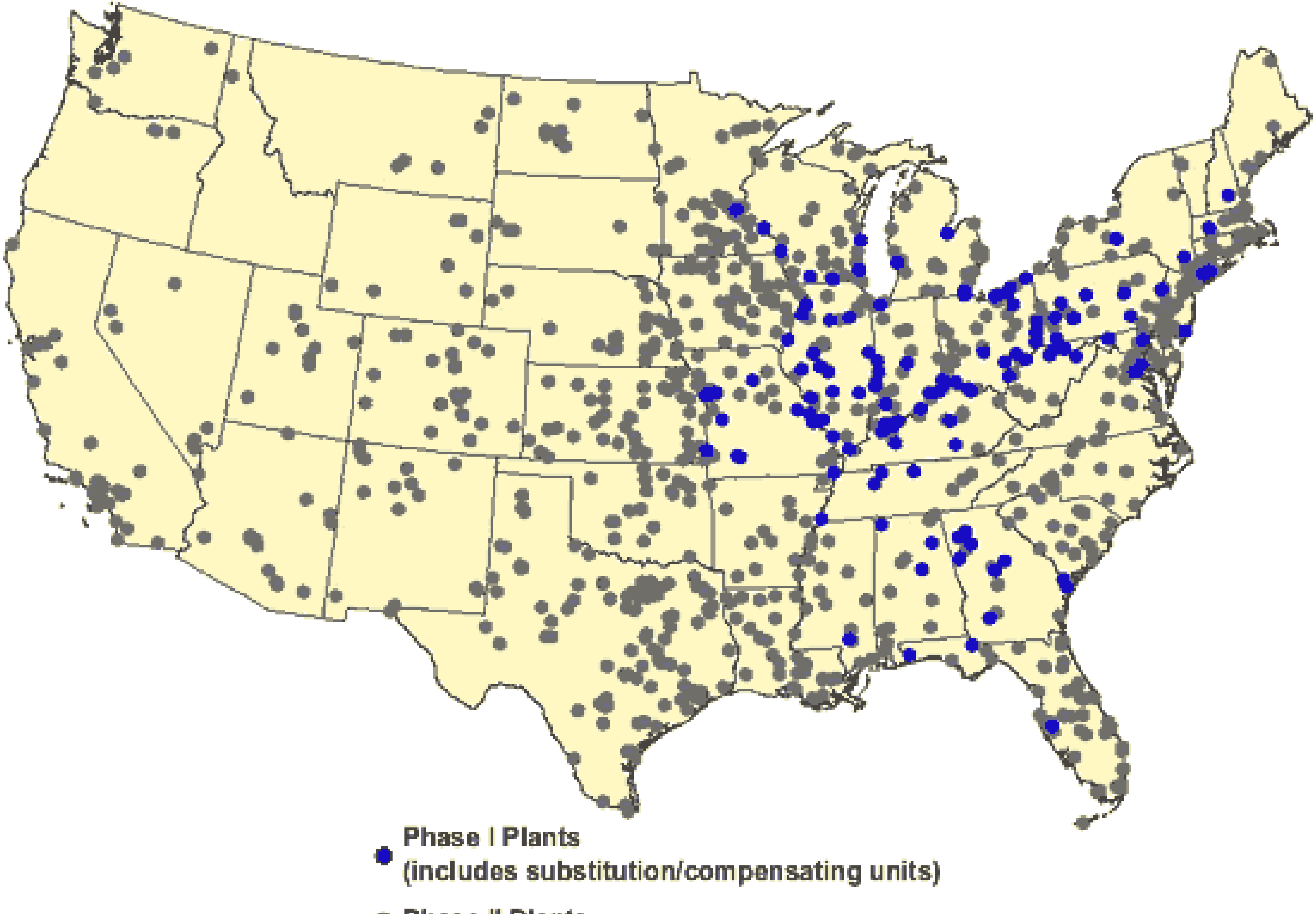
Roughly 440 units are involved in Phase 1

U.S. Market for SO2 Allowances

…In Phase II, which began in 2000, the limits imposed on Phase I plants are tightened, and emissions limits are also imposed on all SO2 emitting units (another 2000 or so units).

EPA allocates allowances to each unit at an emission rate of 1.2 pounds of SO2/mmBtu of heat input, multiplied by the unit's baseline, which effectively places a cap at 8.95 million on the number of allowances issued to units each year.

Market for SO2 Allowances



Market for SO2 Allowances

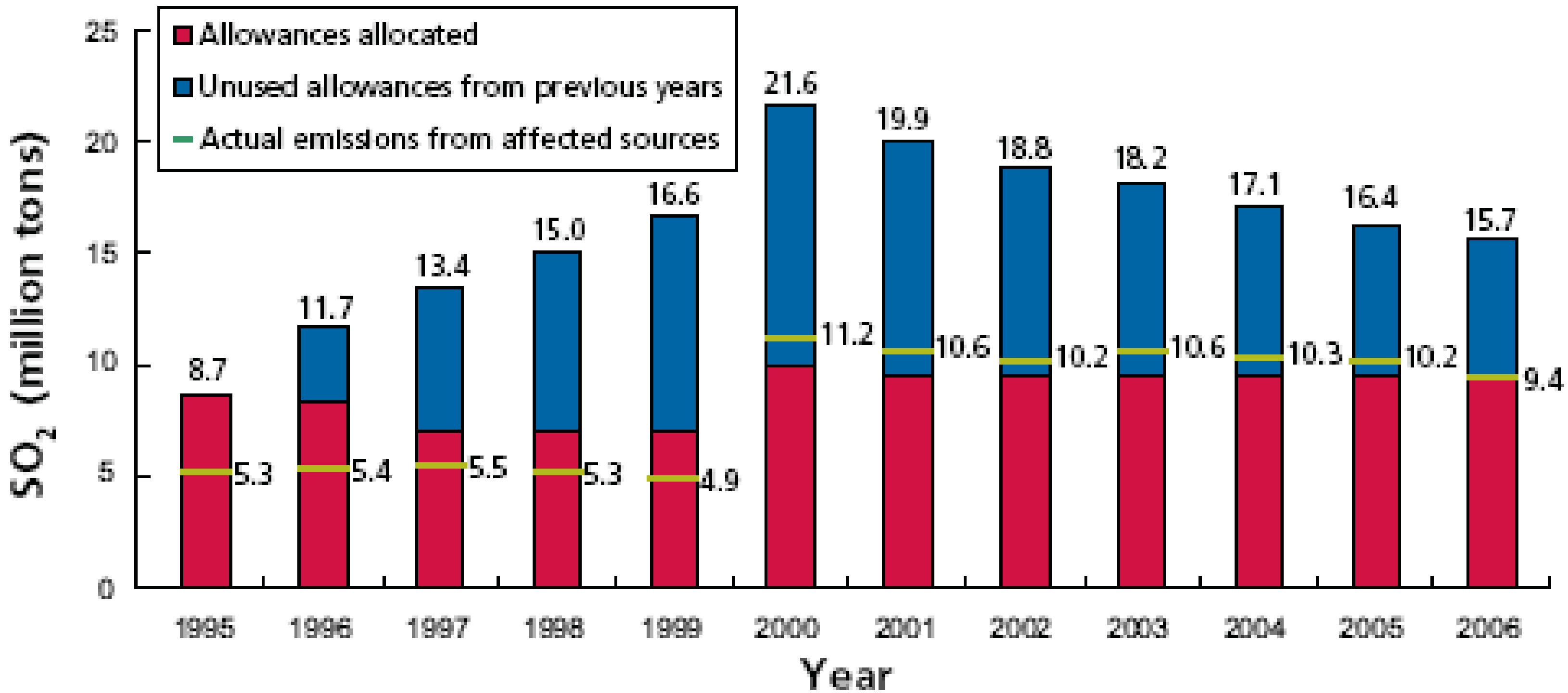
How is Compliance Determined?

At the end of the year, units must hold a quantity of allowances equal to or greater than the amount of SO2 emitted during that year.

What Is The Penalty for Noncompliance?

Penalty for noncompliance: $2000 per ton over, with inflation it is now about $3000 per ton. Firms have a year to offset excess emissions by purchasing permits. Very rarely are firms out of compliance.

Market for SO2 Allowances



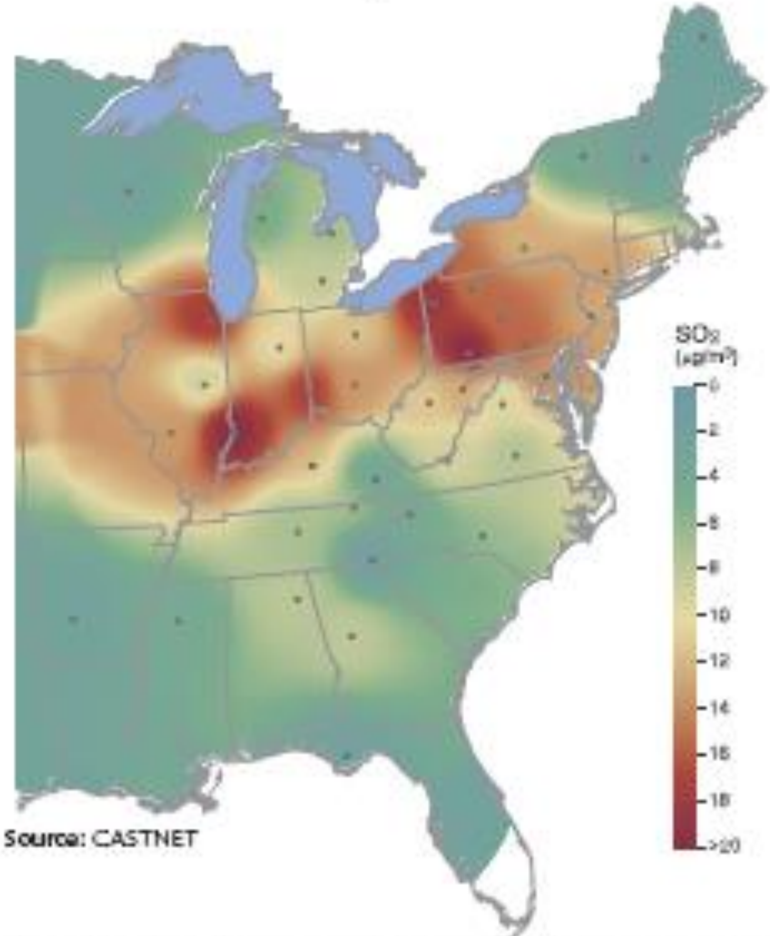
Market for SO_2 Allowances

- Reasons why the program resulted in a robust market
  - There were wide differences in the cost of abating emissions.
  - The policy was flexible.
  - The implementation was simple.
  - The provisions were enforced => the government's property right to no pollution above the standard is secure.

Market for SO_2 Allowances

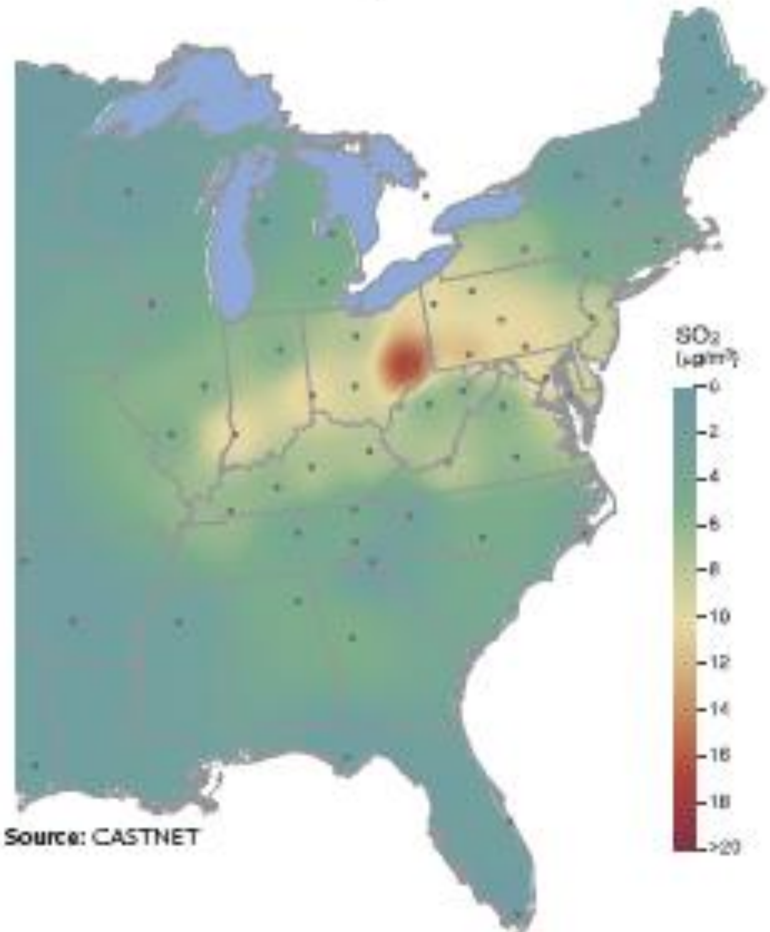
- Two questions:
  - Has the SO_2 trading program reduced emissions?
  - Has the program reduced the cost of emissions control?

Figure 17a: Annual Mean Ambient Sulfur Dioxide Concentration, 1989–1991*



Source: CASTNET

Figure 17b: Annual Mean Ambient Sulfur Dioxide Concentration, 2003–2005



Source: CASTNET

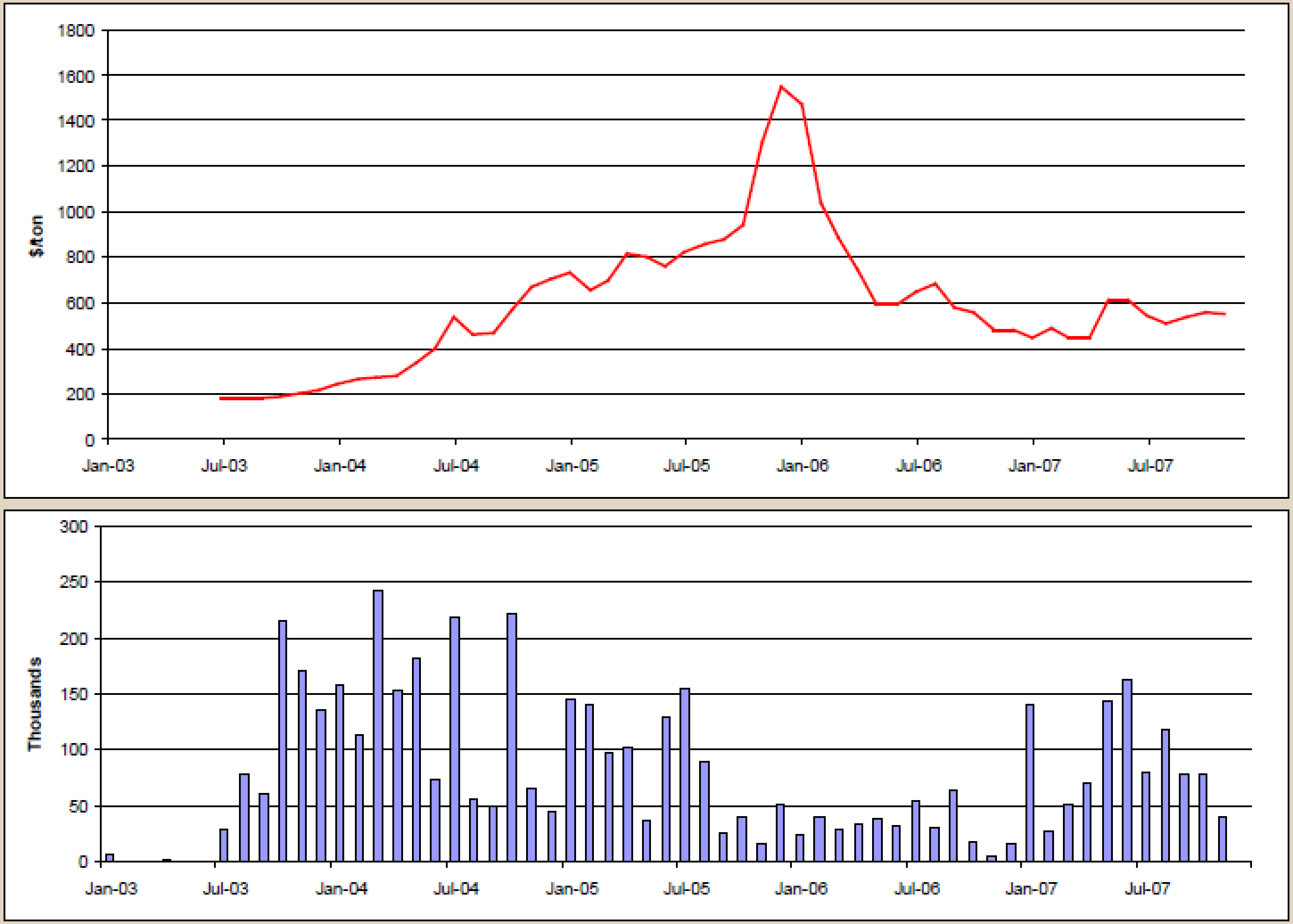
*Dots on all maps represent monitoring sites. Lack of shading for southern Florida on Figures 17a, 18a, and 19a indicates lack of monitoring coverage.

SO2 Trading: That was then…

- 1990: Title IV of the Clean Air Act established trading
- 1995: Phase I (374 large units)
- 2000: Phase II (1,420 more units)

- National Cap with no spatial restrictions on trading. By 2000 emissions were 40% lower than 1980 levels
- Continuous monitoring of emissions1 allowance used for each ton emitted
- Very low transaction costs (0.1% in 1999)

2007)



Source: http://www.cantorco2e.com

Instruments used for industrial pollution

	Country	Region	Instrument
1	CORNARE, Colombia	Latin America	Pollution Charges
2	Rio de Janeiro, Brazil	Latin America	Targeting
3	Juarez, Mexico	Latin America	Technical Assistance
4	Gudalajara, Mexico	Latin America	Assistance in EMS
5	Leguna lake, Philippines	South East Asia	Pollution Charges
6	Philippines	South East Asia	Pollution Rating
7	Indonesia	South East Asia	Pollution Rating
8	China	South East Asia	Pollution Charges
9	Palm Oil producers, Malaysia	South East Asia	Pollution Charges and enforcement of standards
10	Poland	Eastern Europe	Pollution Charges

Colour Labels used in PROPER (program for environmental evaluation and training)

Rating	Technical Requirements
Gold	World Class clean technology Waste minimization & poll^n prevention measures
Green	> legally required standards for env. Protection Good maintenance and environmental work
Blue	At legally required standard
Red	< legally required standards
Black	Serious environmental damage – no poll^n

Change in firms' ratings because of PROPER

Rating	June 1995	Dec. 1995	Dec. 1996	Change (%)
Gold	0	0	0	0
Green	5	4	5	0
Blue	61	72	94	54%
Red	115	108	87	- 24%
Black	6	3	1	- 83%
Total	**187**	**187**	**187**	

Climate Change and Carbon Markets – an Introduction

Presentation Structure

- Background : Climate Change
- How do we measure climate change in the past and present?
- Carbon Markets

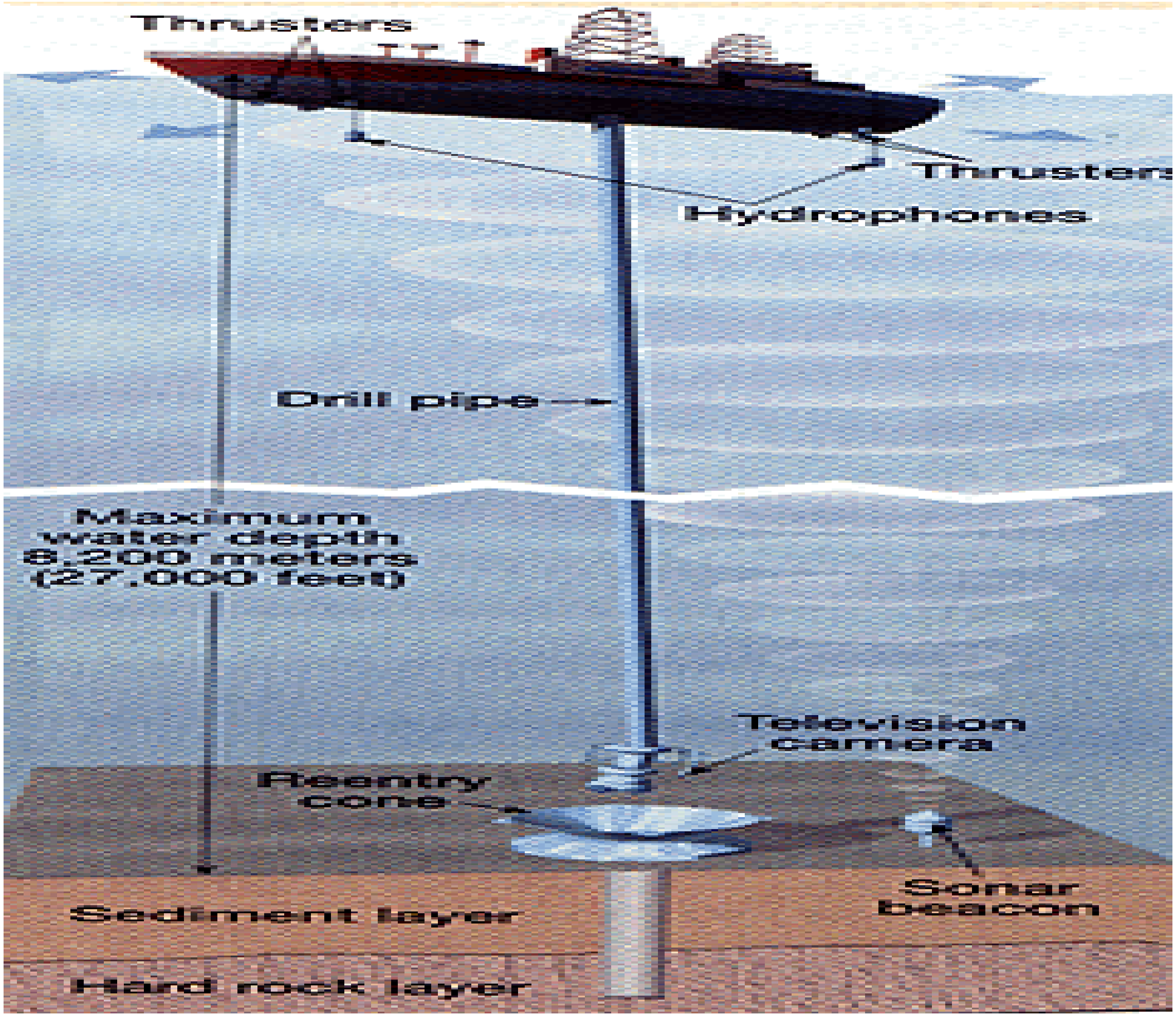
Definition

- **IPCC: Change in climate over time, whether due to natural variability or as a result of human activity.**
-
- Different from UNFCCC: change of climate that is attributed directly or indirectly to human activity that alters the composition of the global atmosphere and that is in addition to natural

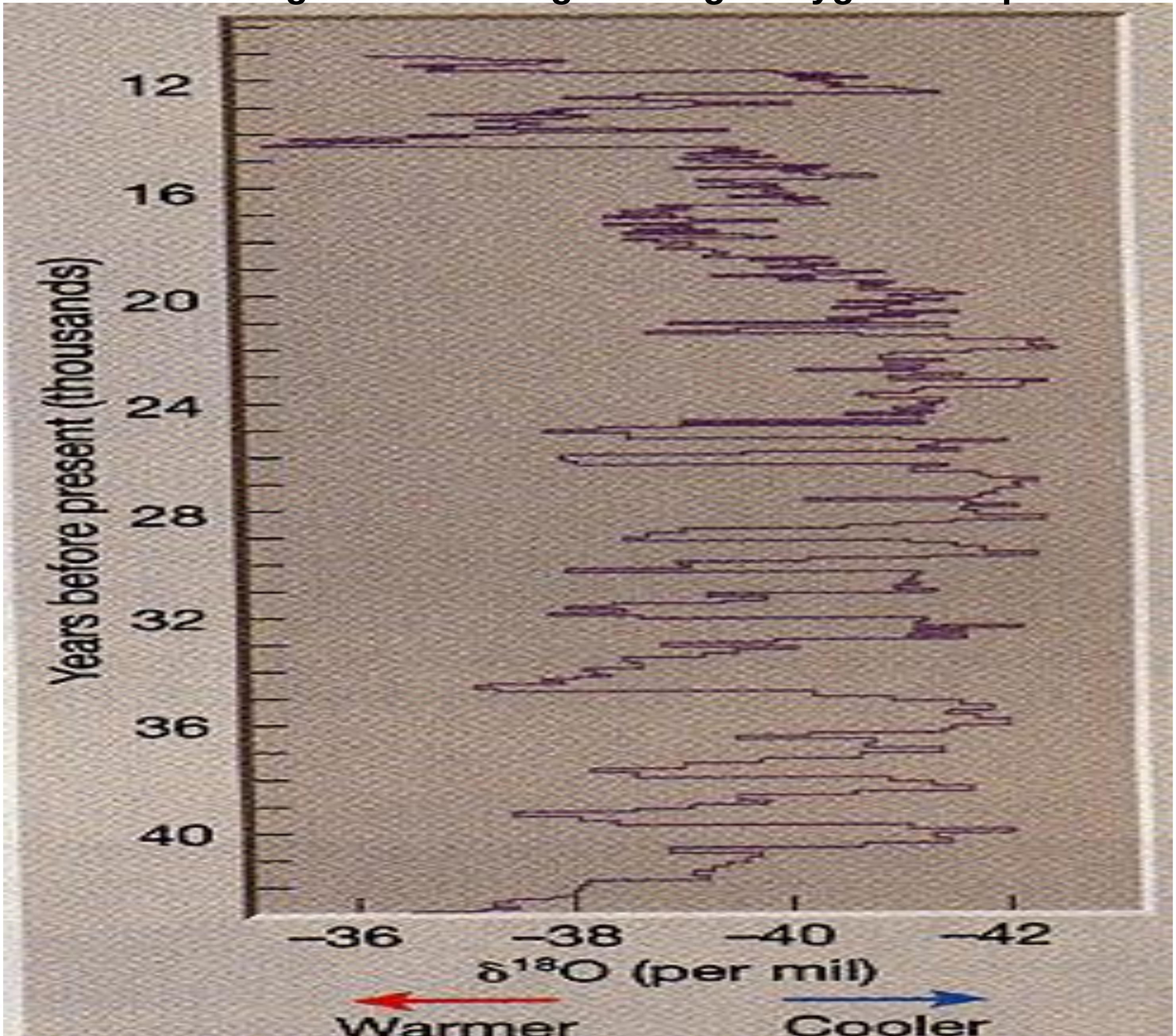
Introduction

- Throughout time, the earth's climate has always been changing
- What physical processes create natural fluctuations in the earth's climate?
- Is there an anthropogenic influence on climate change?????
  - is so, what will its effect be???
- How do we measure climate change in the past?
- How are we predicting climate change in the future?

Determining Past Climate Change - Ocean Floor Sediments



Determing climate change through Oxygen Isotopes



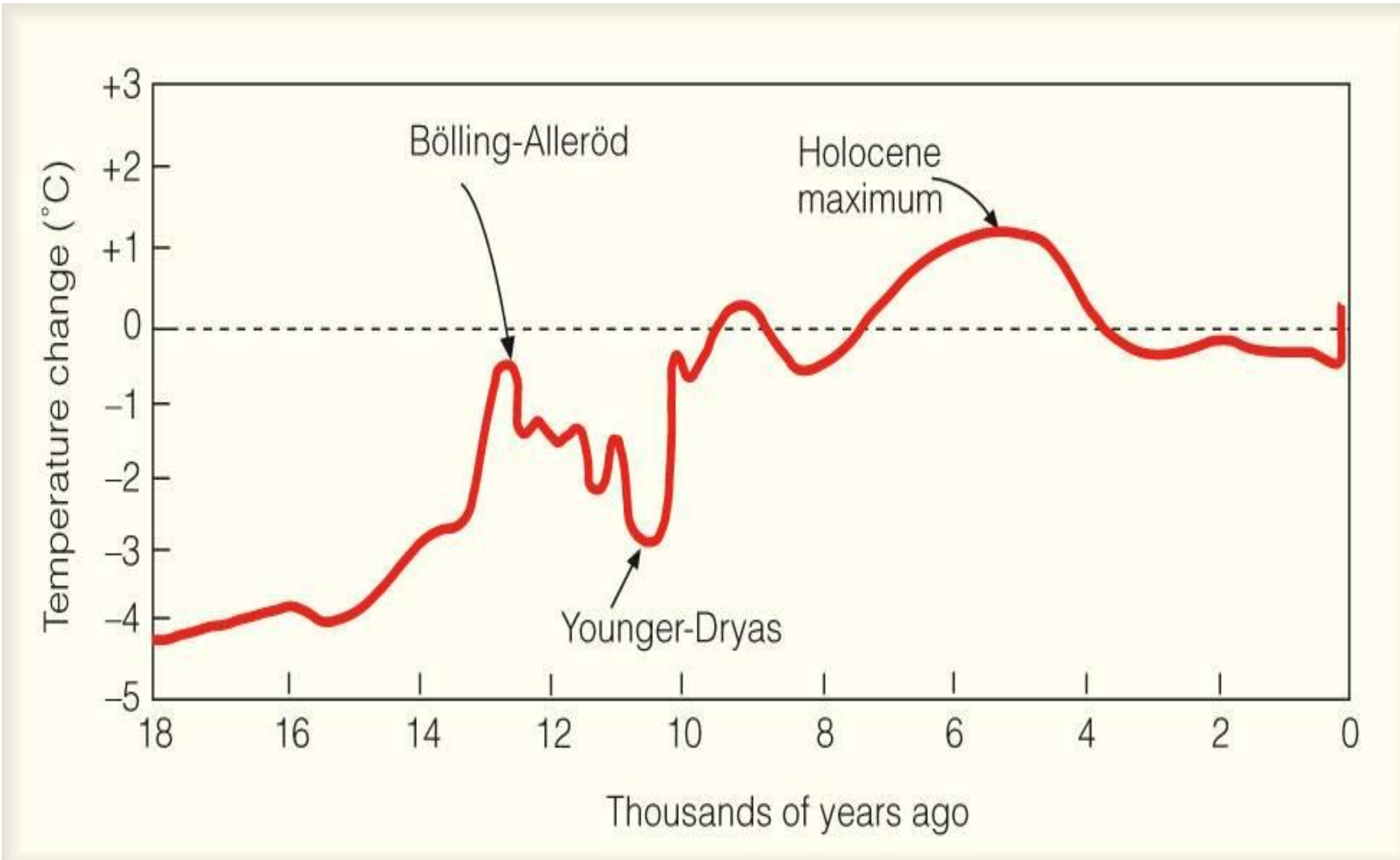
Determining Past Climate Change - Dendrochronology

Other data offering clues about climate change

- Lake-bottom sediment and soil deposits
- Pollens in deep ice caves, soil deposits, and sea sediments
- Geologic evidence in ancient coal beds, sand dunes, and fossils
- Documents concerning droughts, floods, and crop yields

- Knowledge on past climate is still incomplete....., but what do we know from the aforementioned data sources?????

Climate through the Ages



Climate through the last 1000 Years

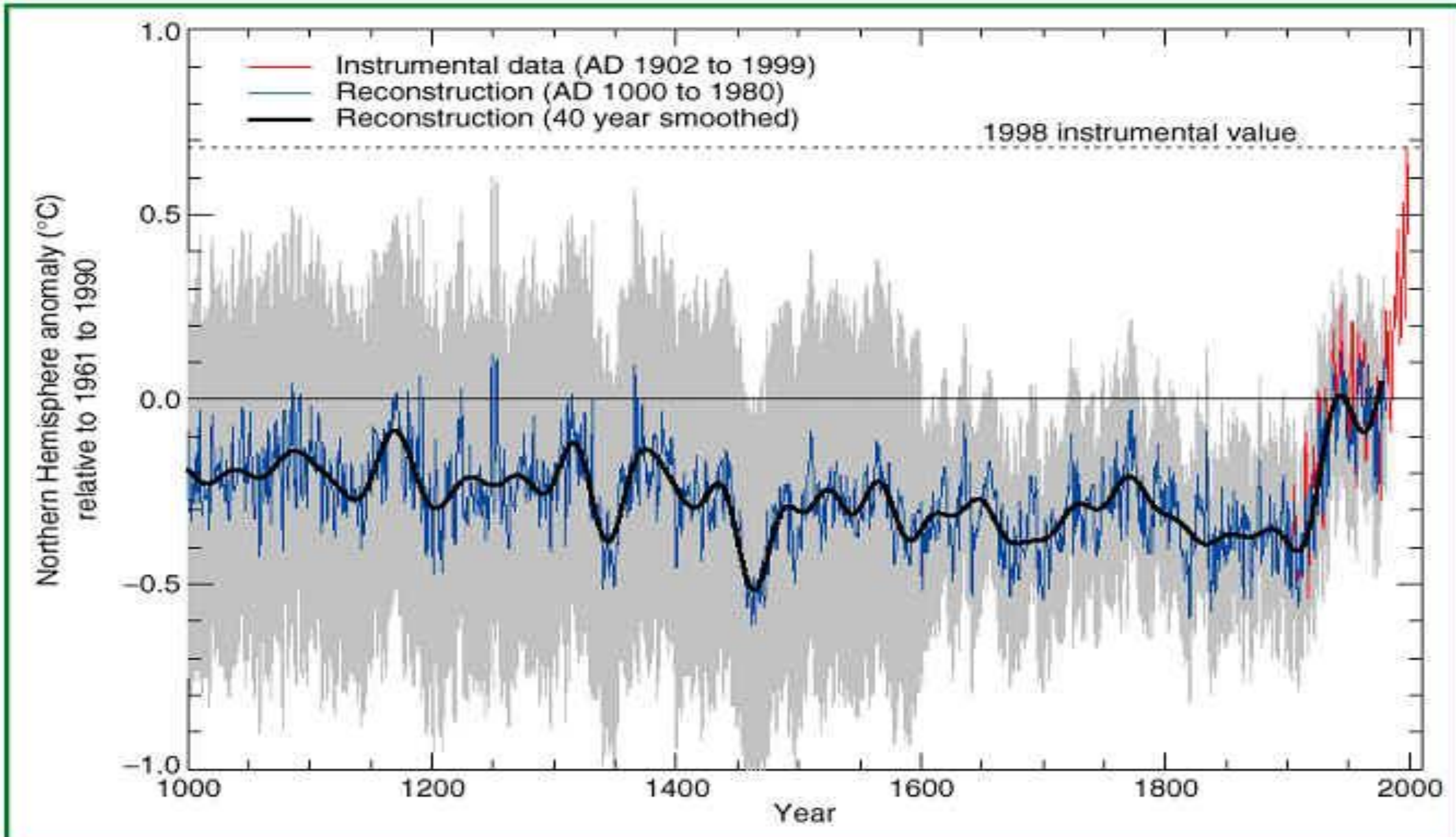


Figure 5: Millennial Northern Hemisphere (NH) temperature reconstruction (blue – tree rings, corals, ice cores, and historical records) and instrumental data (red) from AD 1000 to 1999. Smoother version of NH series (black), and two standard error limits (gray shaded) are shown. [Based on Figure 2.20]

Climate through the Ages - Recent Trends

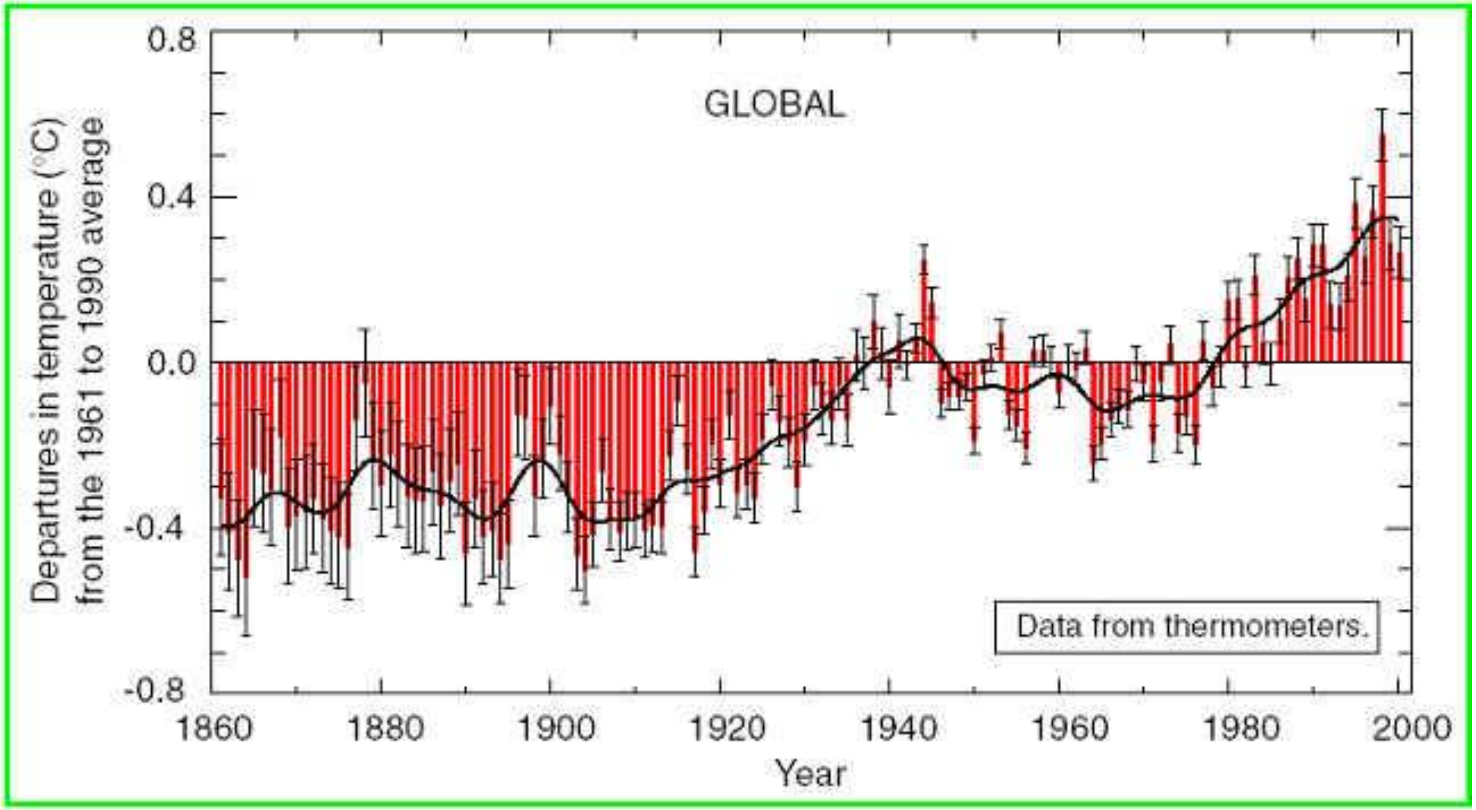
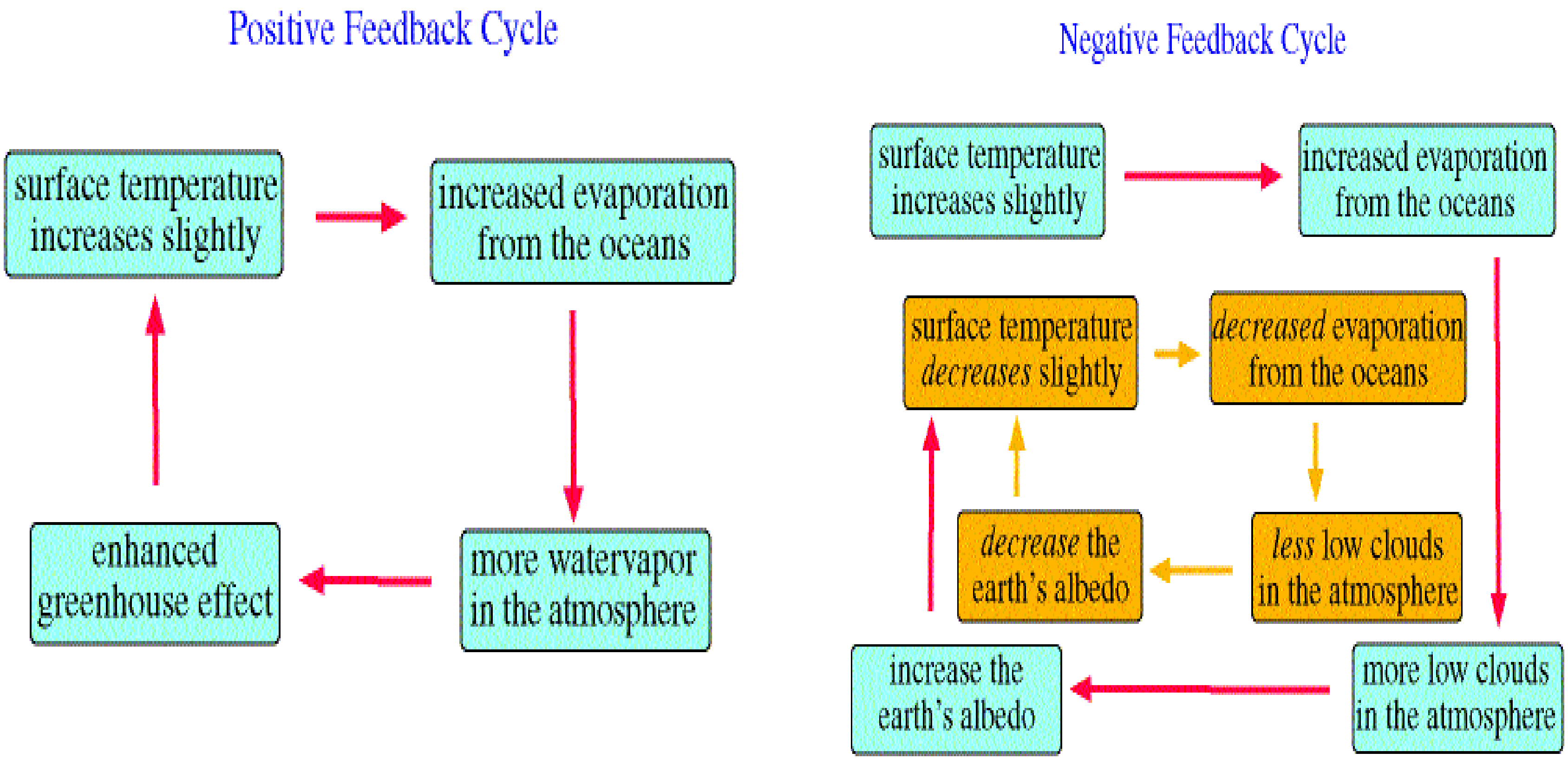


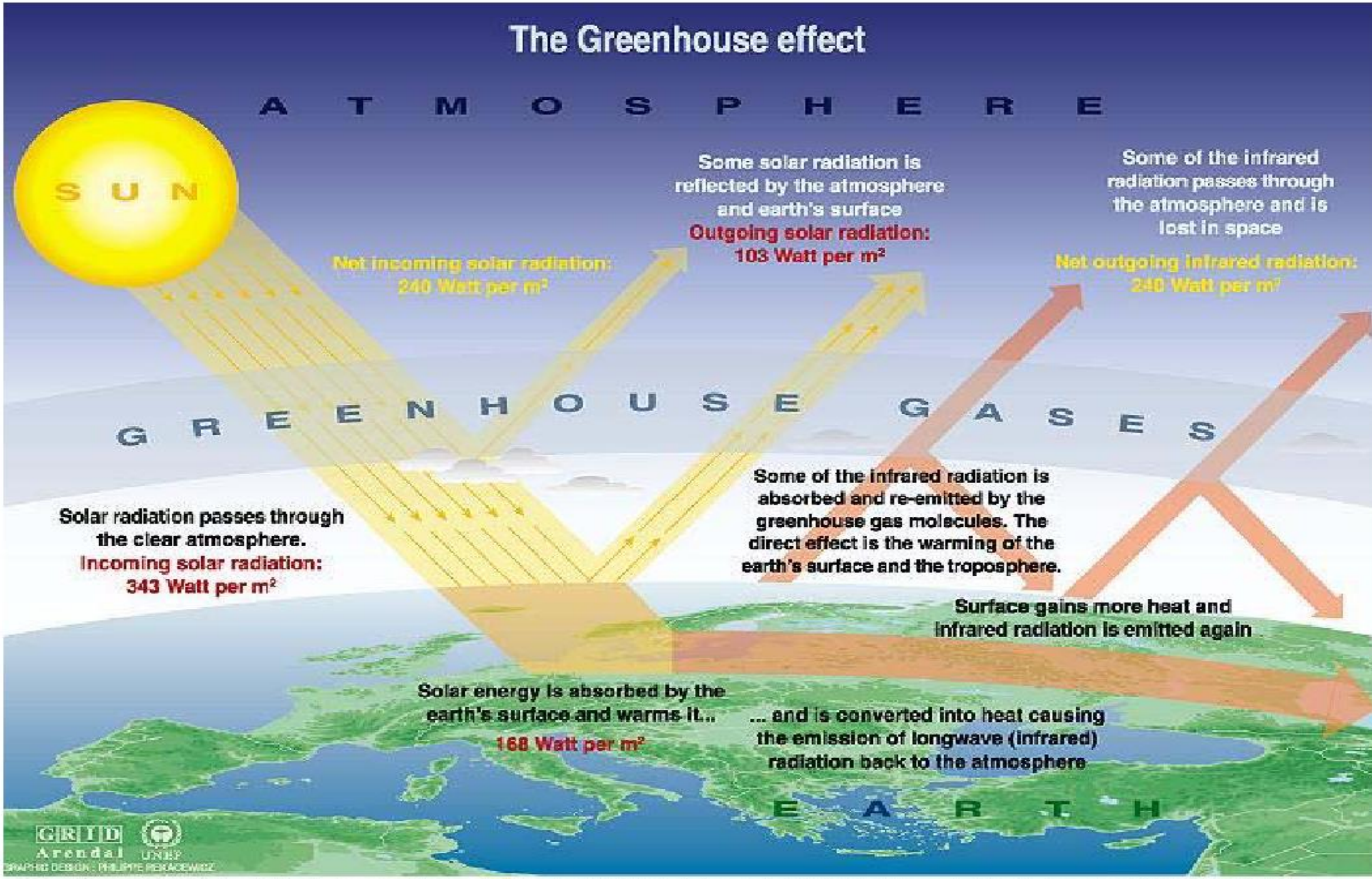
Figure 2: Combined annual land-surface air and sea surface temperature anomalies (°C) 1861 to 2000, relative to 1961 to 1990. Two standard error uncertainties are shown as bars on the annual number. [Based on Figure 2.7c]

Summary of Global Warming Observations

- Rate and duration of warming in 20th century is larger than any other time in last 1000 years.
- Total atmospheric water vapor has likely increased several percent per decade over many regions in NH
- Some subtropical regions are experiencing more dry spells
- Annual land precipitation has increased in the middle and high latitudes of the Northern Hemisphere.
- Increased cloud cover by about 2% since the beginning of the 20th century over the Northern Hemisphere.
- Decreasing snow cover and sea-ice amounts in NH. No significant trends in Antarctic sea-ice are apparent.
- Global mean sea-level rise during 20th century has been observed to be 1.0-2.0 mm per year.
- Increase in heavy and extreme precipitation events in regions where total precipitation has increased.

Climate Change and Feedback Mechanisms

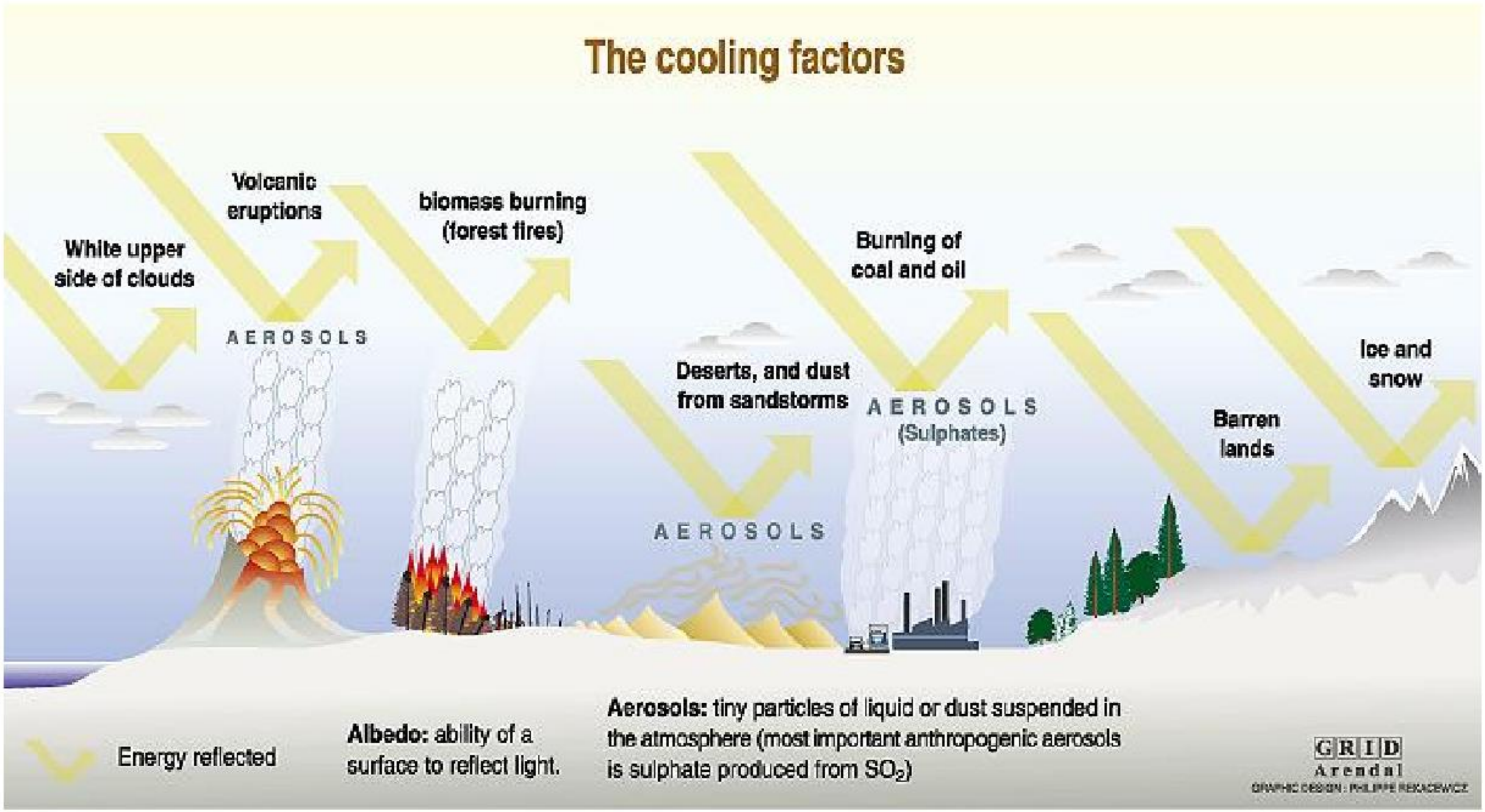




Sources: Okanagan university college in Canada, Department of geography, University of Oxford, school of geography; United States Environmental Protection Agency (EPA), Washington; Climate change 1995, The science of climate change, contribution of working group 1 to the second assessment report of the intergovernmental panel on climate change, UNEP and WMO, Cambridge university press, 1996.

Source:http://www.grida.no/climate/vital/03.htm

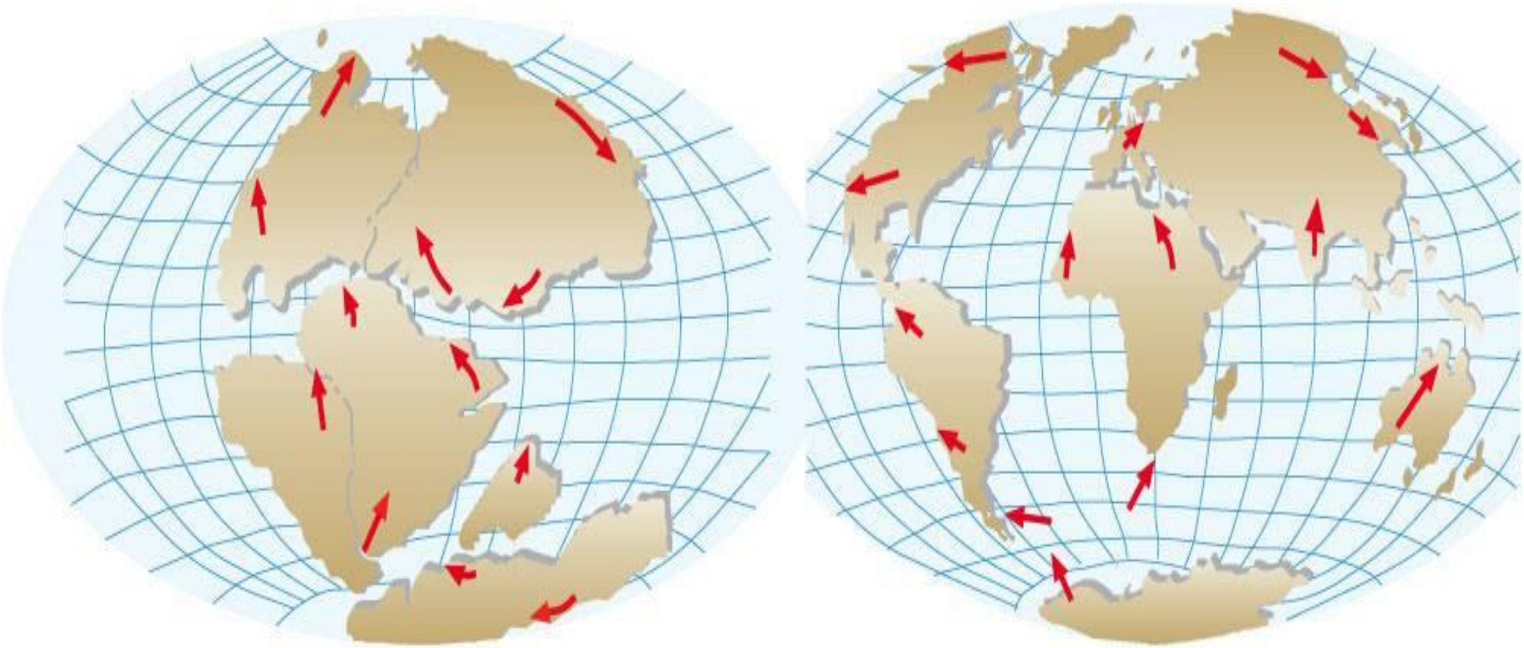
The cooling factors



Sources: Radiative forcing of climate change, the 1994 report of the scientific assessment working group of IPCC, summary for policymakers, WMO, UNEP; L.D. Danny Harvey, Climate and global environmental change, Prentice Hall, pearson Education, Harlow, United Kingdom, 2000.

Source: http://www.grida.no/climate/vital/14.htm

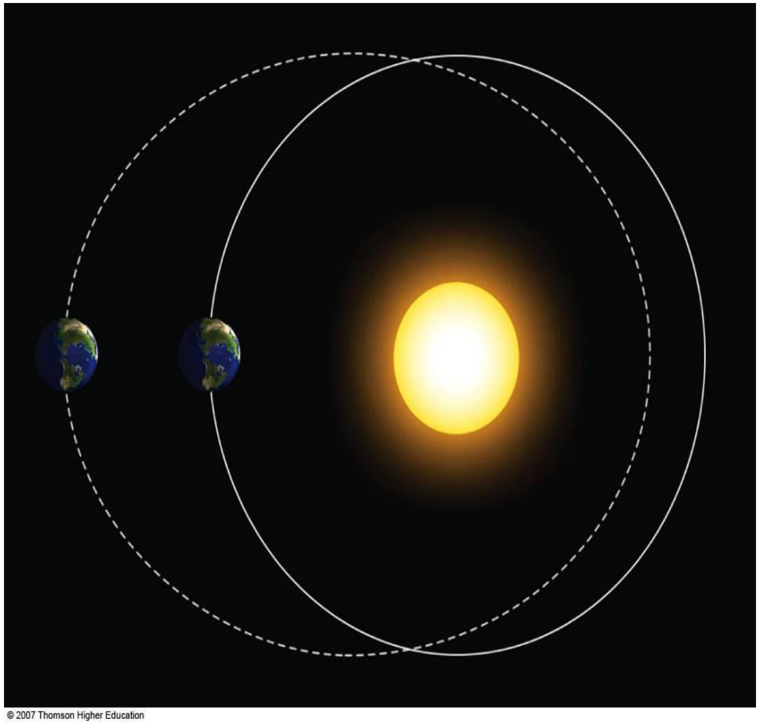
Climate Change - Plate Tectonics



(a) (b)

Climate Change - Milankovitch Theory - Eccentricity Cycle

- Climate change due to variations in the earth's orbit - Milankovitch Theory
- 1) **eccentricity cycle** - the earth's orbit around the sun is elliptical.
- the shape of the ellipse (eccentricity) varies from less elliptical to more elliptical back to less elliptical and take about 100,000 years to complete this cycle.



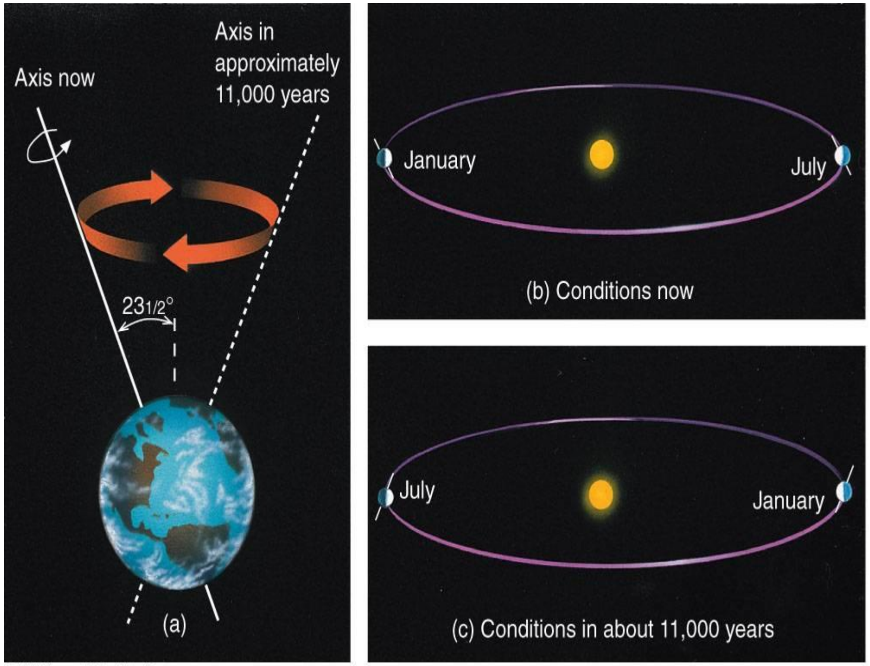
currently, we are in an orbit of low eccentricity (near circular).

Data analysis for the last 800,000 years of deep-ocean sediments show that ice coverage is a maxima every 100,000 years

this matches the Eccentricity cycle period

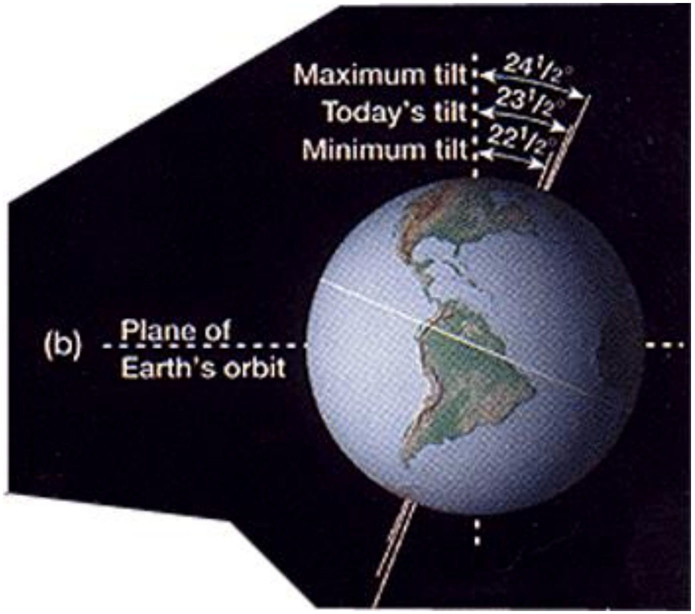
Climate Change - Milankovitch Theory- Precession Cycle

- Climate change due to variations in the earth's orbit - Milankovitch Theory
- 2) **Precession cycle -**
- The earth is wobbling about it's axis of rotation like a spinning top
- To make one complete cycle takes about 23,000 years
- in 11,000 years, the seasons will switch times during year and will be more severe



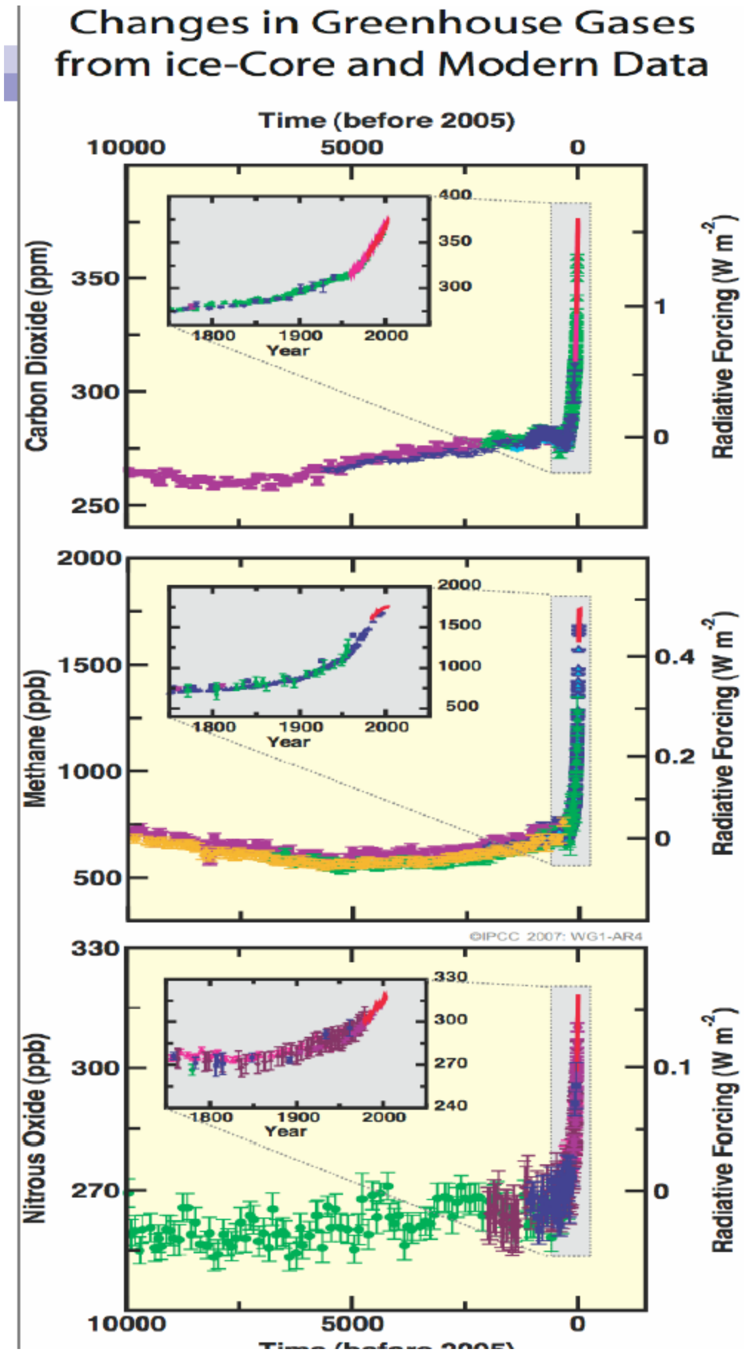
Climate Change - Milankovitch Theory- Tilt Cycle

- 2) **Tilt Cycle -**
- currently, the axis of rotation for the earth is tilted at 23.5°
- However, this value changes from a minimum of 22.5° to a maximum of 24.5° and takes 41,000 years to complete one cycle
- at 22.5° the seasonal variation will be smaller than current?
- at 24.5° the seasonal variation will be larger than current?



The Milankovitch cycles and plate tectonics are not the only natural factors which can affect global climate......, there are other factors to consider:

- ➢amount of dust and aerosols in the atmosphere
- ➢reflectivity of ice sheets
- ➢concentrations of trace gases
- ➢amount of clouds



Human activities have an impact on concentrations of major greenhouse gases

• concentrations relatively constant over the milleniuim before Industrial Revolution

• CO2 increased from 280ppm (parts per million, mole fraction of the gas relative to dry air) in 1750 to 367ppm in 1999. Rate of increase highest in at least 20000 years

• methan (CH4) concentration increased by 150%; emissions and removal of CH4 can be influenced by climate change (e.g. permafrost)

• N2O concentration highest in at least 1000 years

Source:IPCC WGI Fourth Assessment Report

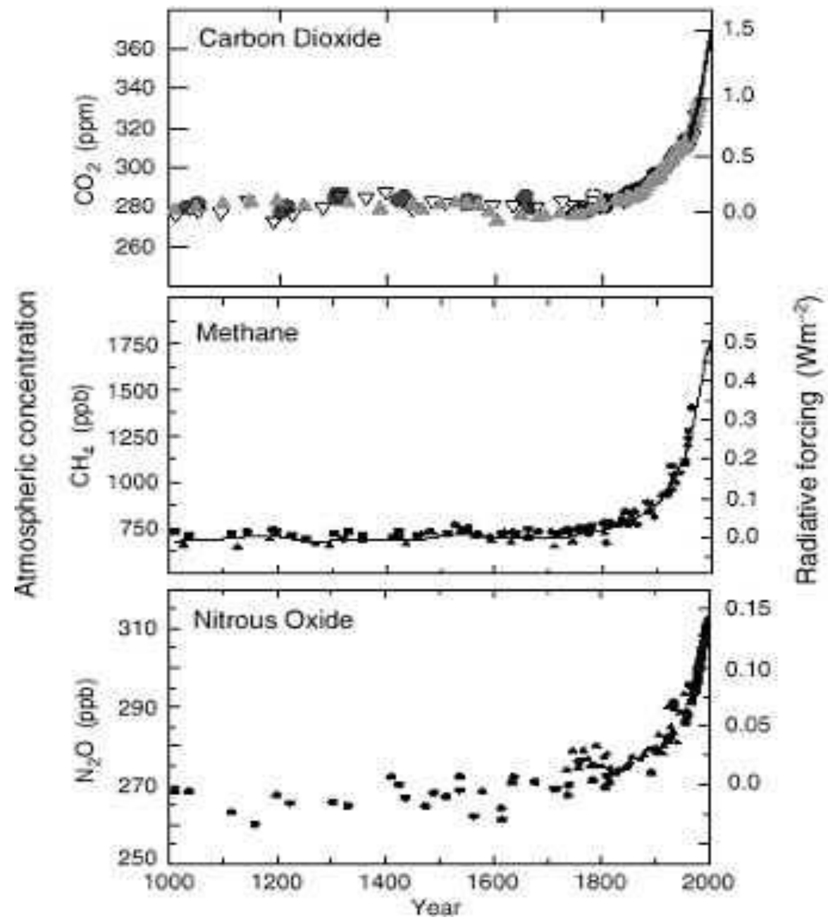
(Summary for Policymakers)

The Source of Global Warming

- is the observed warming over the last 50-100 years due to natural climate variability, human influence, or both?
- "…. natural forcing alone is unlikely to explain the recent observed global warming or the observed changes in vertical temperatures structure of the atmosphere."
- "In light of new evidence and taking into account the remaining uncertainties, most of the observed warming over the last 50 years is likely to have been due to the increase in greenhouse gas concentrations."

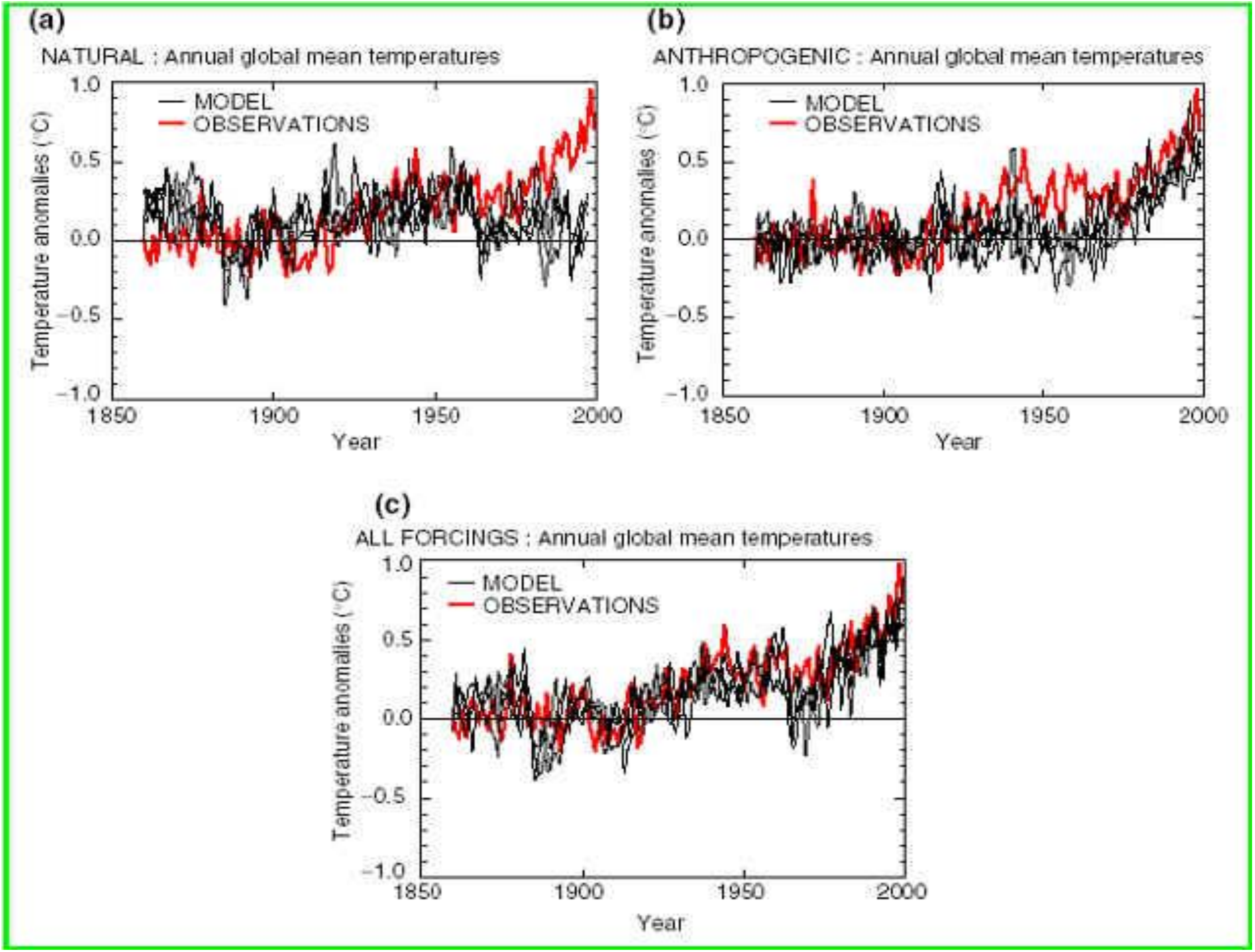
Climate Change - Increasing Concentrations of CO_2 and other Green House Gases

- CO_2 -
  - natural and anthropogenic sources
  - recent increase due to fossil-fuel combustion and deforestation
- CH_4 -
  - natural and anthropogenic sources
  - about 1/2 of current emissions are anthropogenic (land fills, natural gas, agriculture)
- N_2O-
  - natural and anthropogenic sources
  - nitrogen-based fertilizers
- Other important Greenhouse Gases:
  - CFCs
  - Ozone
  - and of course, water vapor!



Climate Change - CO_2 Feedback mechanisms

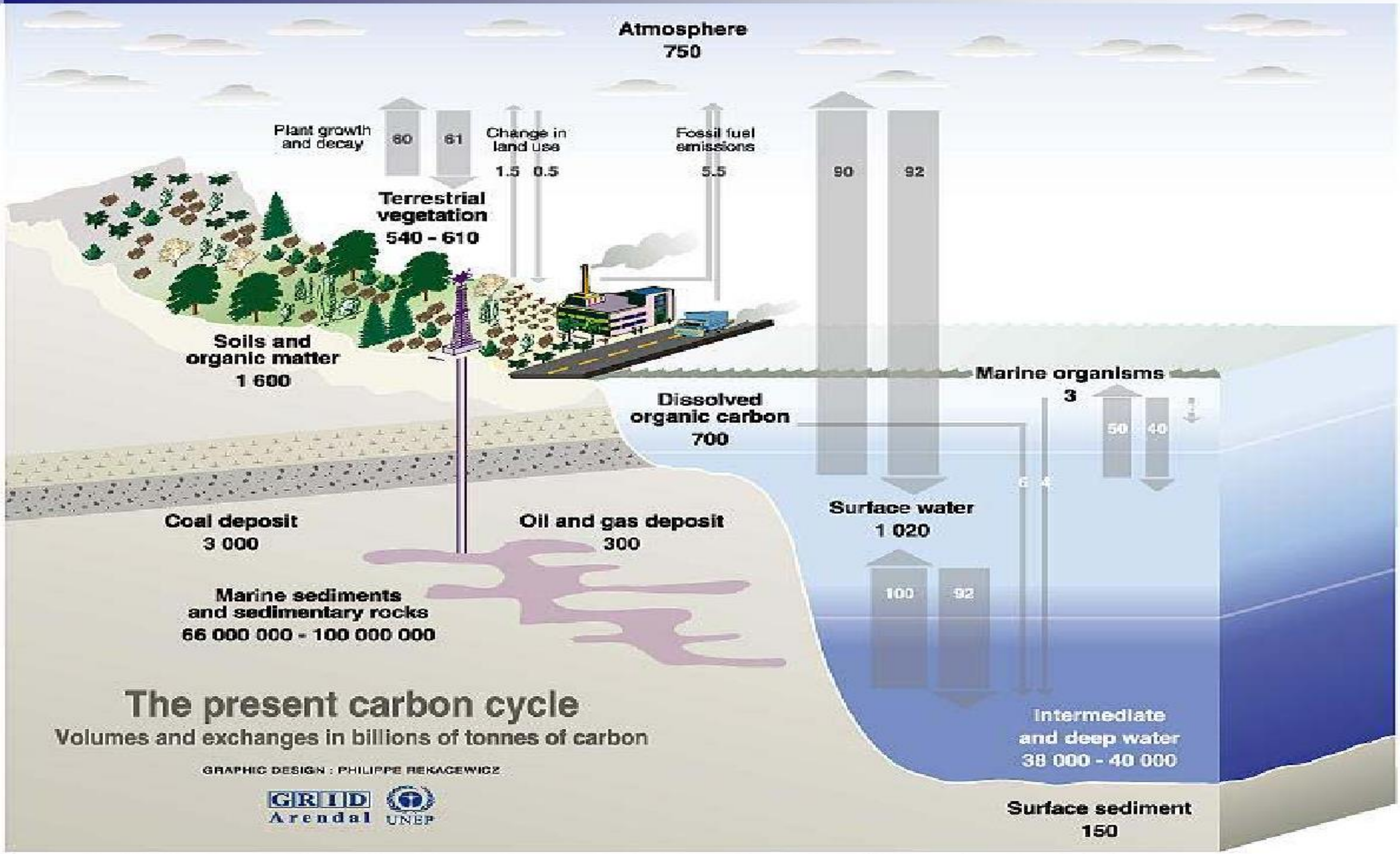
- other trace gases like N_2O, CH_4, and CFCs are also increasing - these are also greenhouse gases
- increased temps will enhance evaporation from oceans -> increased water vapor in atmosphere -> enhanced greenhouse effect
- increased temps will enhance evaporation -> increase amount of low clouds -> increase earth's albedo
- CO_2 will dissolve into the oceans
- Vegetation will remove CO_2 and grow more vigorously


(a)
NATURAL : Annual global mean temperatures
MODEL
OBSERVATIONS
Temperature anomalies (°C)
1.0
0.5
0.0
−0.5
−1.0
1850
1900
1950
2000
Year
(b)
ANTHROPOGENIC : Annual global mean temperatures
MODEL
OBSERVATIONS
Temperature anomalies (°C)
1.0
0.5
0.0
−0.5
−1.0
1850
1900
1950
2000
Year
(c)
ALL FORCINGS : Annual global mean temperatures
MODEL
OBSERVATIONS
Temperature anomalies (°C)
1.0
0.5
0.0
−0.5
−1.0
1850
1900
1950
2000
Year

The Main **Greenhouse Gases**

Greenhouse gases	Chemical formula	Pre-industrial concentration (ppbv)	Concentration in 1994 (ppbv)	Atmospheric lifetime (years)*	Anthropogenic sources	Global warming potential (GWP)**
Carbon dioxide	CO_2	278,000	358,000	Variable	Fossil-fuel combustion Land-use conversion Cement production	1
Methane	CH_4	700	1,721	12.2+/-3	Fossil fuels Rice paddies Waste dumps Livestock	21***
Nitrous oxide	N_2O	275	311	120	Fertilizer Industrial processes Combustion	310
CFC-12	CCl_2F_2	0	0.503	102	Liquid coolants Foams	6,200-7,100****
HCFC-22	$CHClF_2$	0	0.105	12.1	Liquid coolants	1,300-1,400****
Perfluoro-methane	CF_4	0	0.070	50,000	Production of aluminum	6,500
Sulfur hexa-fluoride	SF_6	0	0.032	3,200	Dielectric fluid	23,900

Source: PEW Center (www.pewclimate.org/global-warming-basics)



Sources: Center for climatic research, Institute for environmental studies, university of Wisconsin at Madison; Okanagan university college in Canada, Department of geography; World Watch, November-December 1998; Climate change 1995, The science of climate change, contribution of working group 1 to the second assessment report of the intergovernmental panel on climate change, UNEP and WMO, Cambridge press university, 1996.

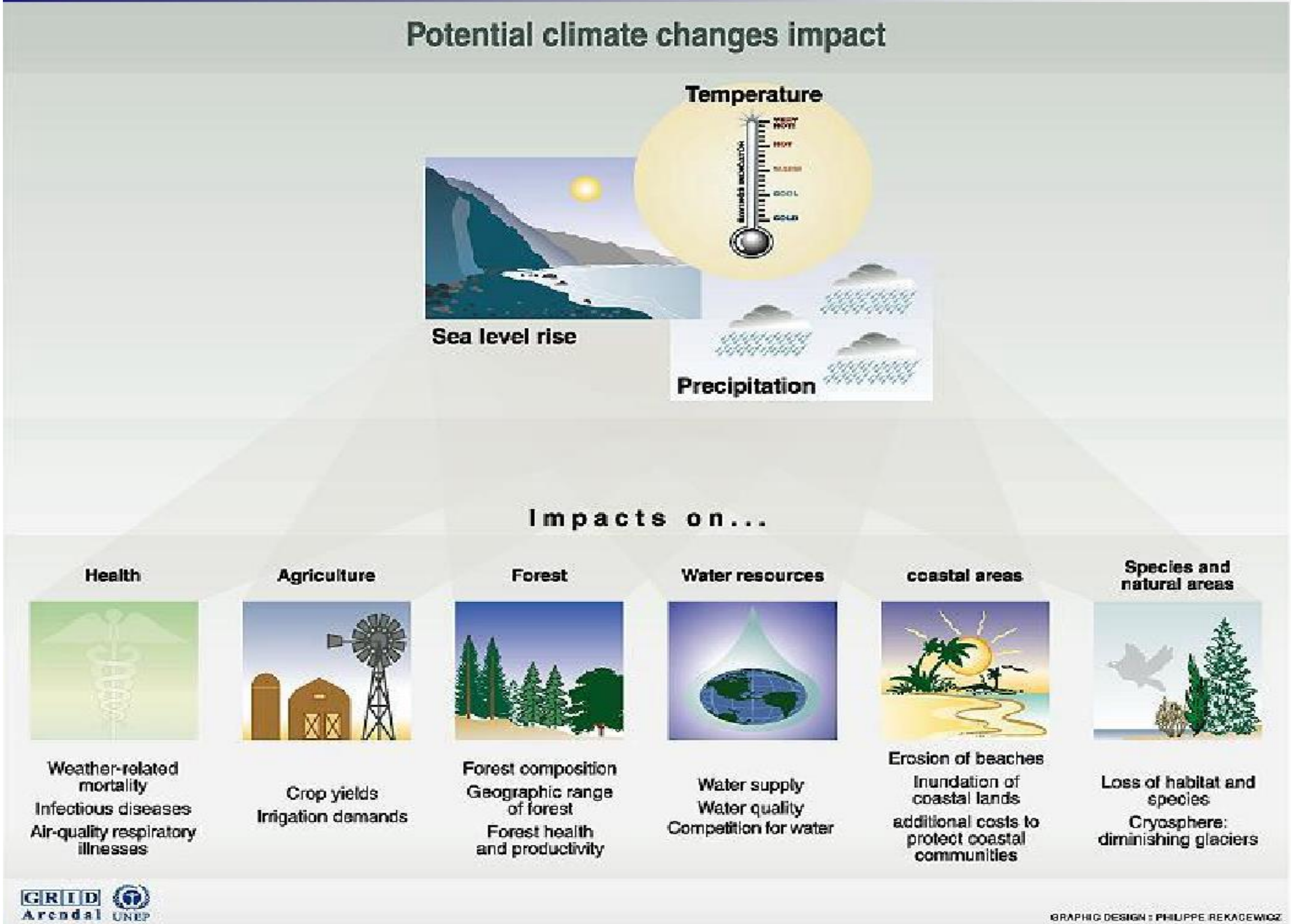
Source:http://www.grida.no/climate/vital/13.htm

Climate Change - Possible Consequences of Global Warming

- Temperature:
- Globally averaged surface temperature is projected to increase by 1.4 to 5.8 ºC over the period 1990-2100.
- Greater relative warming in the higher latitudes
- Land areas will warm more and faster than ocean areas
- "The projected rate of warming is much larger than the observed changes during the 20th century and is very likely to be without precedent during at least the last 10,000 years, based on palaeoclimate data".
- Precipitation:
- Globally averaged water vapor, evaporation and precipitation are projected to increase.
- NOTE: At the regional scale, increases and decreases will be observed.

Climate Change - Possible Consequences of Global Warming

- Extreme Weather:
- More hot days and heat waves are likely over nearly all land areas
- Frost days and cold waves are very likely to become fewer
- Frequency of extreme precipitation events is projected to increase
- General drying (increased drought frequency) of all mid-continental areas during summer.
- Glaciers and Snow Cover:
- Glaciers and ice caps will continue their widespread retreat during the 21st century and NH snow cover and sea ice are projected to decrease further.
- NOTE: The Antarctic ice sheet is likely to gain mass because of greater precipitation
- Sea Level Rise:
- A sea level rise of 0.09 – 0.88 meters is projected for 1990-2100.
  - Thermal expansion of the oceans
  - Loss of mass from glaciers and ice caps



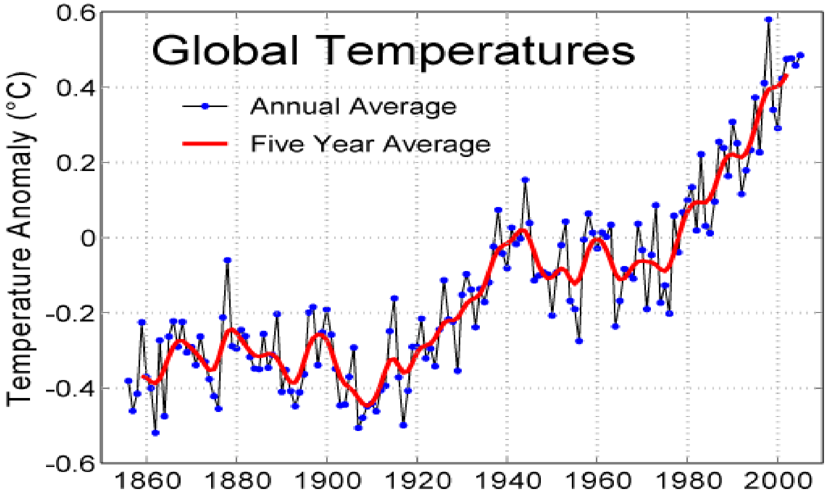
Source: United States environmental protection agency (EPA).

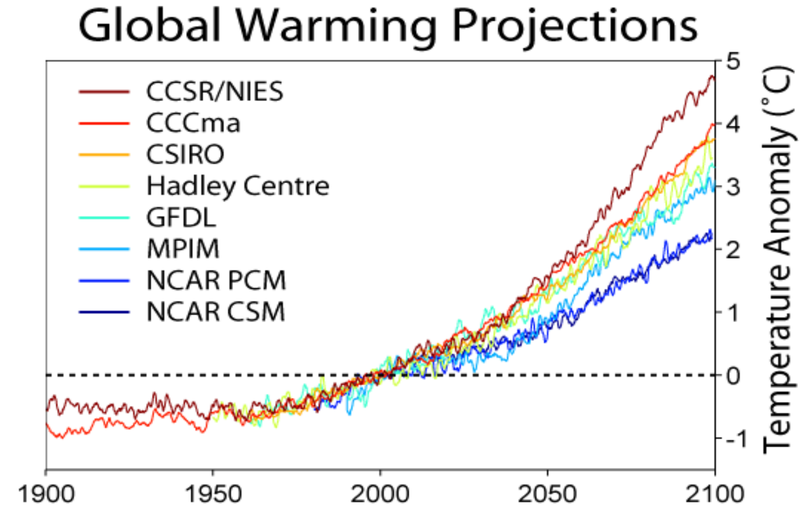
Source:http://www.grida.no/climate/vital/20.htm

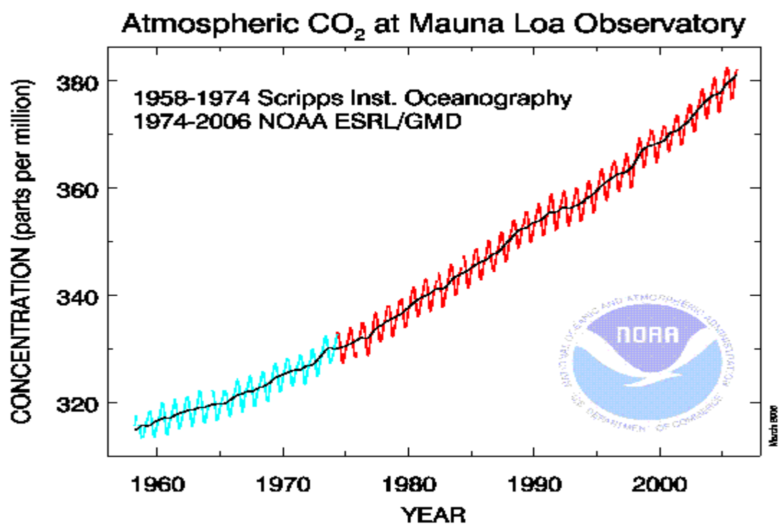
Evidence for changes in the climate

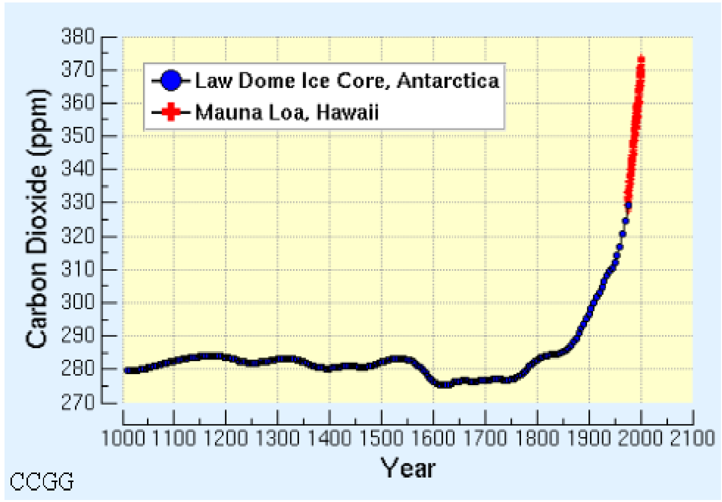
- Widespread retreat of mountain glaciers in non-polar regions
- Northern hemisphere sea-ice extent decreased by 10-15% since 1950s
- Thawing of permafrost
- Global average sea level has risen and ocean heat content has increased
- Changes in precipitation (increases over mid- and high-latitudes of Northern Hemisphere, decreases in subtropical regions)
- Changes in cloud cover
- No compelling evidence to indicate that characteristics of tropical and extratropical storms have changed (IPCC2001)

How Real Is Global Warming ?



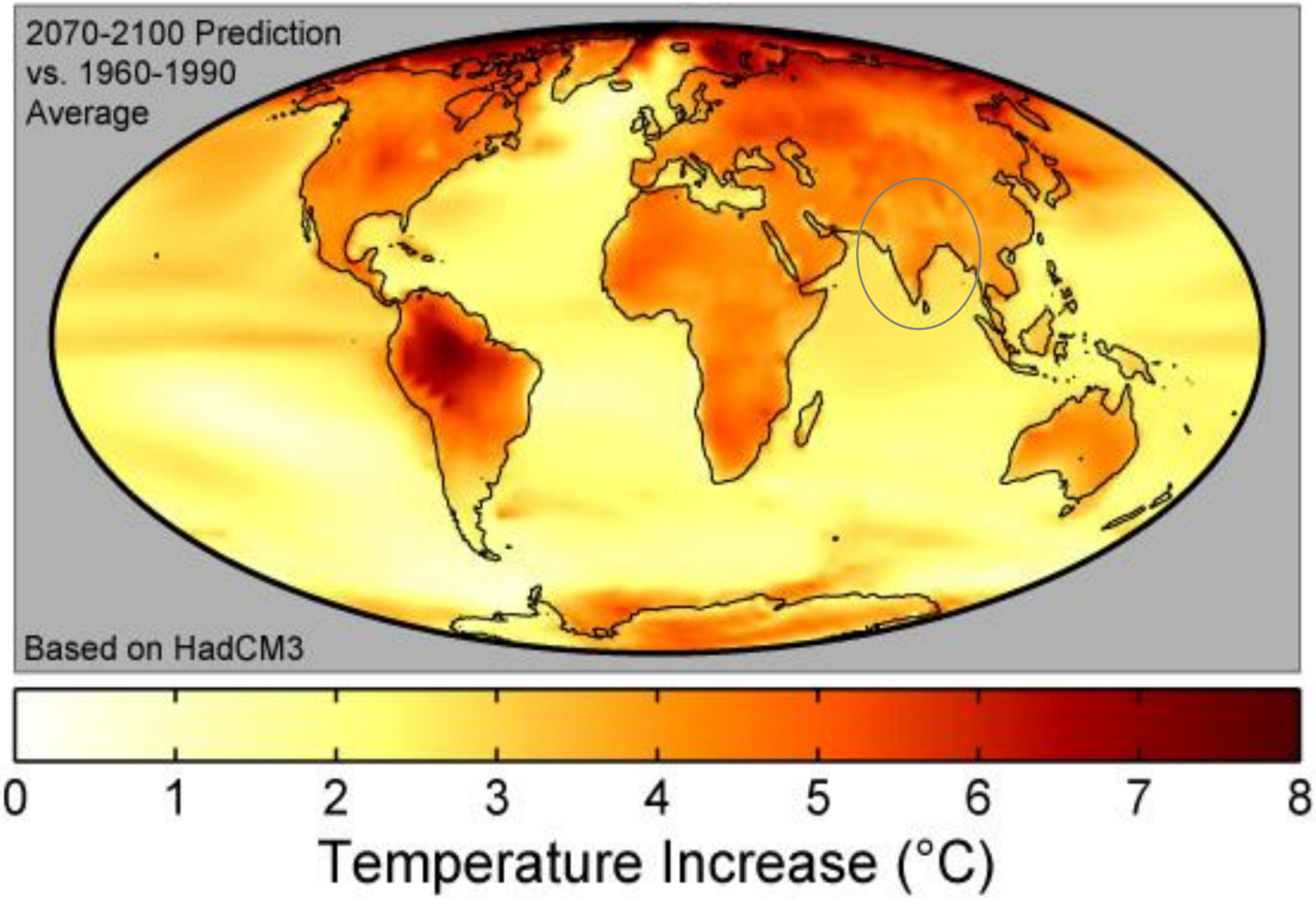






Source : Compilations from IPCC & other sources, as above

What do "Average Temperature Increases" Conceal ?



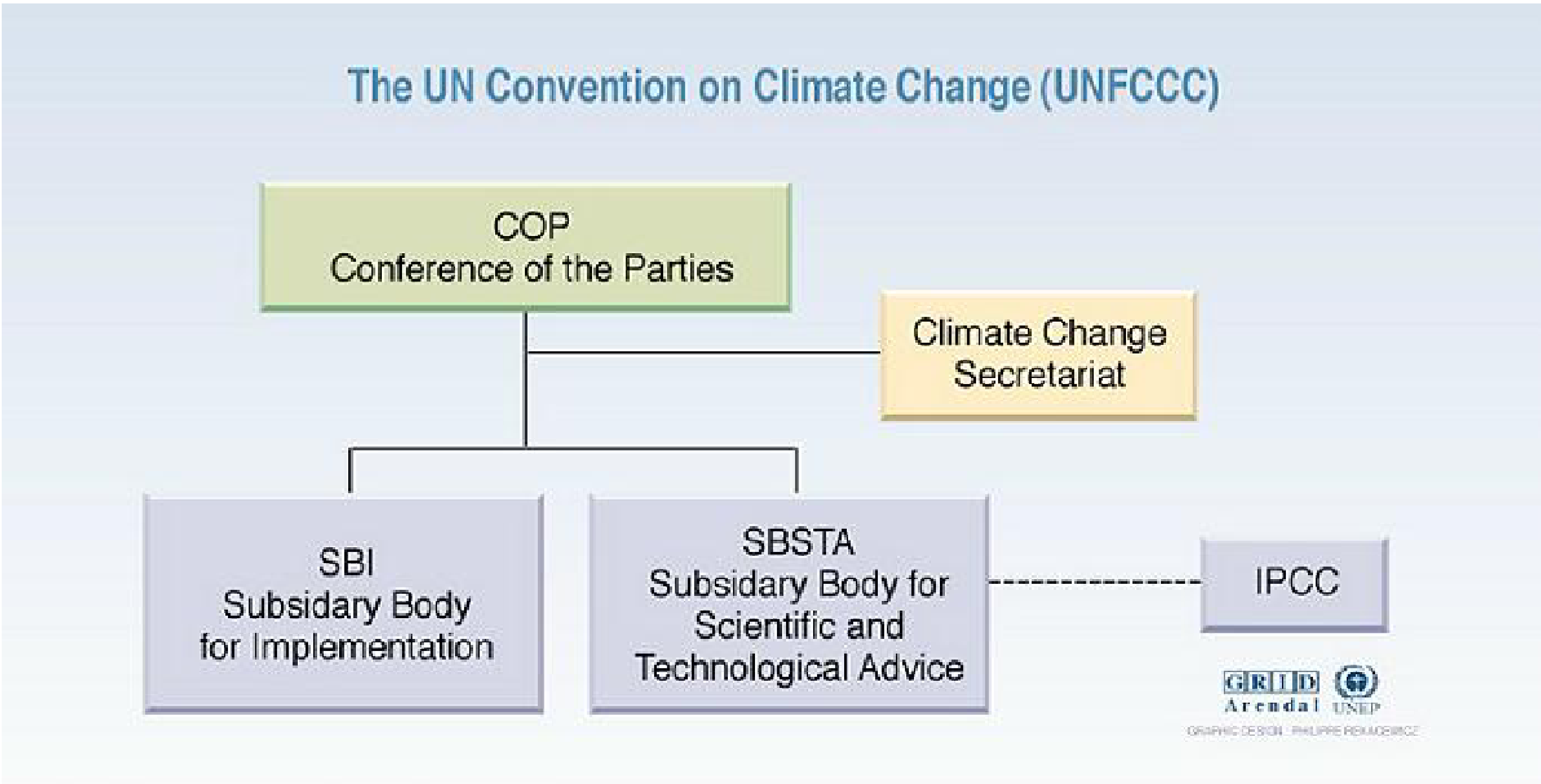
Over land, warming is higher than global "average" …

Policy implications

- Two categories of instruments need to be considered:
- Domestic policy instruments to enable individual nations to achieve their specific targets/goals
- International policy instruments to allocate responsibility among nations
- Main criteria
- Cost effectiveness (Minimum aggregate costs)
- Dynamic incentives for technology innovation and diffusion
- Adaptability to economic and social changes
- Distributional equity
- Institutional (political and administrative) feasibility

How to combat?

- Green house gas emission reduction - a collective target
- Collective responsibility
- Bilateral or multilateral policy
- Voluntary agreement leads to free riding and leakages
- Free riding arises when countries that do not contribute to reduction also benefits
- Emission leakage occurs when abatement by cooperating countries alters world relative prices and leads noncooperating countries to increase emissions
- Kyoto Protocol – an agreement made under UNFCCC
- Countries that ratify agreed to reduce the emissions below a target level or engage in emissions trading if they maintain or increase the green house gases
- Objective: Stabilization of the greenhouse gas concentrations in the atmosphere at a level that would prevent dangerous anthropogenic interference with the climate system



Source: United Nations framework convention on climate change (UNFCCC).

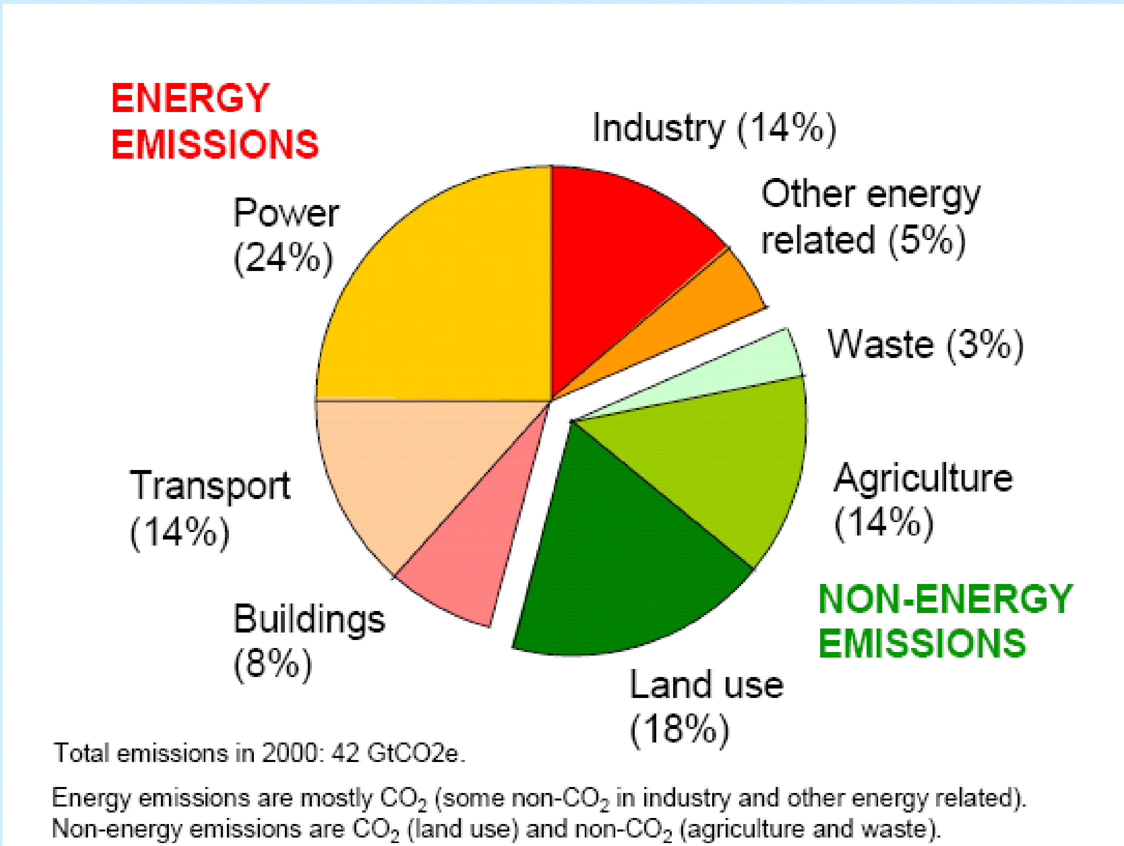
- UNFCCC (1992) as foundation of global efforts to combat global warming
- Objective: "stabilization of greenhouse gas concentrations in the atmosphere at a level that would prevent dangerous anthropogenic human-induced interference with the climate system.“
- Ratified by 180 states

Source: http://www.grida.no/climate/vital/15.htm

Carbon Markets

The main sources of Greenhouse Gases… globally

Figure 1 Greenhouse-gas emissions in 2000, by source



Source: Prepared by Stern Review, from data drawn from World Resources Institute Climate Analysis Indicators Tool (CAIT) on-line database version 3.0.

of global GHG emissions,

65 % are CO_2 emissions from using fossil fuels (29 GigaTonnes)

18 % are CO_2 releases due to deforestation (8 GigaTonnes)

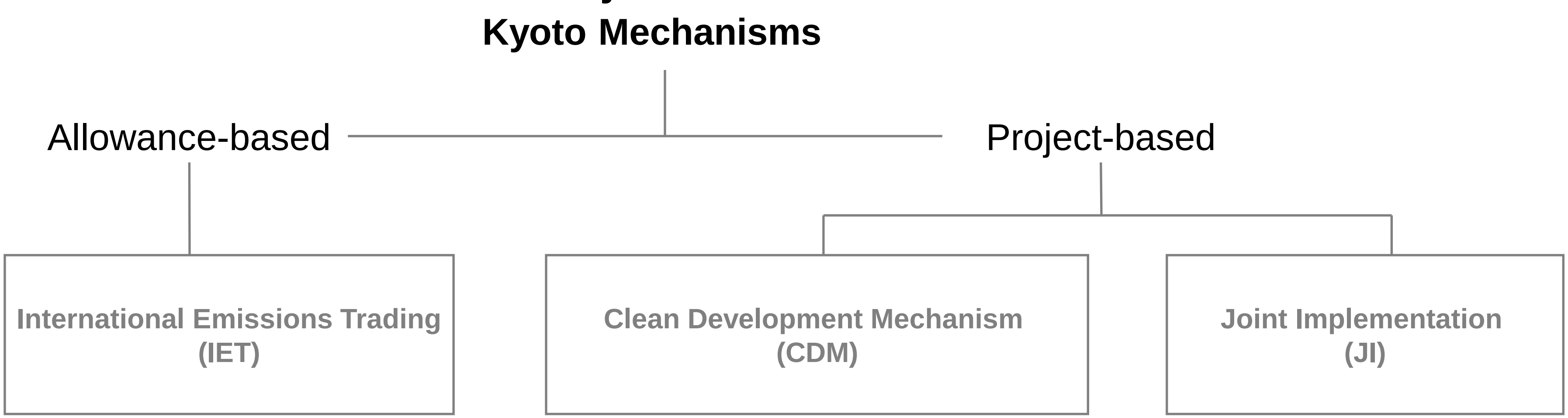
Source : Stern Review

Introduction to Carbon Markets

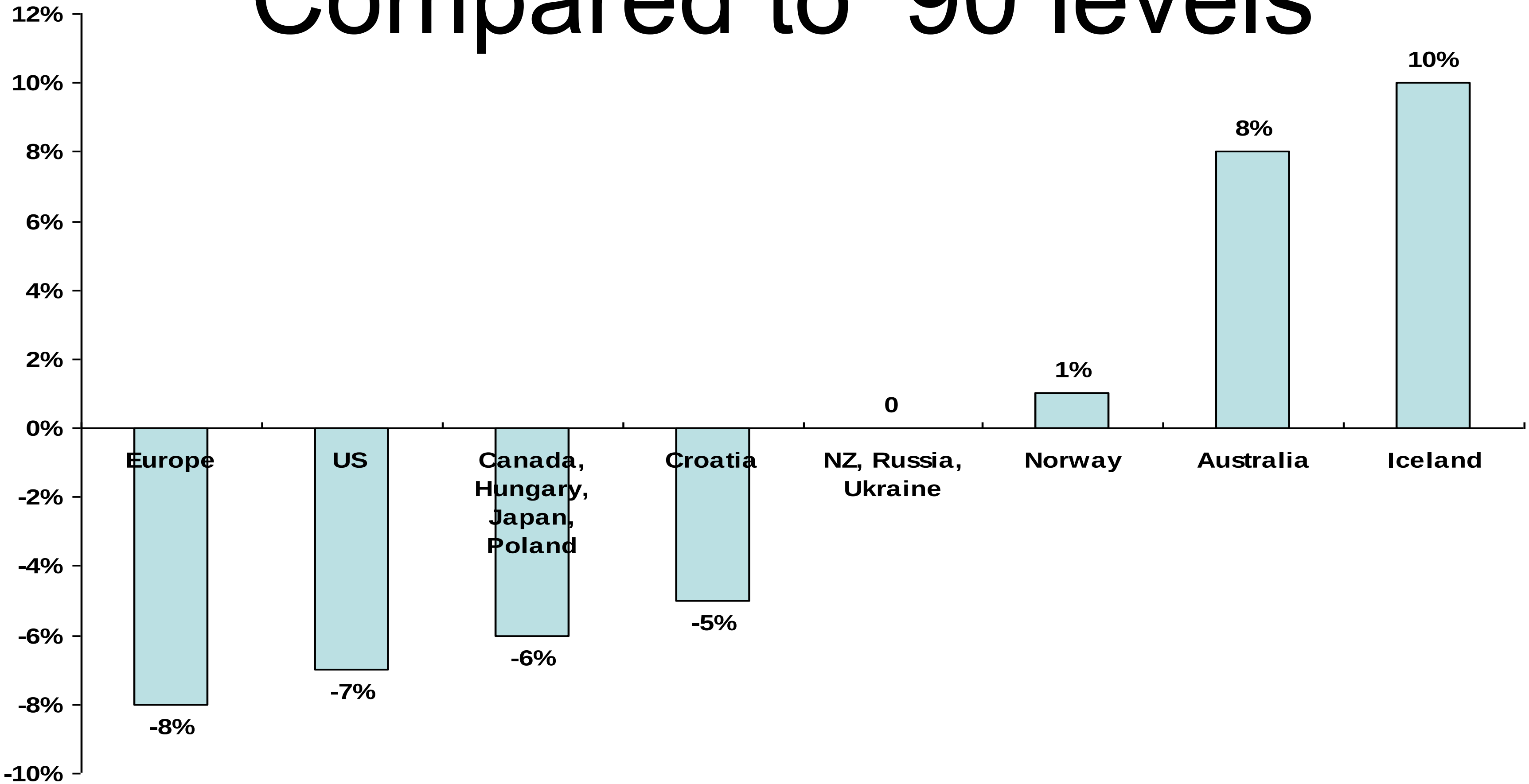
- **Carbon dioxide is the new kid on the block in commodity trading.**
- **Fundamental difference – people are buying and selling a *lack or absence* of the commodity in question.**
- **Typical sellers fall into two categories.**
- **Sellers within Annex I countries will produce or expect to produce less than they are allowed to produce, so that they may sell the unused right to emit to someone who emits more than their allocated amount.**
- **In CDM countries, sellers will gain credits from project based reductions and sell these to Annex I parties who emit more than their allocated amount.**
- **The emissions markets today are geared to achieving levels specified under the Kyoto Protocol by the UNFCCC.**

Introduction to Kyoto Protocol

- Voluntary treaty signed by 141 countries, including the EU, Japan and Canada
- Target for reducing GHG emission by 5.2% below 1990 levels by ’12
- US, with over a fourth of global emissions, is not a party to the agreement.
- 2 types of transactions are allowed under Kyoto – Allowance-based, and Project-based



Kyoto Emissions Reductions – Compared to ’90 levels



NOTES:
•Europe includes EU-15 includes Bulgaria, Czech Republic, Estonia, Latvia,Liechtenstein, Lithuania, Monaco, Romania, Slovakia, Slovenia, Switzerland
•The EU’s 15 member States will redistribute their targets among themselves, taking advantage of a scheme under the Protocol known as a “bubble”. The EU has already reached agreement on how its targets will be redistributed
•Some EITs have a baseline other than 1990.
•US has not ratified the Kyoto Protocol

Source: UNFCCC